Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)
```

```
    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_rabbit

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_rabbit

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

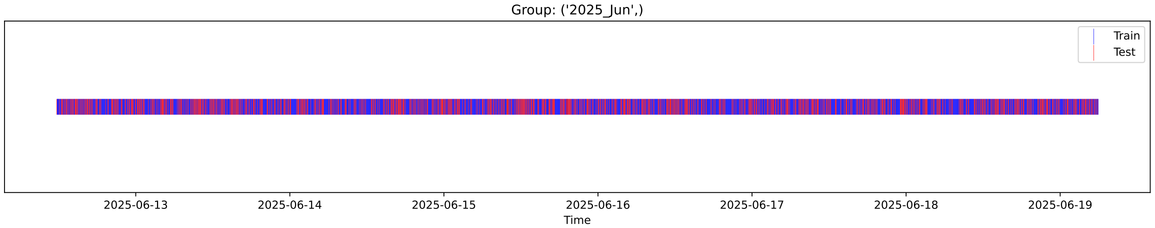
[Green] r_test > 1.05 (over performance of the test)

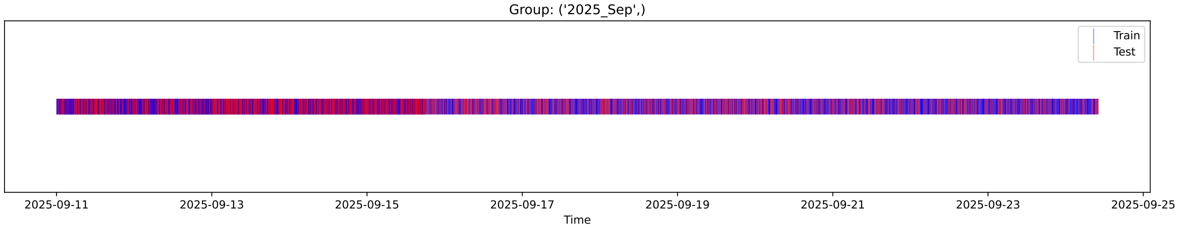
[Red] r_train not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

External Split: Train: 5717 samples Test: 2451 samples

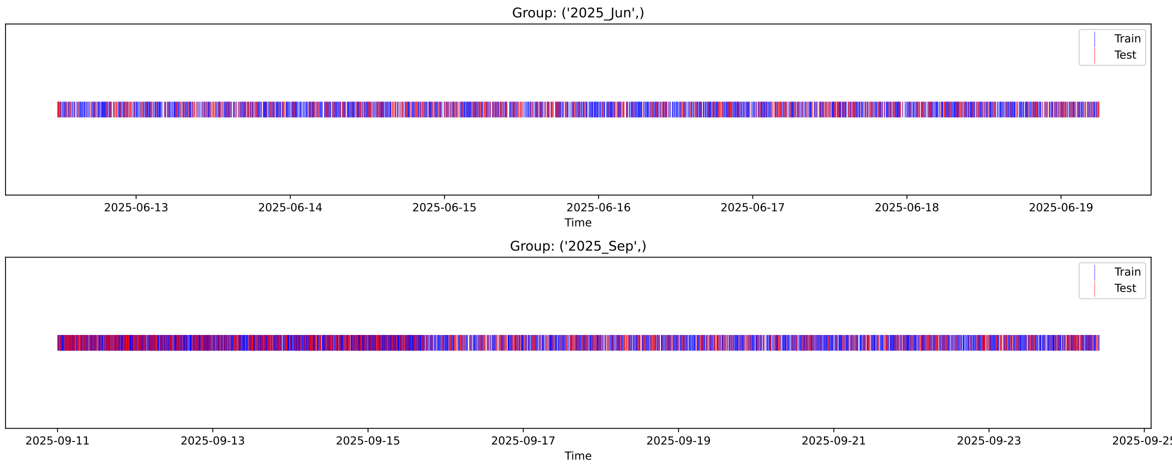




2.2.2 Internal CV Folds

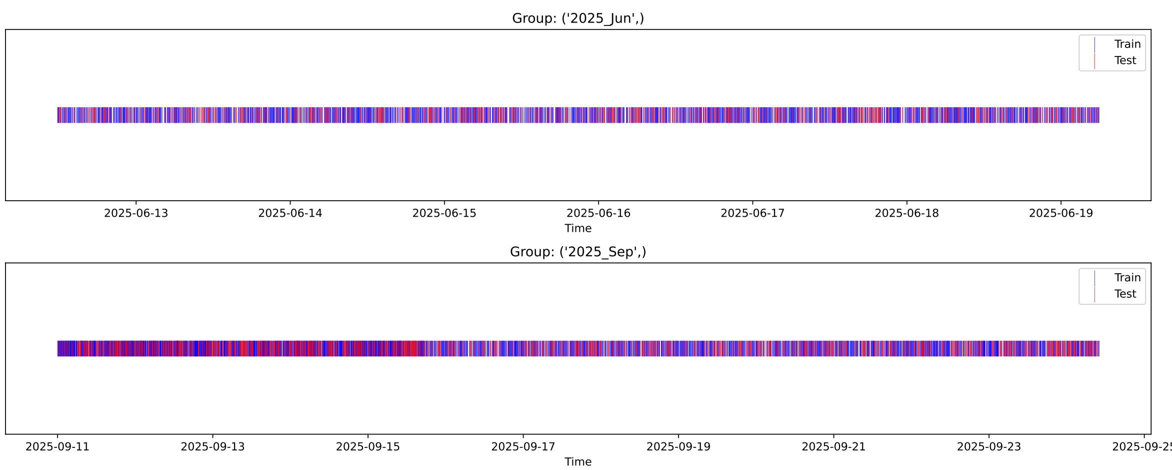
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



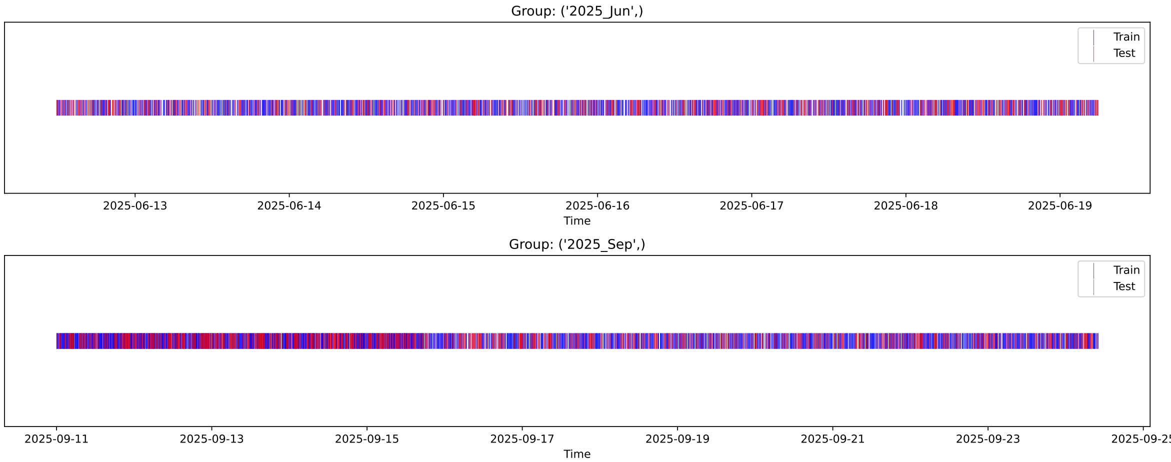
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



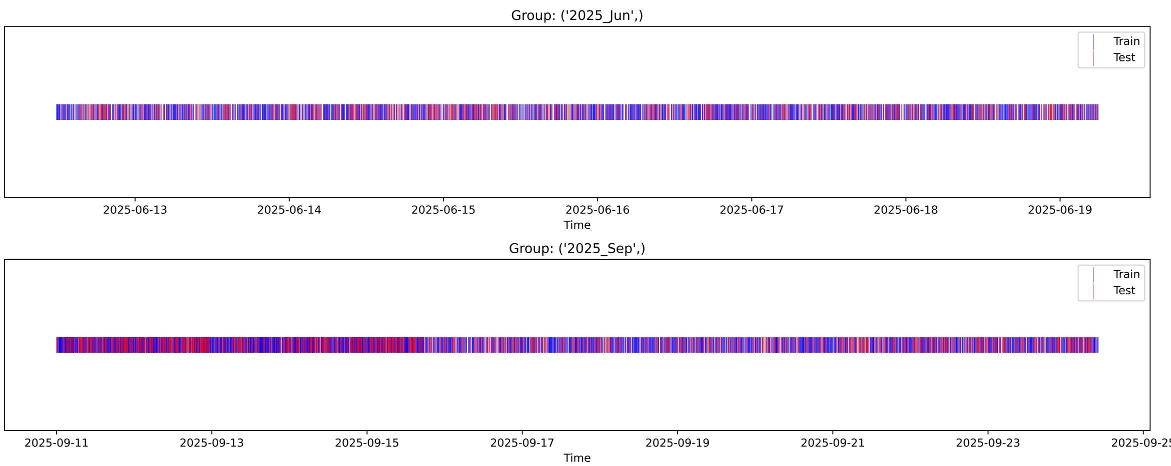
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



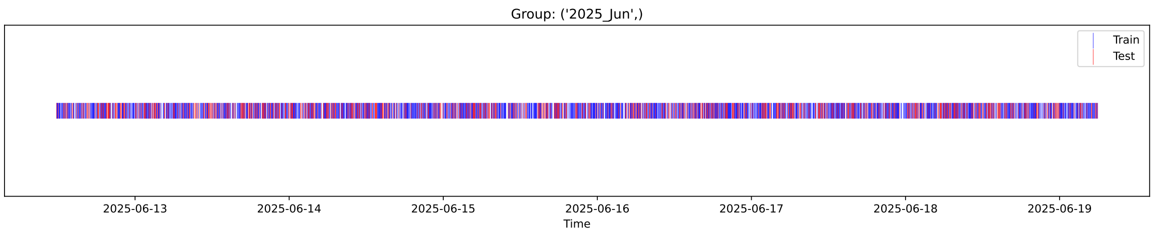
Fold 4/5

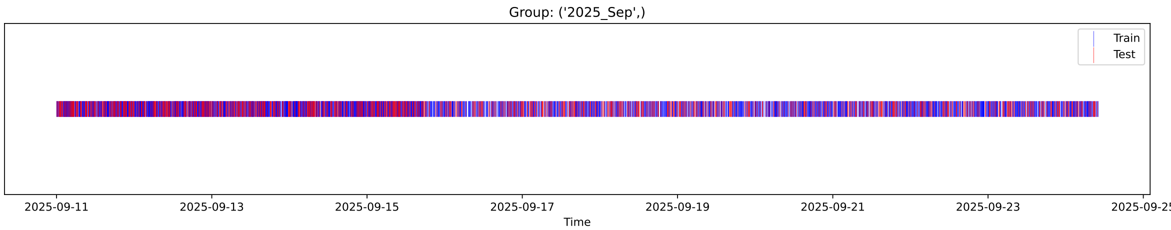
Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h





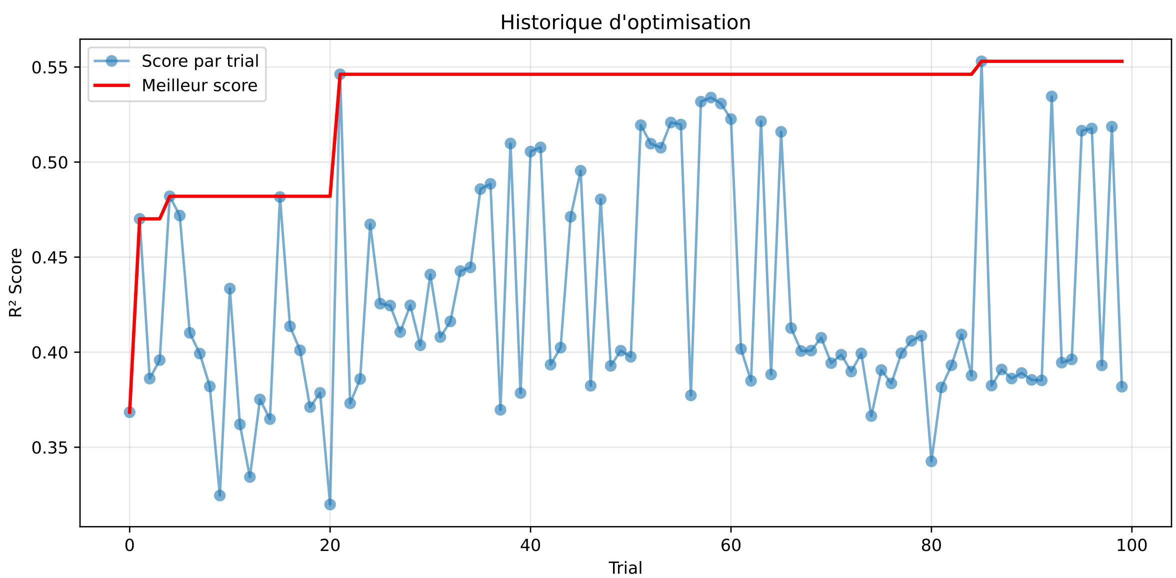
2.3 Model : LGBM

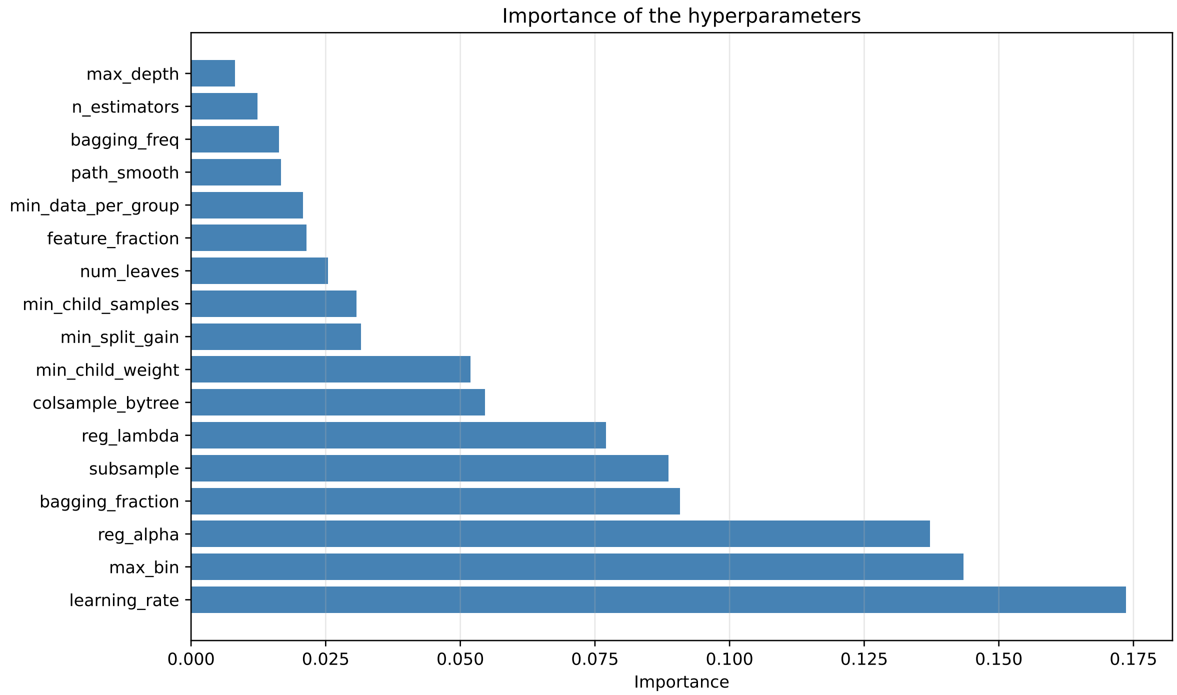
2.3.1 Gridsearch — Group: 1

Len(group) : 795, **Len(training)** : 557, **Len(test)** : 238

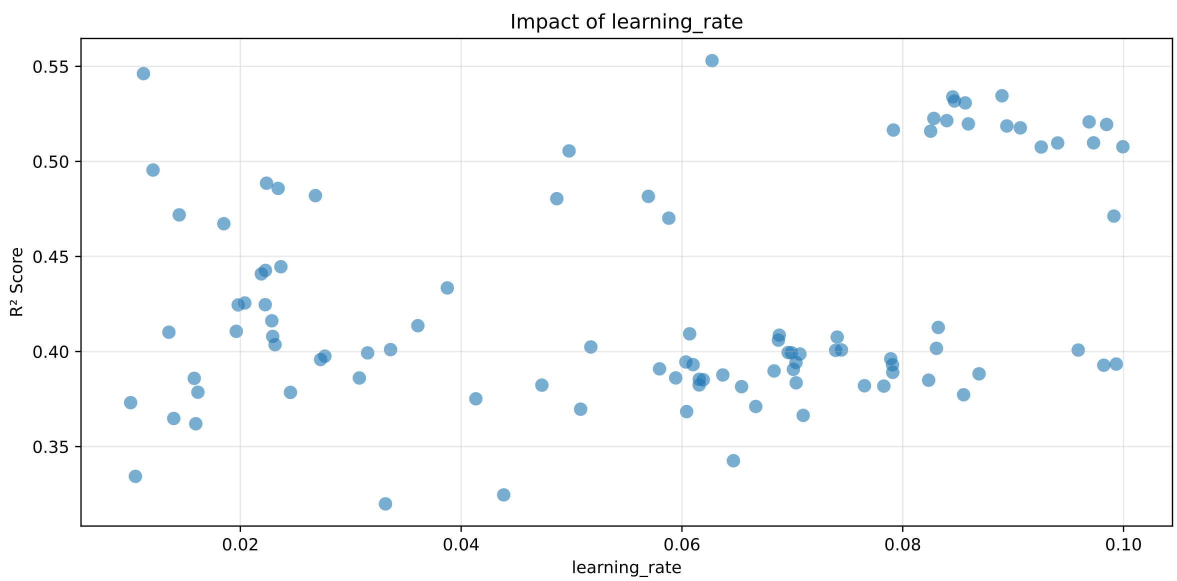
Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986 y_test: mean=2.492, std=0.625, min=1.466, max=5.447

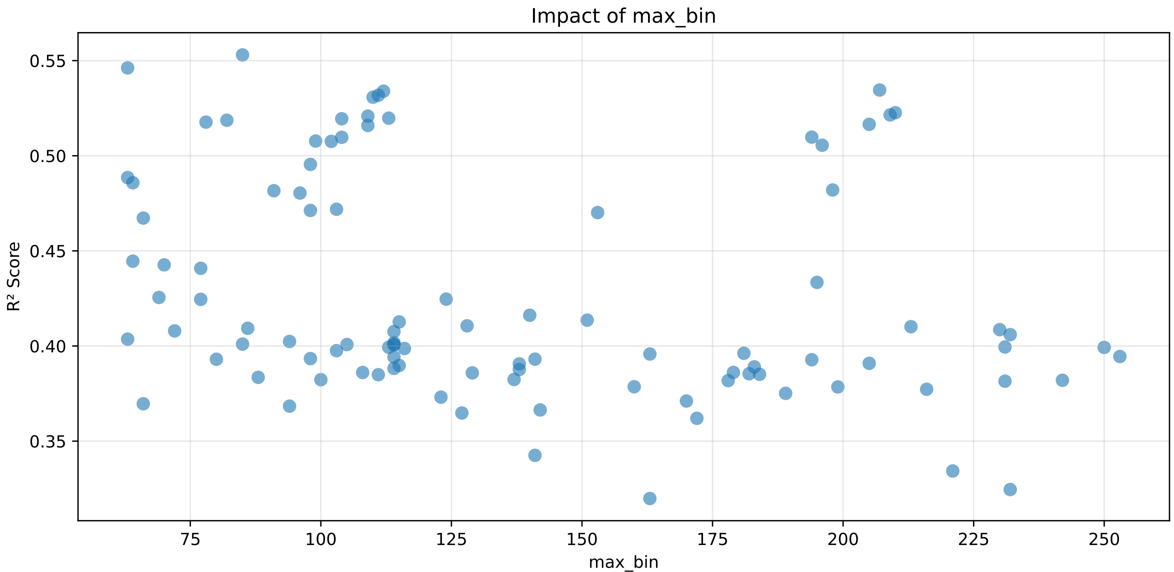
==================== TOP Importance FEATURES ==================== feature importance r_temperature 49 r_umidity 30 r_CO2 18 r_THI 9 r_NH3 3

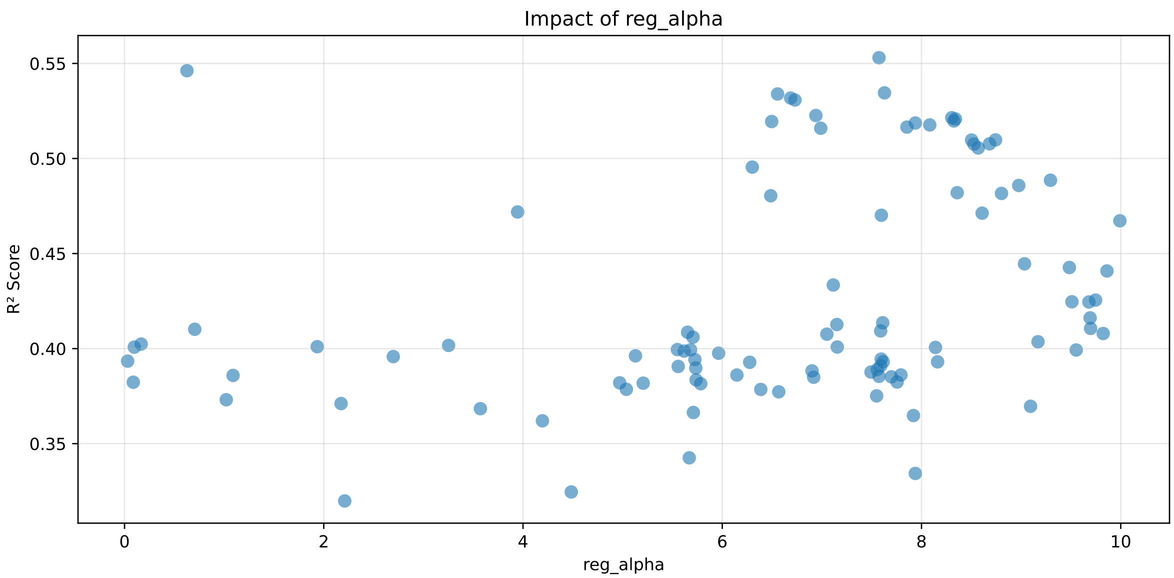


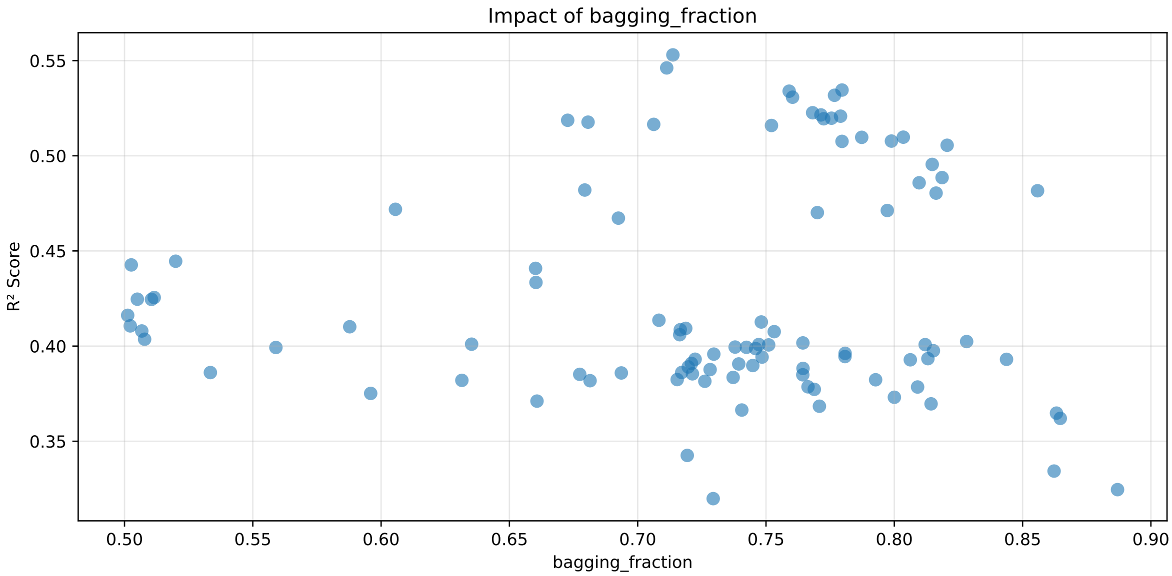


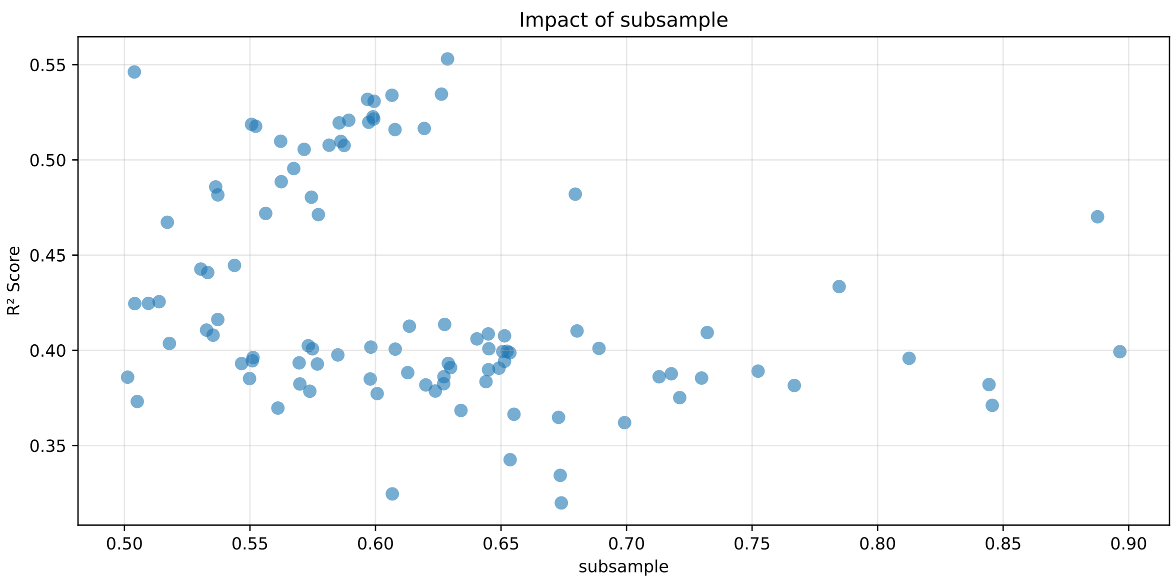
Paramètres avec >5% d'importance : ['learning_rate', 'max_bin', 'reg_alpha', 'bagging_fraction', 'subsample', 'reg_lambda', 'colsample_bytree', 'min_child_weight']

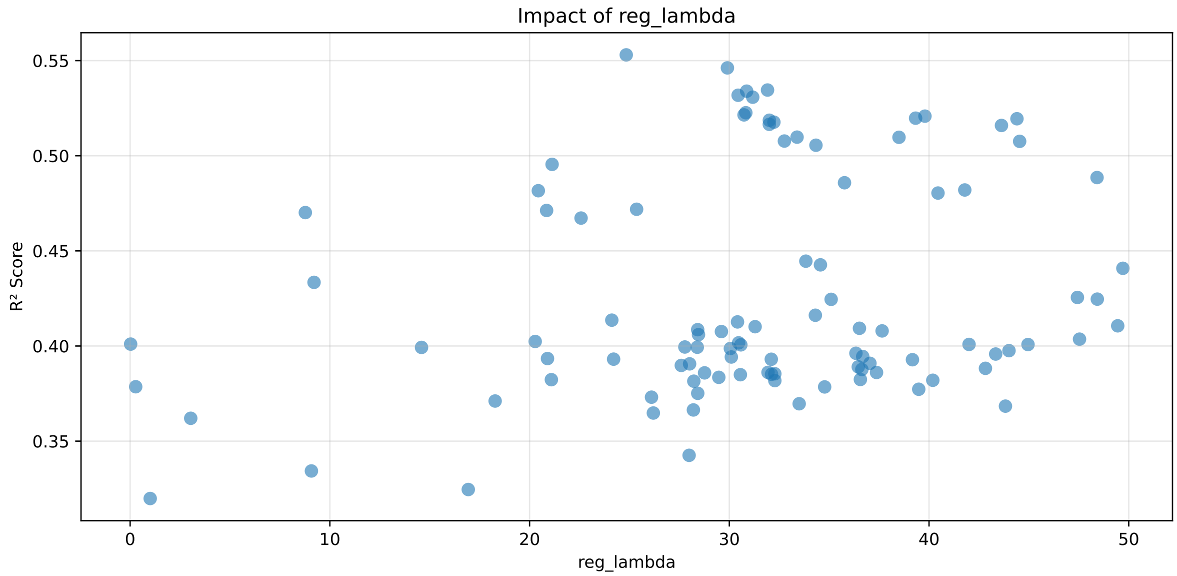



Impact of max_bin
R² Score
0.55
0.50
0.45
0.40
0.35
75
100
125
150
175
200
225
250
max_bin

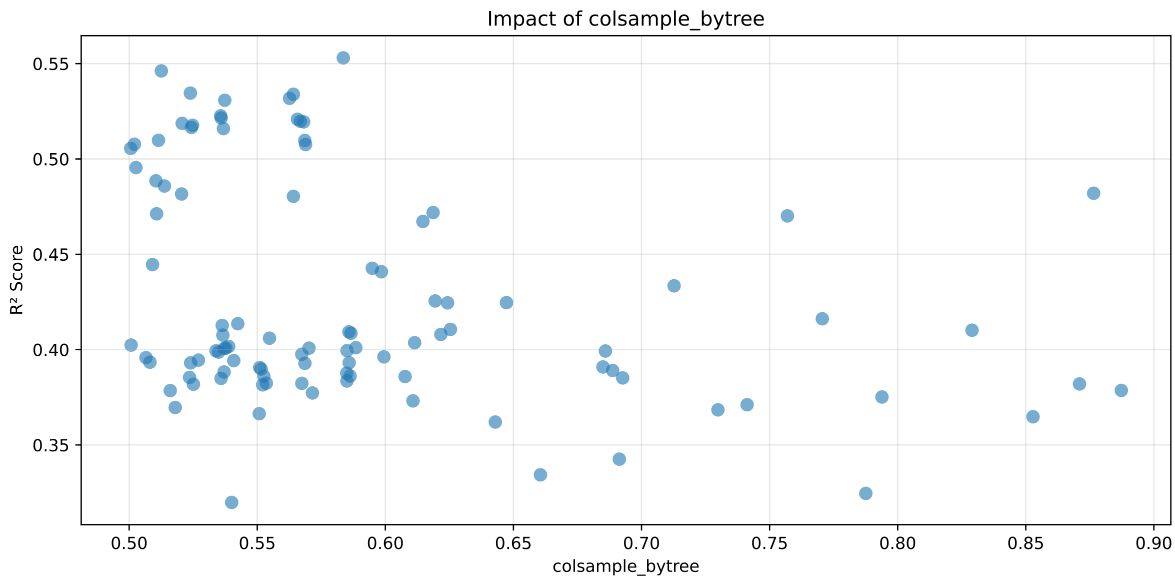

Impact of reg_alpha
R² Score
0.55
0.50
0.45
0.40
0.35
0
2
4
6
8
10
reg_alpha


Impact of bagging_fraction
R² Score
0.55
0.50
0.45
0.40
0.35
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
bagging_fraction

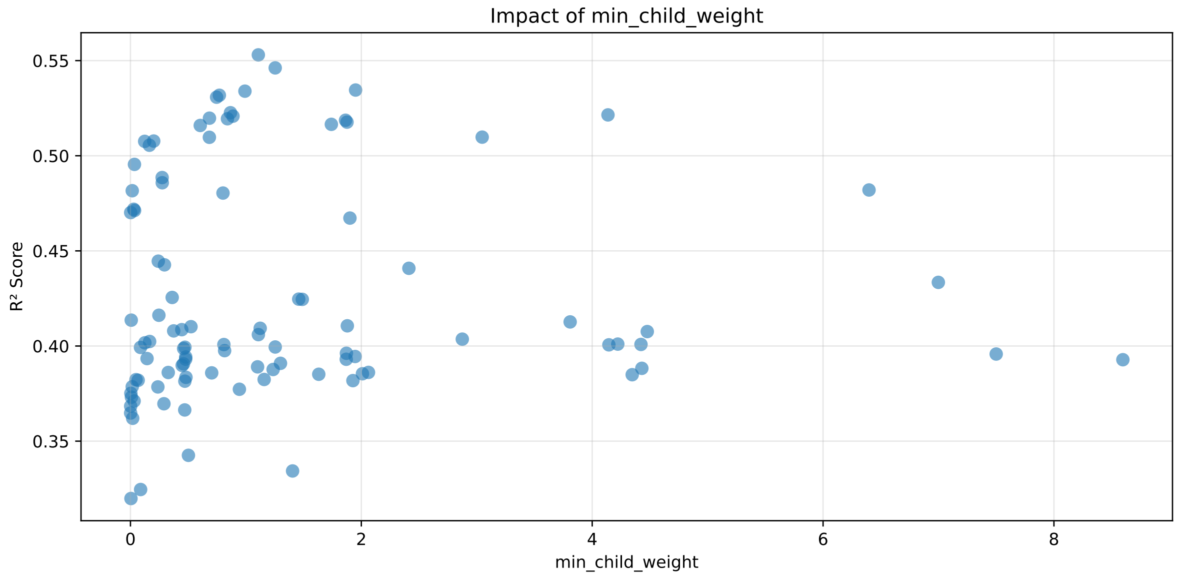

Impact of subsample
R² Score
0.55
0.50
0.45
0.40
0.35
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
subsample



Impact of reg_lambda
R² Score
reg_lambda
0.55
0.50
0.45
0.40
0.35
0
10
20
30
40
50

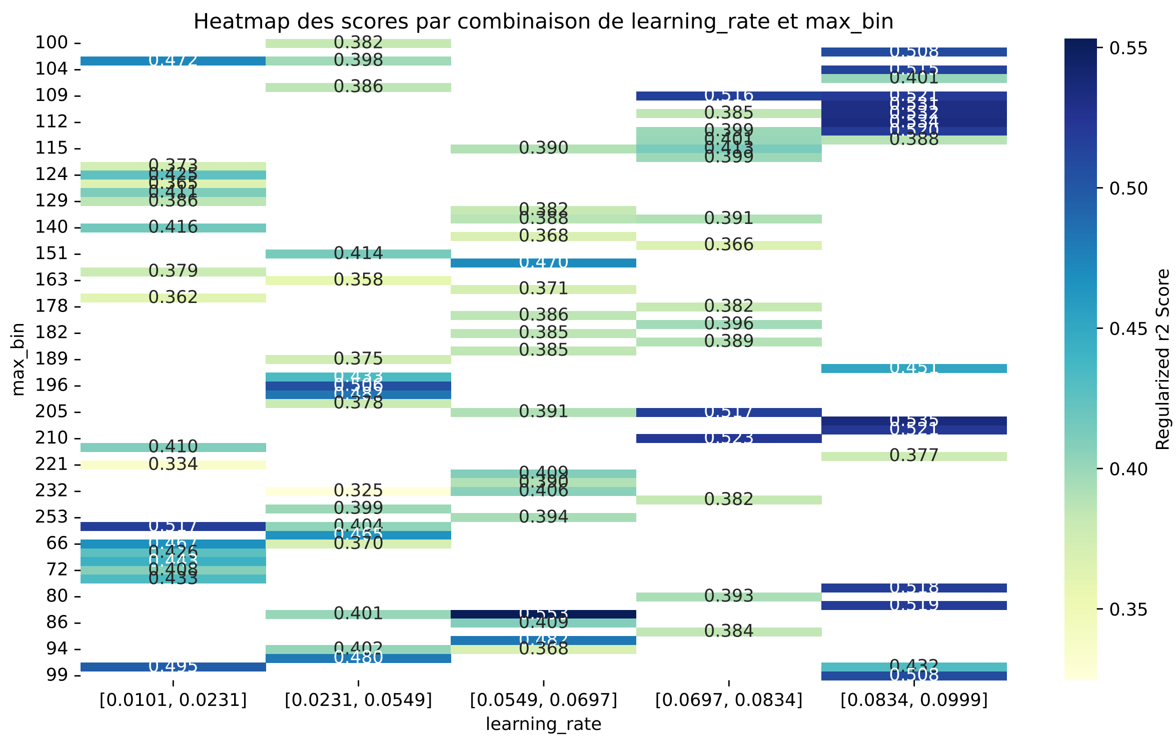


Impact of colsample_bytree
R² Score
colsample_bytree
0.55
0.50
0.45
0.40
0.35
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5805	0.553	0.5158	0.0341	0.553	0.5365	-0.0165	0.9702	1.0498
2	0.5734	0.5462	0.5055	0.0346	0.5462	0.5339	-0.0123	0.9774	1.0498
3	0.5596	0.5345	0.4951	0.0355	0.5345	0.5241	-0.0104	0.9805	1.0469
4	0.558	0.5339	0.4921	0.0393	0.5339	0.5185	-0.0154	0.9711	1.045
5	0.5564	0.5318	0.4907	0.0362	0.5318	0.5137	-0.0181	0.966	1.0462
6	0.5549	0.5308	0.4887	0.0367	0.5308	0.5191	-0.0116	0.9781	1.0454
7	0.5469	0.5226	0.4826	0.0276	0.5226	0.5139	-0.0088	0.9832	1.0464
8	0.5444	0.5215	0.4802	0.0359	0.5215	0.5101	-0.0114	0.9782	1.044
9	0.5439	0.5208	0.4791	0.0294	0.5208	0.5147	-0.0061	0.9883	1.0444
10	0.5418	0.5198	0.4778	0.0365	0.5198	0.5091	-0.0107	0.9795	1.0423
---	---	---	---	---	---	---	---	---	---
100	0.631	0.5791	0.5579	0.0471	0.3198	0.5766	-0.0026	0.9956	1.0895

Hyperparameters:

Rank 1: {'reg___n_estimators': 400, 'reg___max_depth': 7, 'reg___learning_rate': 0.06272817294918391, 'reg___num_leaves': 45, 'reg___subsample': 0.6287004301525789, 'reg___colsample_bytree': 0.5835930246612872, 'reg___min_child_samples': 44, 'reg___reg_alpha': 7.573332840263241, 'reg___reg_lambda': 24.839952644969983, 'reg___min_split_gain': 0.24938995050811208, 'reg___path_smooth': 3.3894816585428957, 'reg___min_child_weight': 1.1077788096944916, 'reg___feature_fraction': 0.5535190513906822, 'reg___bagging_fraction': 0.7137169643012954, 'reg___bagging_freq': 3, 'reg___max_bin': 85, 'reg___min_data_per_group': 122}

Rank 2: {'reg___n_estimators': 800, 'reg___max_depth': 5, 'reg___learning_rate': 0.011221536598457305, 'reg___num_leaves': 41, 'reg___subsample': 0.5039574868628074, 'reg___colsample_bytree': 0.512557649975261, 'reg___min_child_samples': 10, 'reg___reg_alpha': 0.6266358028056338, 'reg___reg_lambda': 29.905400924535726, 'reg___min_split_gain': 1.7025851216612762, 'reg___path_smooth': 3.103028304678947, 'reg___min_child_weight': 1.2539395324496603, 'reg___feature_fraction': 0.6892247813186118, 'reg___bagging_fraction': 0.7113051108577778, 'reg___bagging_freq': 7, 'reg___max_bin': 63, 'reg___min_data_per_group': 143}

Rank 3: {'reg___n_estimators': 500, 'reg___max_depth': 7, 'reg___learning_rate': 0.08898943094109063, 'reg___num_leaves': 49, 'reg___subsample': 0.626274801528322, 'reg___colsample_bytree': 0.5238918499320148, 'reg___min_child_samples': 11, 'reg___reg_alpha': 7.628408330915827, 'reg___reg_lambda': 31.914553486319395, 'reg___min_split_gain': 0.9177507220459149, 'reg___path_smooth': 1.9802081547106993, 'reg___min_child_weight': 1.9502851801105177, 'reg___feature_fraction': 0.550747944455333, 'reg___bagging_fraction': 0.7796278273579682, 'reg___bagging_freq': 3, 'reg___max_bin': 207, 'reg___min_data_per_group': 78}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.0845303939092783, 'reg__num_leaves': 47, 'reg__subsample': 0.606562664208635, 'reg__colsample_bytree': 0.5641180817302598, 'reg__min_child_samples': 13, 'reg__reg_alpha': 6.555137431295735, 'reg__reg_lambda': 30.86977033870043, 'reg__min_split_gain': 1.1508622265720643, 'reg__path_smooth': 2.7457955480513268, 'reg__min_child_weight': 0.9916689978344385, 'reg__feature_fraction': 0.5559494055879783, 'reg__bagging_fraction': 0.759025870991372, 'reg__bagging_freq': 3, 'reg__max_bin': 112, 'reg__min_data_per_group': 102}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.08468306012232182, 'reg__num_leaves': 47, 'reg__subsample': 0.5968259952051883, 'reg__colsample_bytree': 0.562614759487728, 'reg__min_child_samples': 13, 'reg__reg_alpha': 6.686304685653563, 'reg__reg_lambda': 30.44088536821826, 'reg__min_split_gain': 1.122383099993832, 'reg__path_smooth': 2.789867192593643, 'reg__min_child_weight': 0.7701665753896945, 'reg__feature_fraction': 0.5602775498475647, 'reg__bagging_fraction': 0.7766971090405271, 'reg__bagging_freq': 3, 'reg__max_bin': 111, 'reg__min_data_per_group': 103}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.08566212957447013, 'reg__num_leaves': 47, 'reg__subsample': 0.5994622468877459, 'reg__colsample_bytree': 0.5373168279376777, 'reg__min_child_samples': 13, 'reg__reg_alpha': 6.730130386147749, 'reg__reg_lambda': 31.17287355102652, 'reg__min_split_gain': 1.1066944439497202, 'reg__path_smooth': 2.721890992366802, 'reg__min_child_weight': 0.7462625753131764, 'reg__feature_fraction': 0.5496743806135888, 'reg__bagging_fraction': 0.7603480114339237, 'reg__bagging_freq': 3, 'reg__max_bin': 110, 'reg__min_data_per_group': 102}

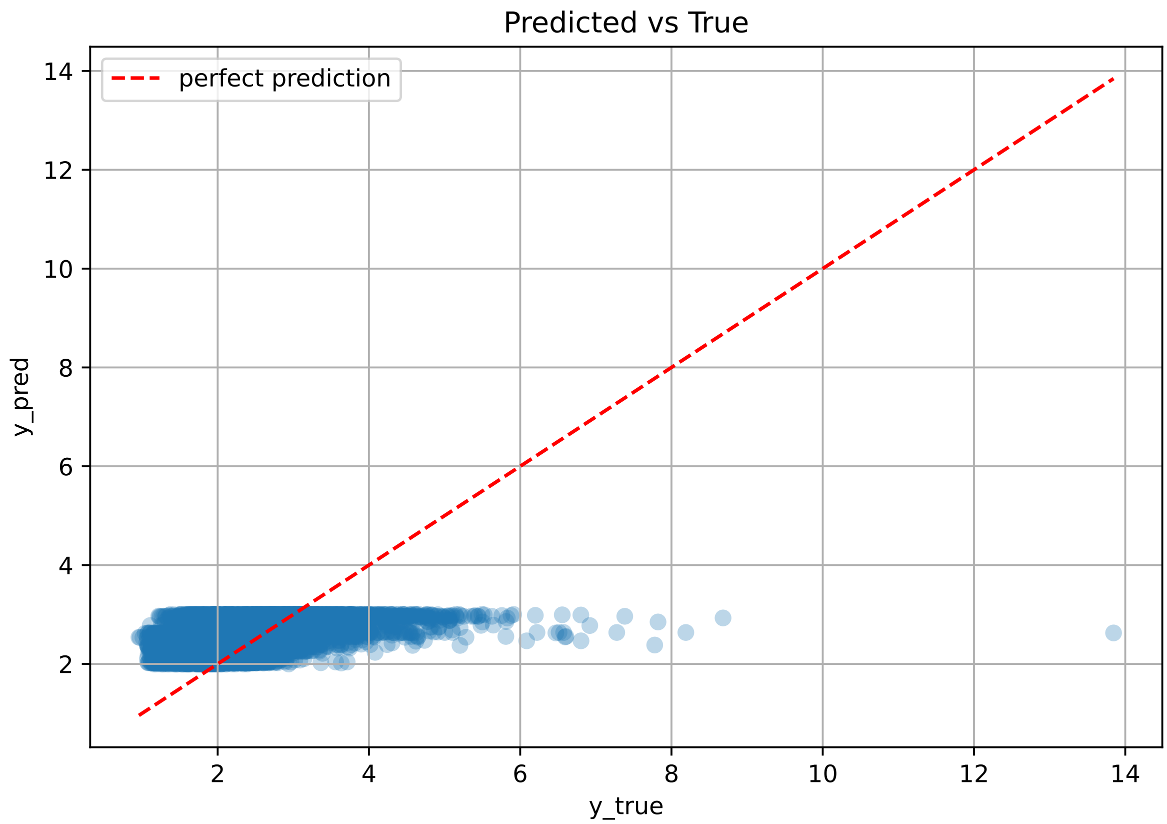
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.08281723731436669, 'reg__num_leaves': 47, 'reg__subsample': 0.5990018398460866, 'reg__colsample_bytree': 0.5357241480007308, 'reg__min_child_samples': 13, 'reg__reg_alpha': 6.939569238554977, 'reg__reg_lambda': 30.827534662397596, 'reg__min_split_gain': 1.110834844374017, 'reg__path_smooth': 2.6883616611979897, 'reg__min_child_weight': 0.8662410014538201, 'reg__feature_fraction': 0.5392633330747707, 'reg__bagging_fraction': 0.7681683517471278, 'reg__bagging_freq': 3, 'reg__max_bin': 210, 'reg__min_data_per_group': 105}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.08399437663253789, 'reg__num_leaves': 47, 'reg__subsample': 0.5992968239445742, 'reg__colsample_bytree': 0.5359112733099589, 'reg__min_child_samples': 13, 'reg__reg_alpha': 8.303479633043654, 'reg__reg_lambda': 30.737037095301368, 'reg__min_split_gain': 1.1215201329566318, 'reg__path_smooth': 2.833410590474889, 'reg__min_child_weight': 4.136709998826744, 'reg__feature_fraction': 0.5418815428347776, 'reg__bagging_fraction': 0.7714056358566763, 'reg__bagging_freq': 3, 'reg__max_bin': 209, 'reg__min_data_per_group': 103}

Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.09688379856664003, 'reg__num_leaves': 48, 'reg__subsample': 0.5893443381480586, 'reg__colsample_bytree': 0.5657132911632919, 'reg__min_child_samples': 13, 'reg__reg_alpha': 8.340493425120485, 'reg__reg_lambda': 39.7958130049281, 'reg__min_split_gain': 1.0670905166060929, 'reg__path_smooth': 2.448674744476249, 'reg__min_child_weight': 0.8882640058598056, 'reg__feature_fraction': 0.5764391957725823, 'reg__bagging_fraction': 0.7790519300402851, 'reg__bagging_freq': 3, 'reg__max_bin': 109, 'reg__min_data_per_group': 100}

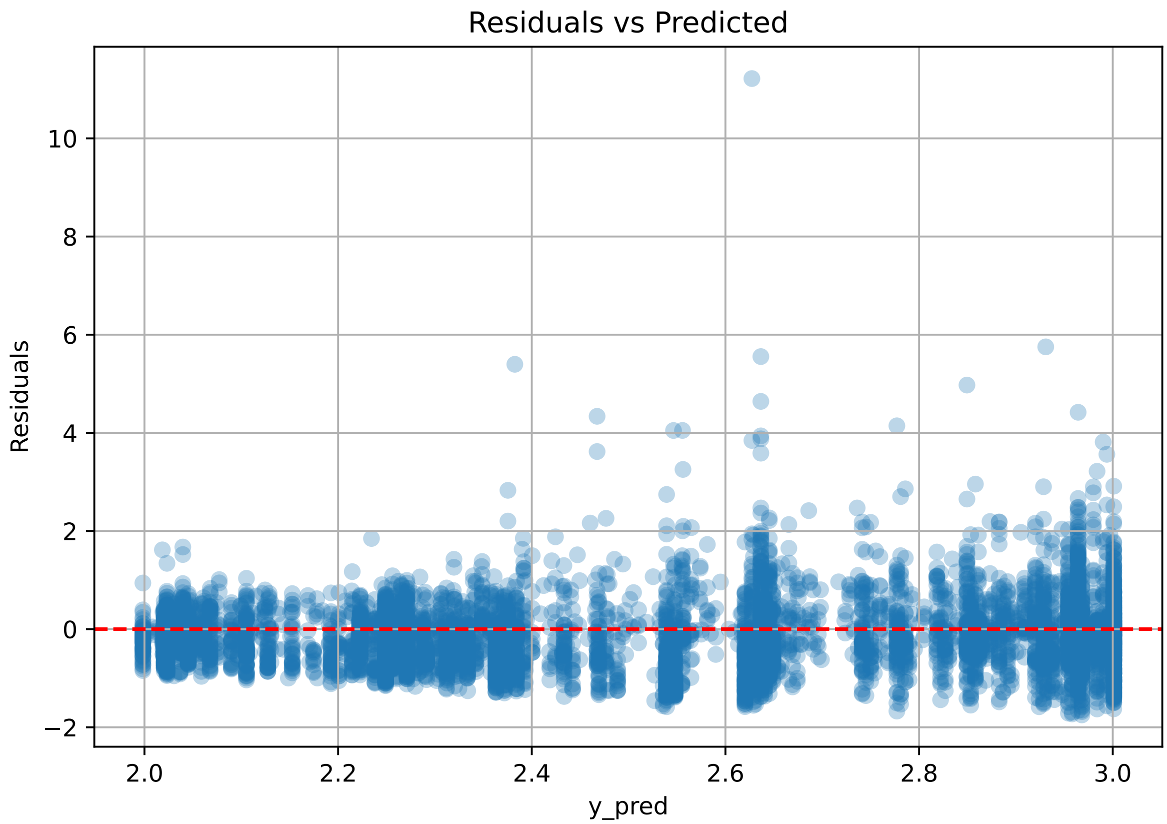
Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.08594526629096315, 'reg__num_leaves': 48, 'reg__subsample': 0.5973159173687883, 'reg__colsample_bytree': 0.5668300496238188, 'reg__min_child_samples': 13, 'reg__reg_alpha': 8.325653761601957, 'reg__reg_lambda': 39.32739599828173, 'reg__min_split_gain': 1.1109277188957647, 'reg__path_smooth': 2.643760840834663, 'reg__min_child_weight': 0.6844926429893615, 'reg__feature_fraction': 0.5681222228298644, 'reg__bagging_fraction': 0.7755300696479765, 'reg__bagging_freq': 3, 'reg__max_bin': 113, 'reg__min_data_per_group': 103}

2.3.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

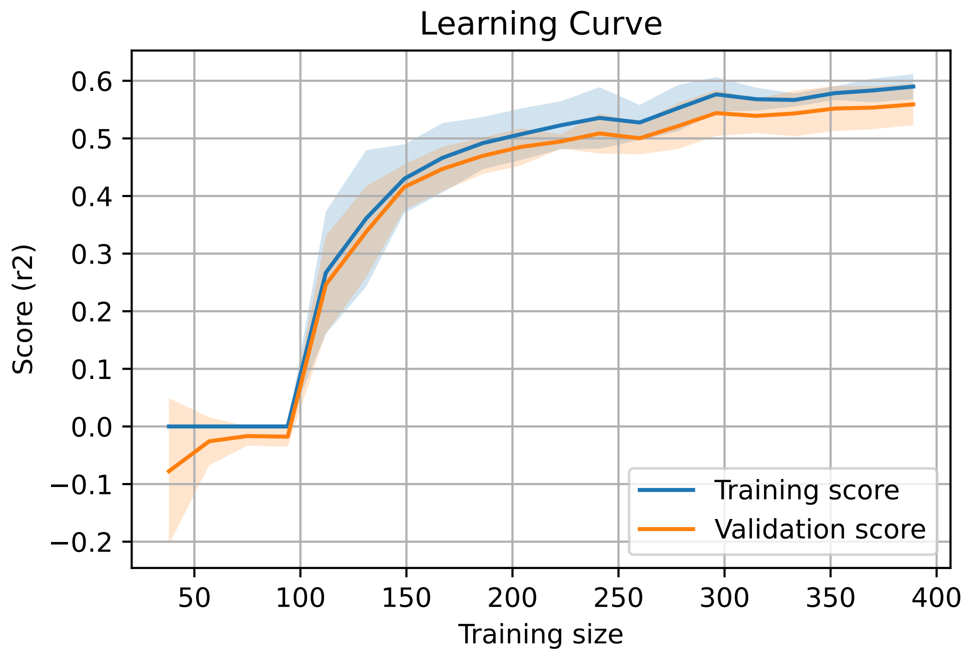
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 1

$\mathbf{Len(group)} : 795, \mathbf{Len(training)} : 557,$

$\mathbf{Len(training\_real)} : 390, \mathbf{Len(test\_of\_training)} : 167, \mathbf{Len(test)} : 238$

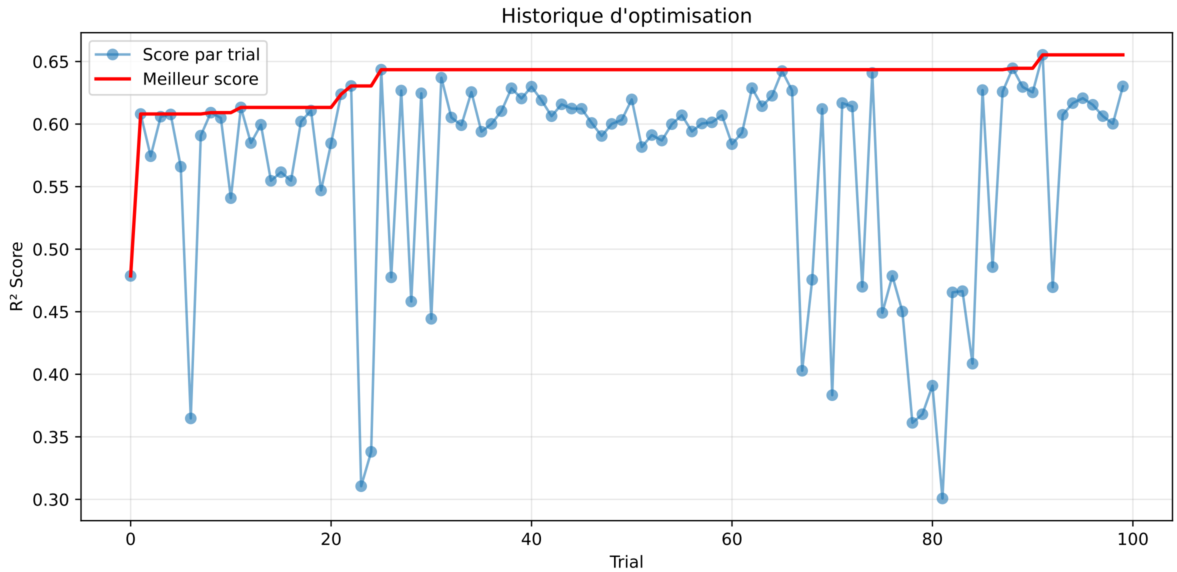


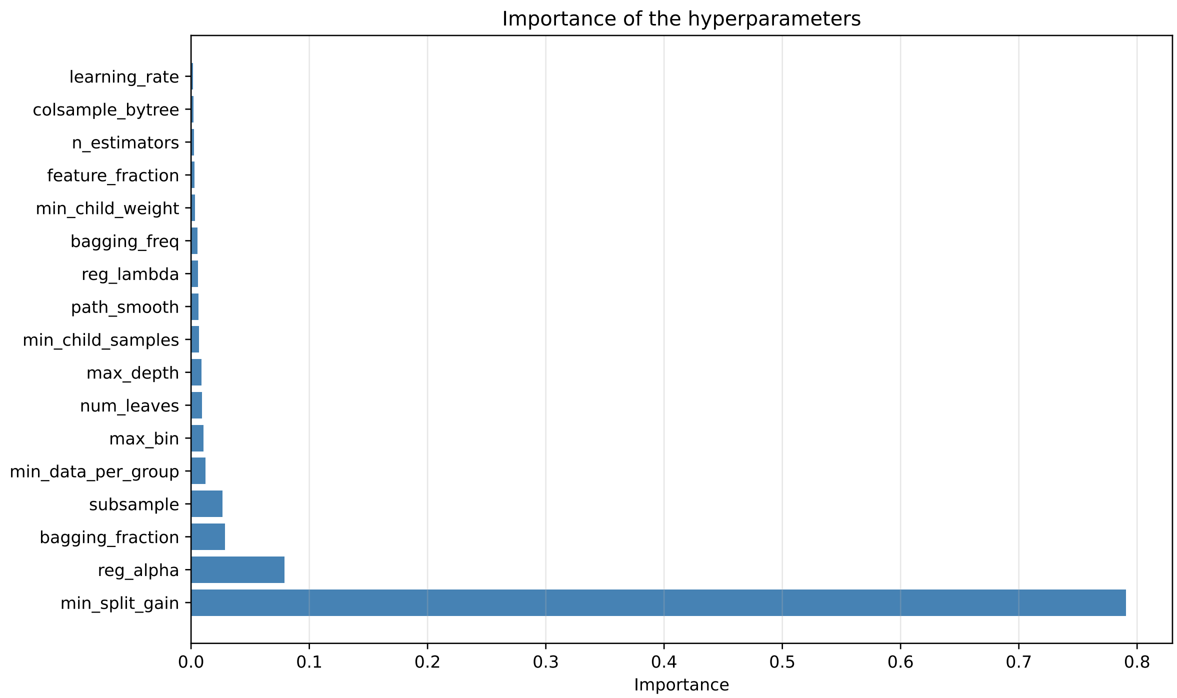
2.3.5 Gridsearch — Group: 2

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

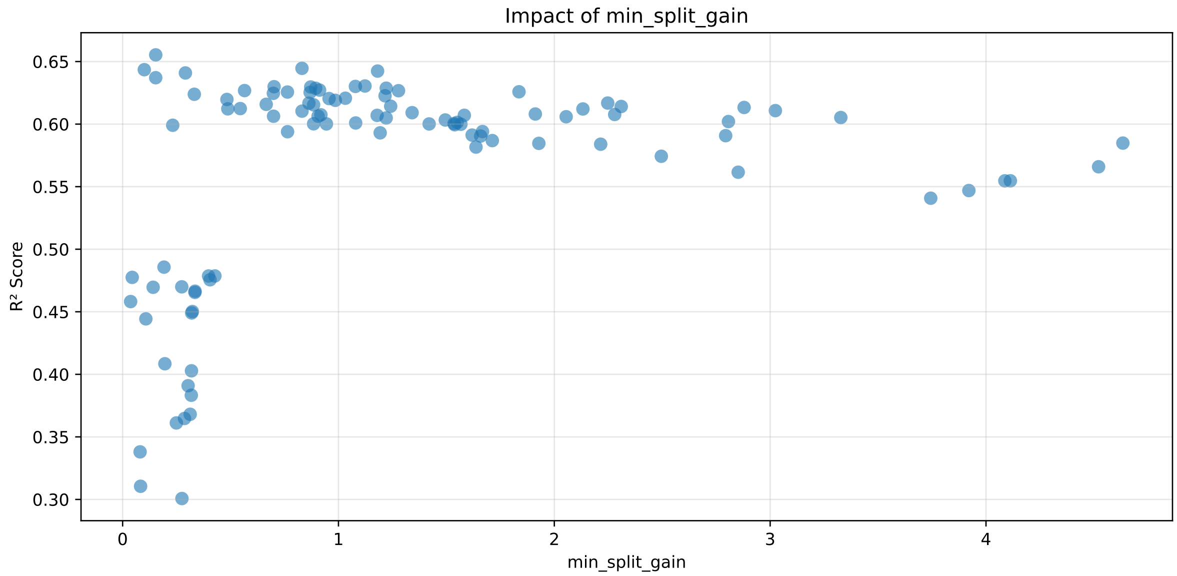
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682 y_test: mean=2.561, std=0.947, min=1.120, max=13.845

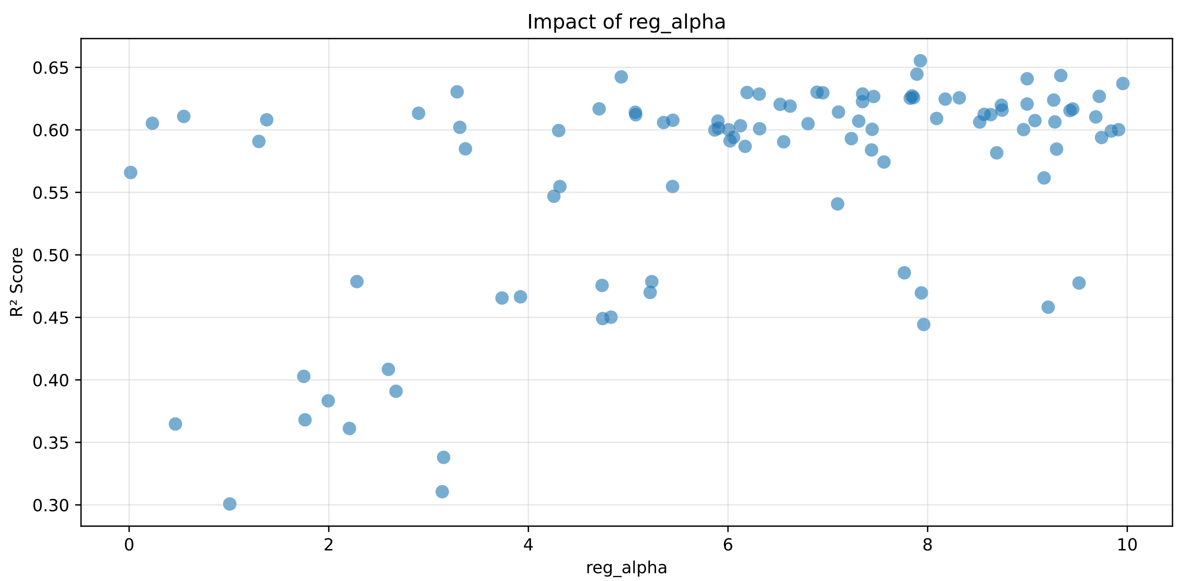
==================== TOP Importance FEATURES ==================== feature importance r_umidity 292 r_CO2 260 r_NH3 250 r_THI 190 r_temperature 143





Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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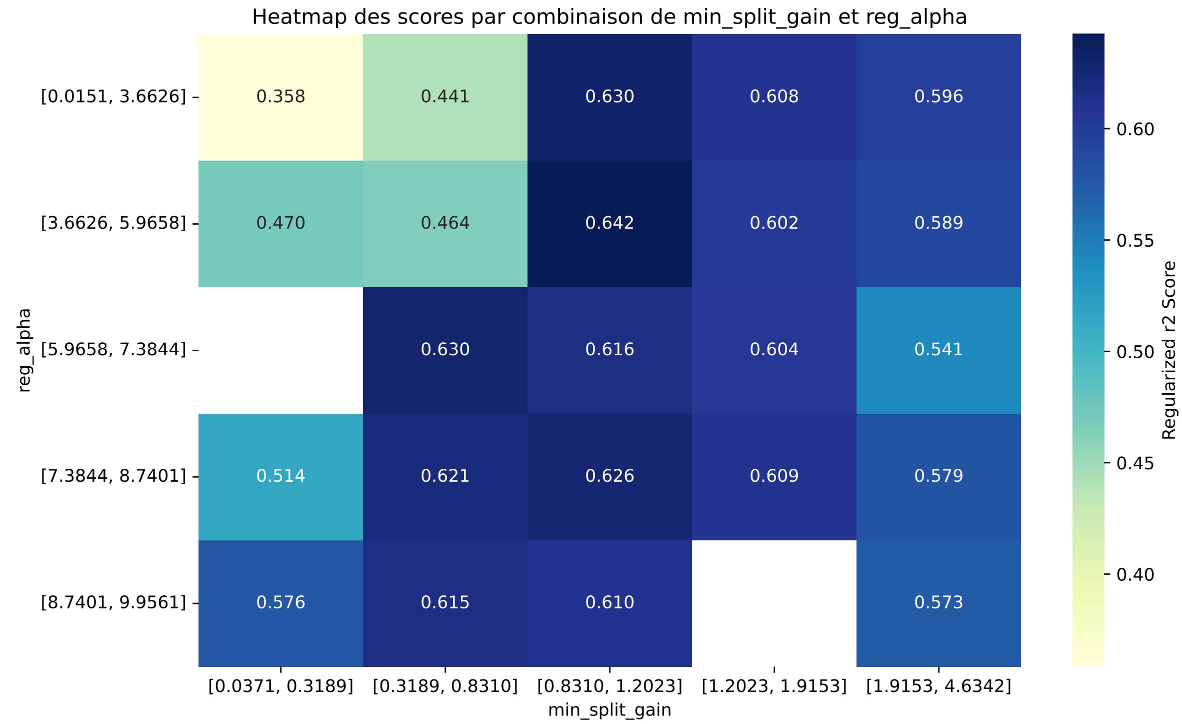
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6848	0.6553	0.5583	0.0129	0.6553	0.5743	-0.0811	0.8763	1.0449
2	0.6695	0.6446	0.552	0.0119	0.6446	0.567	-0.0776	0.8797	1.0386
3	0.6728	0.6435	0.5559	0.0127	0.6435	0.57	-0.0735	0.8858	1.0455
4	0.6637	0.6424	0.5499	0.0149	0.6424	0.5656	-0.0768	0.8805	1.0332
5	0.6687	0.6409	0.5485	0.0107	0.6409	0.5632	-0.0777	0.8788	1.0434
6	0.6646	0.6371	0.5478	0.0126	0.6371	0.5615	-0.0756	0.8814	1.0431
7	0.6549	0.6305	0.5449	0.0121	0.6305	0.5585	-0.0719	0.8859	1.0387
8	0.6517	0.6301	0.5381	0.011	0.6301	0.5531	-0.077	0.8778	1.0342
9	0.6515	0.6299	0.5397	0.0109	0.6299	0.556	-0.0739	0.8827	1.0343
10	0.6477	0.6297	0.5353	0.0122	0.6297	0.5478	-0.0818	0.87	1.0286
---	---	---	---	---	---	---	---	---	---
100	0.7618	0.685	0.5768	0.0119	0.3007	0.5896	-0.0954	0.8608	1.1122

Hyperparameters:

Rank 1: {'reg___n_estimators': 400, 'reg___max_depth': 4, 'reg___learning_rate': 0.047083168478878175, 'reg___num_leaves': 56, 'reg___subsample': 0.8203422744278107, 'reg___colsample_bytree': 0.5576258022433572, 'reg___min_child_samples': 28, 'reg___reg_alpha': 7.927233796467711, 'reg___reg_lambda': 3.129154947336146, 'reg___min_split_gain': 0.15334146364359014, 'reg___path_smooth': 4.50108843846595, 'reg___min_child_weight':

0.23312595872063369, 'reg___feature_fraction': 0.6924858392599109, 'reg___bagging_fraction': 0.7774138973894658, 'reg___bagging_freq': 5, 'reg___max_bin': 244, 'reg___min_data_per_group': 163}

Rank 2: {'reg___n_estimators': 300, 'reg___max_depth': 8, 'reg___learning_rate': 0.07554187733446482, 'reg___num_leaves': 61, 'reg___subsample': 0.7489656312756469, 'reg___colsample_bytree': 0.6235366736083746, 'reg___min_child_samples': 38, 'reg___reg_alpha': 7.8924327902597415, 'reg___reg_lambda': 7.18541995408023, 'reg___min_split_gain': 0.8309495728311087, 'reg___path_smooth': 2.1362357463521664, 'reg___min_child_weight': 0.002080085424566918, 'reg___feature_fraction': 0.7615266175831013, 'reg___bagging_fraction': 0.8523648832618681, 'reg___bagging_freq': 1, 'reg___max_bin': 244, 'reg___min_data_per_group': 182}

Rank 3: {'reg___n_estimators': 800, 'reg___max_depth': 7, 'reg___learning_rate': 0.044476068593762956, 'reg___num_leaves': 63, 'reg___subsample': 0.7598798915989848, 'reg___colsample_bytree': 0.8196715058059023, 'reg___min_child_samples': 42, 'reg___reg_alpha': 9.334345647085367, 'reg___reg_lambda': 49.22059885532676, 'reg___min_split_gain': 0.10043464069676666, 'reg___path_smooth': 3.9069049780814167, 'reg___min_child_weight': 0.0020529952675529014, 'reg___feature_fraction': 0.673892039660861, 'reg___bagging_fraction': 0.7738039498278091, 'reg___bagging_freq': 1, 'reg___max_bin': 69, 'reg___min_data_per_group': 97}

Rank 4: {'reg___n_estimators': 800, 'reg___max_depth': 7, 'reg___learning_rate': 0.04274077820306479, 'reg___num_leaves': 59, 'reg___subsample': 0.8373534672310101, 'reg___colsample_bytree': 0.6401781926469612, 'reg___min_child_samples': 47, 'reg___reg_alpha': 4.930542477411138, 'reg___reg_lambda': 6.35397056264021, 'reg___min_split_gain': 1.1812392777751648, 'reg___path_smooth': 3.2200729298754145, 'reg___min_child_weight': 0.16001994280993448, 'reg___feature_fraction': 0.7257284133834381, 'reg___bagging_fraction': 0.727076480285295, 'reg___bagging_freq': 1, 'reg___max_bin': 215, 'reg___min_data_per_group': 182}

Rank 5: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.06322453081210382, 'reg___num_leaves': 63, 'reg___subsample': 0.7462141155773319, 'reg___colsample_bytree': 0.5168069141738104, 'reg___min_child_samples': 47, 'reg___reg_alpha': 8.997226602400879, 'reg___reg_lambda': 8.04704107154248, 'reg___min_split_gain': 0.29090970585154696, 'reg___path_smooth': 1.6646072012114708, 'reg___min_child_weight': 0.0010042321897007474, 'reg___feature_fraction': 0.6935039435390623, 'reg___bagging_fraction': 0.8088787817961797, 'reg___bagging_freq': 1, 'reg___max_bin': 222, 'reg___min_data_per_group': 90}

Rank 6: {'reg___n_estimators': 200, 'reg___max_depth': 7, 'reg___learning_rate': 0.09991383665672156, 'reg___num_leaves': 56, 'reg___subsample': 0.7793599838258171, 'reg___colsample_bytree': 0.6025154376666688, 'reg___min_child_samples': 50, 'reg___reg_alpha': 9.956124461306684, 'reg___reg_lambda': 14.208758981016963, 'reg___min_split_gain': 0.15325289055643268, 'reg___path_smooth': 2.026520565931215, 'reg___min_child_weight':

0.029816115735251093, 'reg___feature_fraction': 0.6180702251630174, 'reg___bagging_fraction': 0.7306478471697989, 'reg___bagging_freq': 1, 'reg___max_bin': 245, 'reg___min_data_per_group': 149}

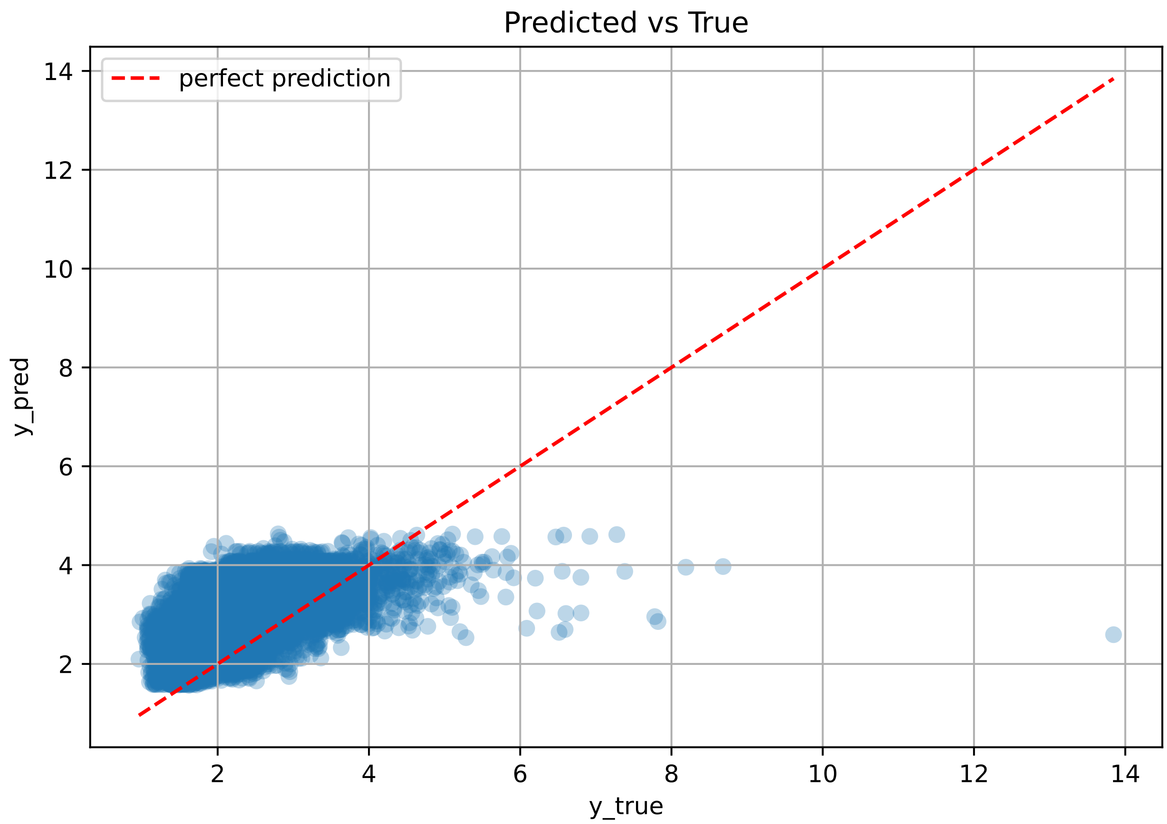
Rank 7: {'reg___n_estimators': 800, 'reg___max_depth': 8, 'reg___learning_rate': 0.060290266523413336, 'reg___num_leaves': 62, 'reg___subsample': 0.8969890305332666, 'reg___colsample_bytree': 0.8797804863460488, 'reg___min_child_samples': 46, 'reg___reg_alpha': 3.28613544300141, 'reg___reg_lambda': 46.22933246650023, 'reg___min_split_gain': 1.122837899315742, 'reg___path_smooth': 2.5111874418428357, 'reg___min_child_weight': 0.0013648411575345998, 'reg___feature_fraction': 0.6770035733846376, 'reg___bagging_fraction': 0.7796422635559317, 'reg___bagging_freq': 1, 'reg___max_bin': 246, 'reg___min_data_per_group': 136}

Rank 8: {'reg___n_estimators': 400, 'reg___max_depth': 4, 'reg___learning_rate': 0.035184064610698756, 'reg___num_leaves': 55, 'reg___subsample': 0.7845571875865188, 'reg___colsample_bytree': 0.6214166271481073, 'reg___min_child_samples': 32, 'reg___reg_alpha': 6.891961492981601, 'reg___reg_lambda': 3.845310848237283, 'reg___min_split_gain': 1.0784307452423347, 'reg___path_smooth': 1.7500004580013084, 'reg___min_child_weight': 0.0020576057806979244, 'reg___feature_fraction': 0.6892168554684891, 'reg___bagging_fraction': 0.8968204132459355, 'reg___bagging_freq': 7, 'reg___max_bin': 240, 'reg___min_data_per_group': 181}

Rank 9: {'reg___n_estimators': 600, 'reg___max_depth': 7, 'reg___learning_rate': 0.04904351407100182, 'reg___num_leaves': 52, 'reg___subsample': 0.8008244793175674, 'reg___colsample_bytree': 0.5810925060518418, 'reg___min_child_samples': 44, 'reg___reg_alpha': 6.191023387875417, 'reg___reg_lambda': 0.3649261944857445, 'reg___min_split_gain': 0.7017965485737753, 'reg___path_smooth': 2.489403697784905, 'reg___min_child_weight': 0.007868401992107806, 'reg___feature_fraction': 0.5767289788968896, 'reg___bagging_fraction': 0.6749189167554731, 'reg___bagging_freq': 2, 'reg___max_bin': 235, 'reg___min_data_per_group': 170}

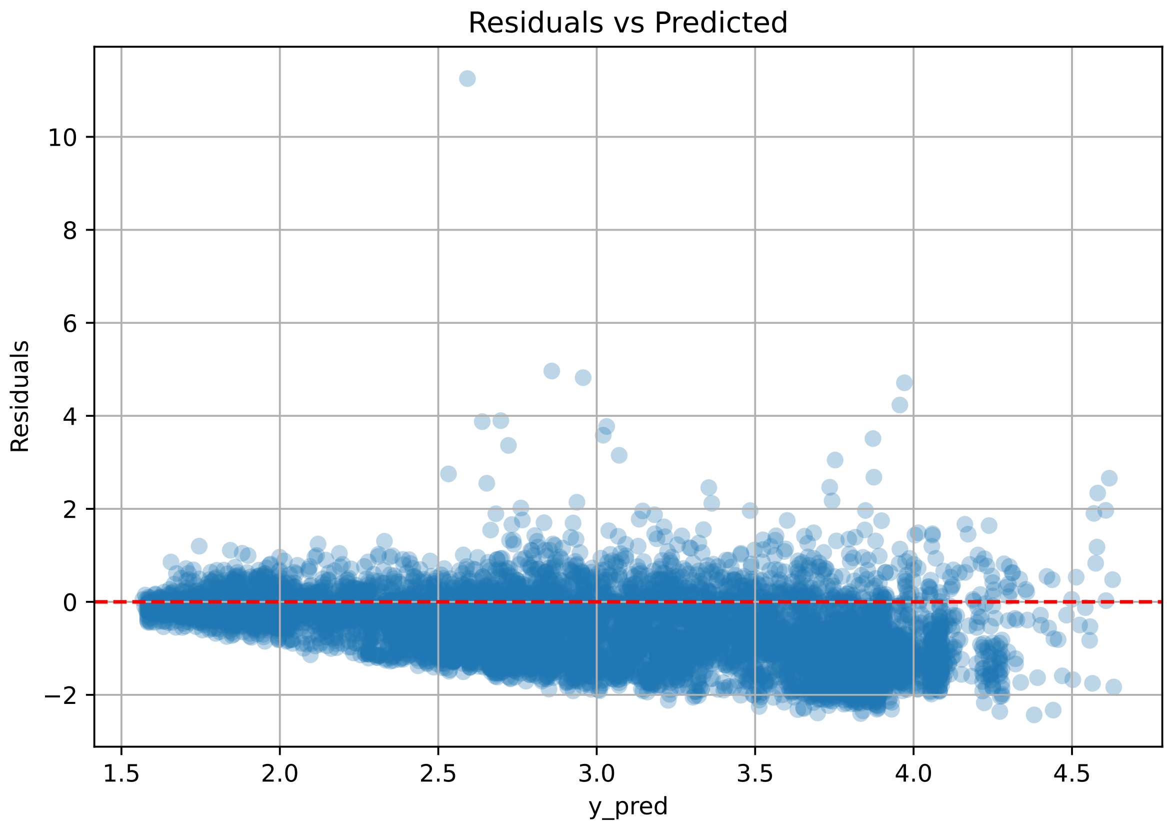
Rank 10: {'reg___n_estimators': 400, 'reg___max_depth': 8, 'reg___learning_rate': 0.046849106045179105, 'reg___num_leaves': 61, 'reg___subsample': 0.8855225165059726, 'reg___colsample_bytree': 0.5578571281414479, 'reg___min_child_samples': 49, 'reg___reg_alpha': 6.950129188514774, 'reg___reg_lambda': 6.870433051749654, 'reg___min_split_gain': 0.8713953101596357, 'reg___path_smooth': 1.7581912870854786, 'reg___min_child_weight': 0.002109623268355398, 'reg___feature_fraction': 0.7641106483169395, 'reg___bagging_fraction': 0.7808041032083742, 'reg___bagging_freq': 7, 'reg___max_bin': 243, 'reg___min_data_per_group': 184}

2.3.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

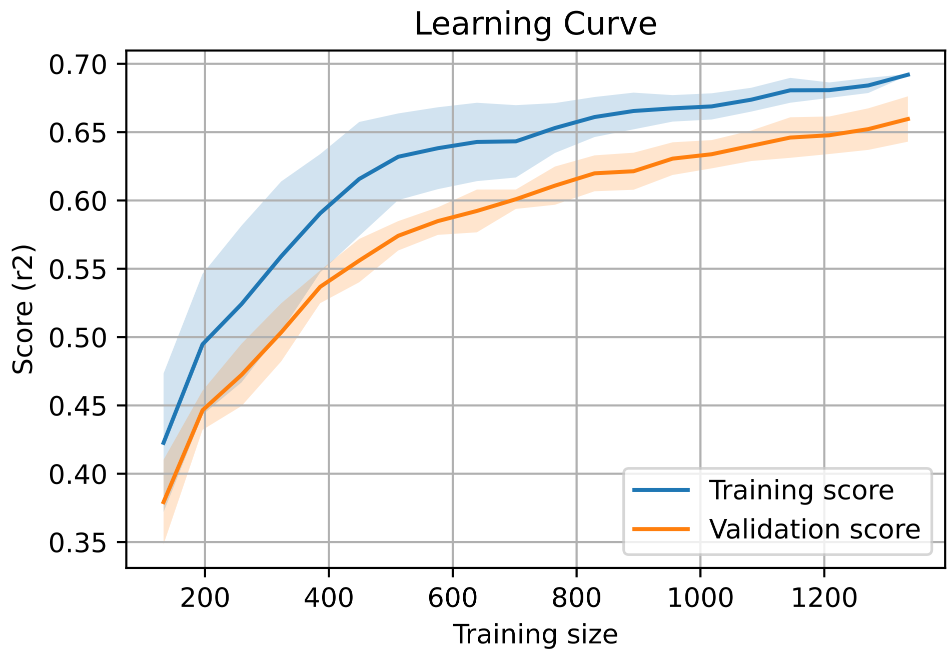
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817

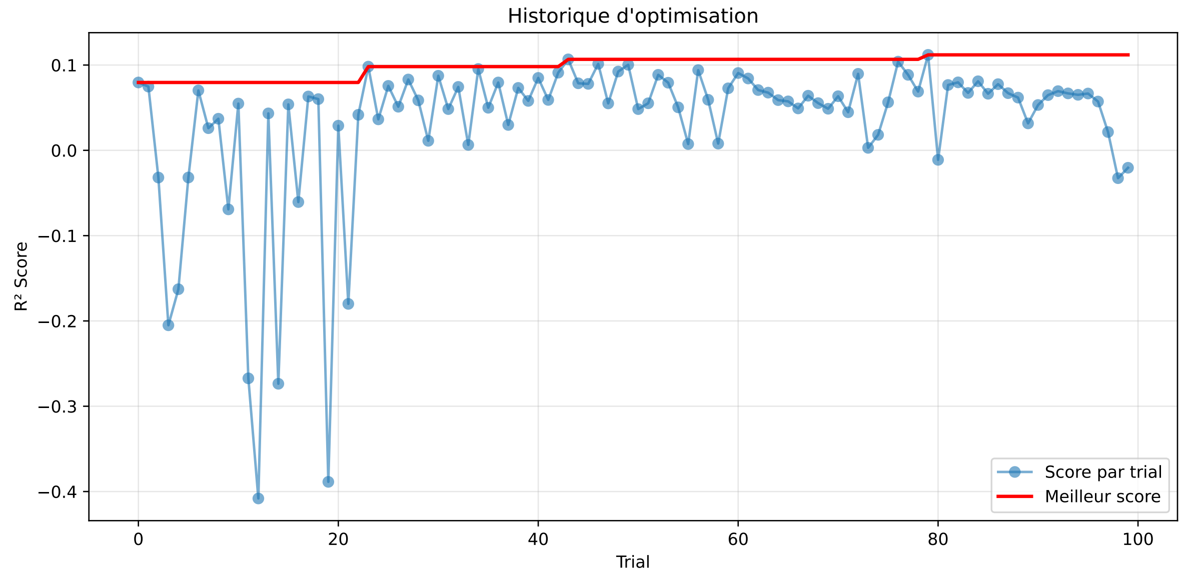


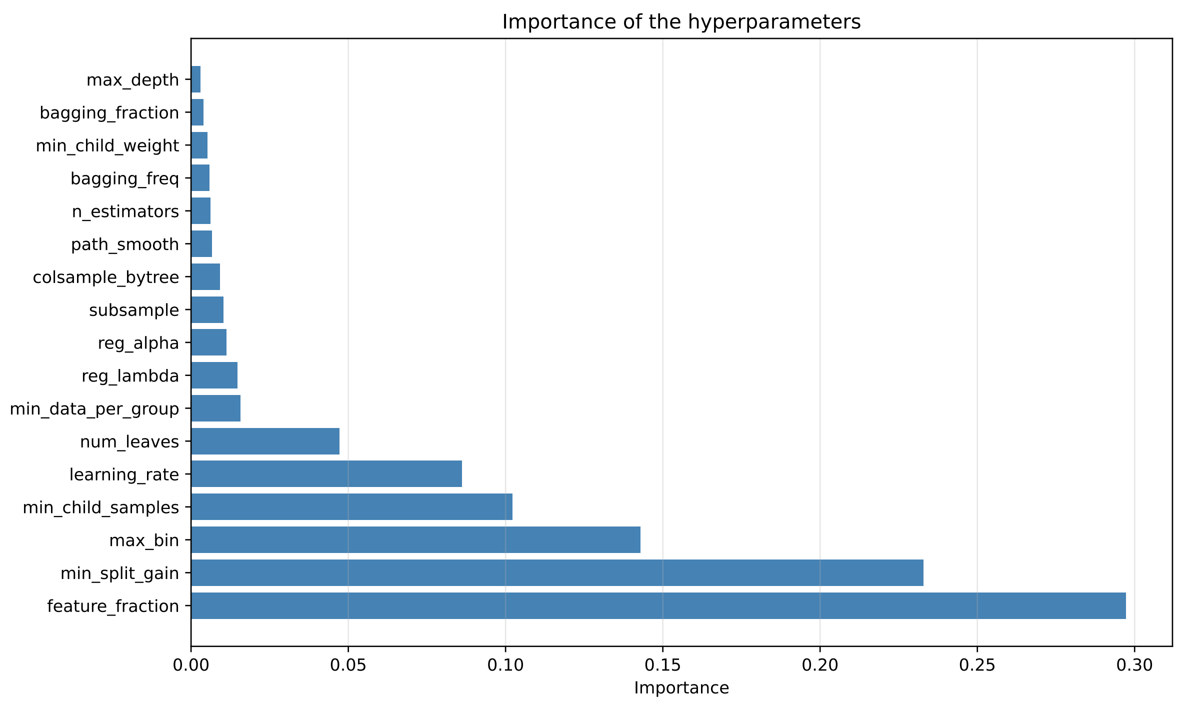
2.3.9 Gridsearch — Group: 3

Len(group) : 568, **Len(training)** : 398, **Len(test)** : 170

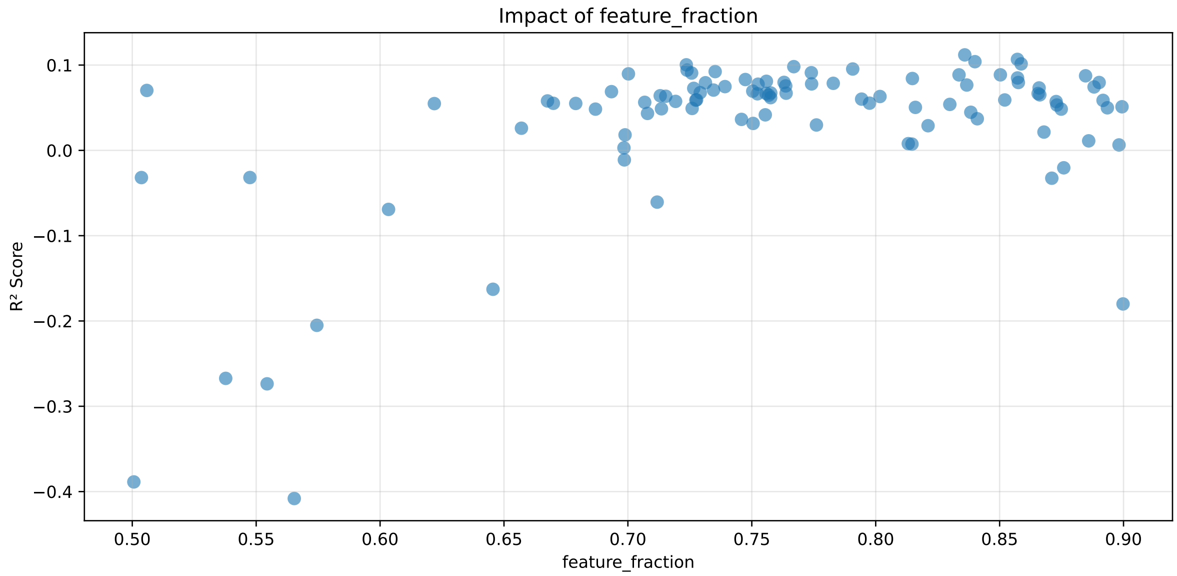
Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804 y_test: mean=1.833, std=0.729, min=1.098, max=7.822

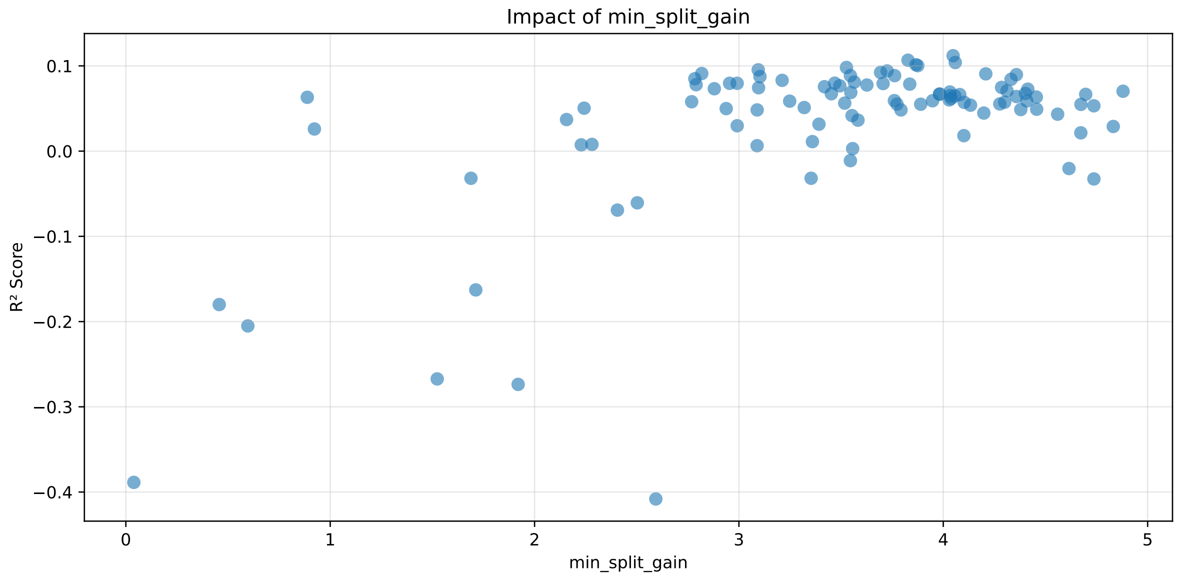
==================== TOP Importance FEATURES ==================== feature importance r_umidity 15 r_NH3 3 r_THI 3 r_temperature 0 r_CO2 0

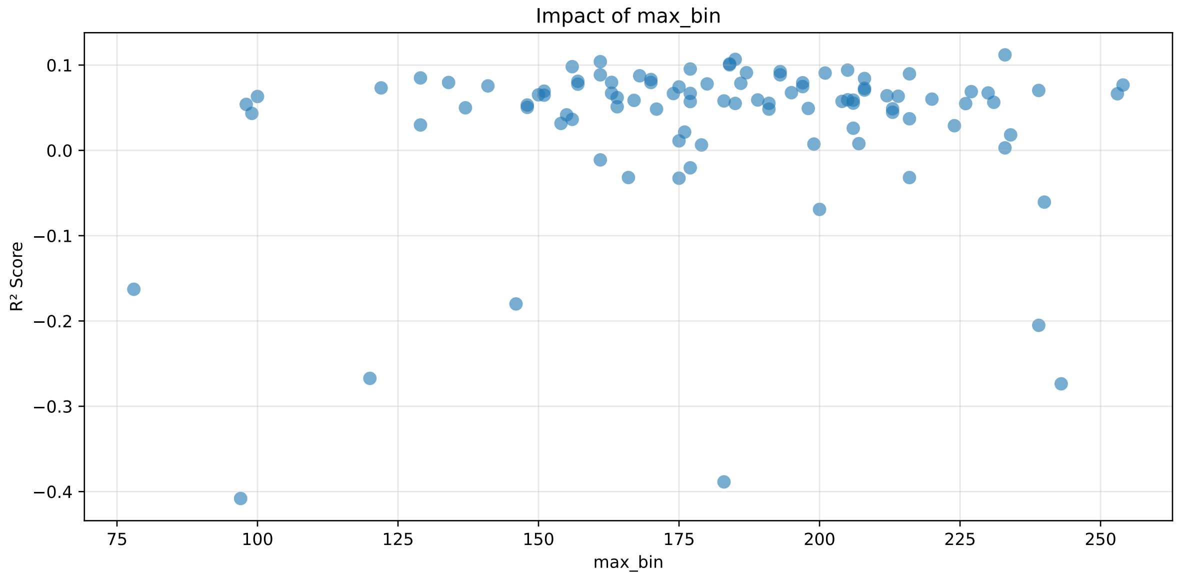


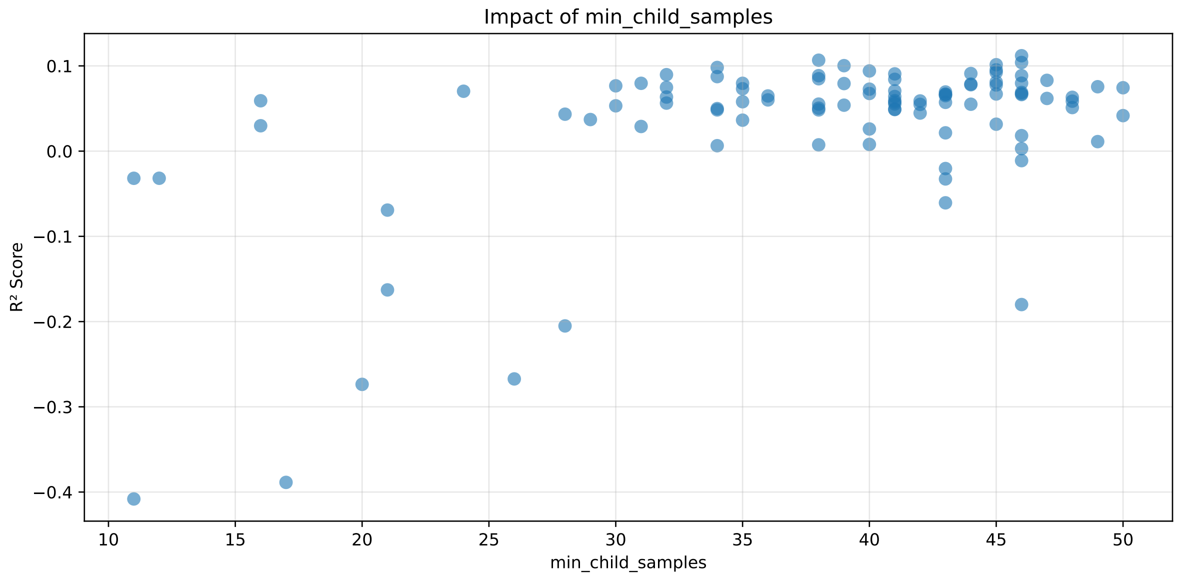


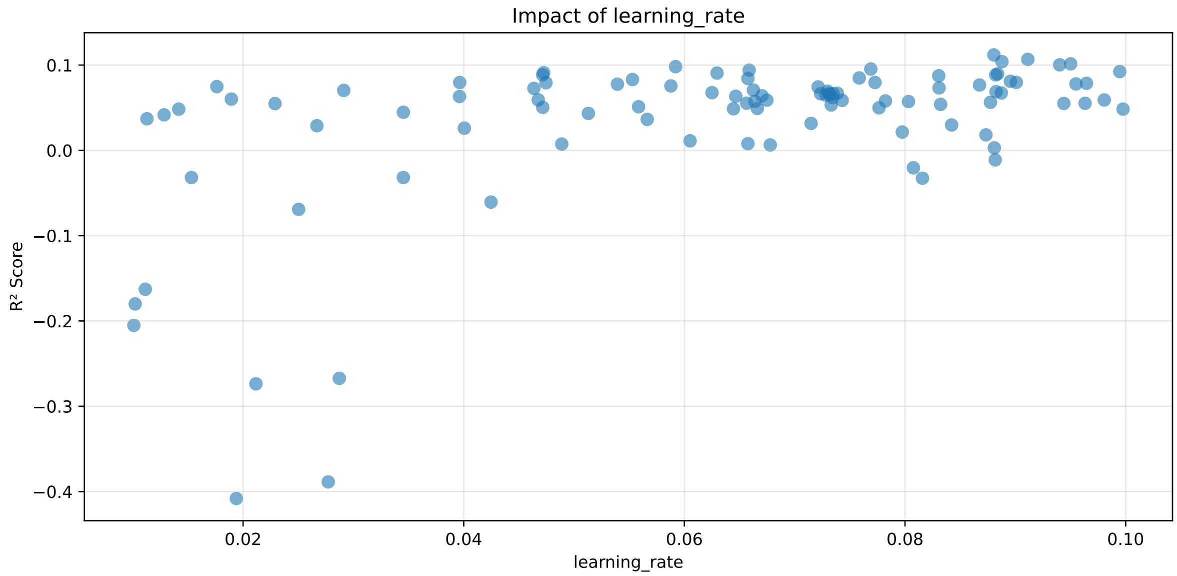
Paramètres avec >5% d'importance : ['feature_fraction', 'min_split_gain', 'max_bin', 'min_child_samples', 'learning_rate']


Impact of feature_fraction
R² Score
feature_fraction
0.1
0.0
−0.1
−0.2
−0.3
−0.4
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90


Impact of min_split_gain
R² Score
min_split_gain
0.1
0.0
−0.1
−0.2
−0.3
−0.4
0
1
2
3
4
5


Impact of max_bin
R² Score
max_bin
0.1
0.0
−0.1
−0.2
−0.3
−0.4
75
100
125
150
175
200
225
250


Impact of min_child_samples
R² Score
min_child_samples
0.1
0.0
−0.1
−0.2
−0.3
−0.4
10
15
20
25
30
35
40
45
50

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C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```
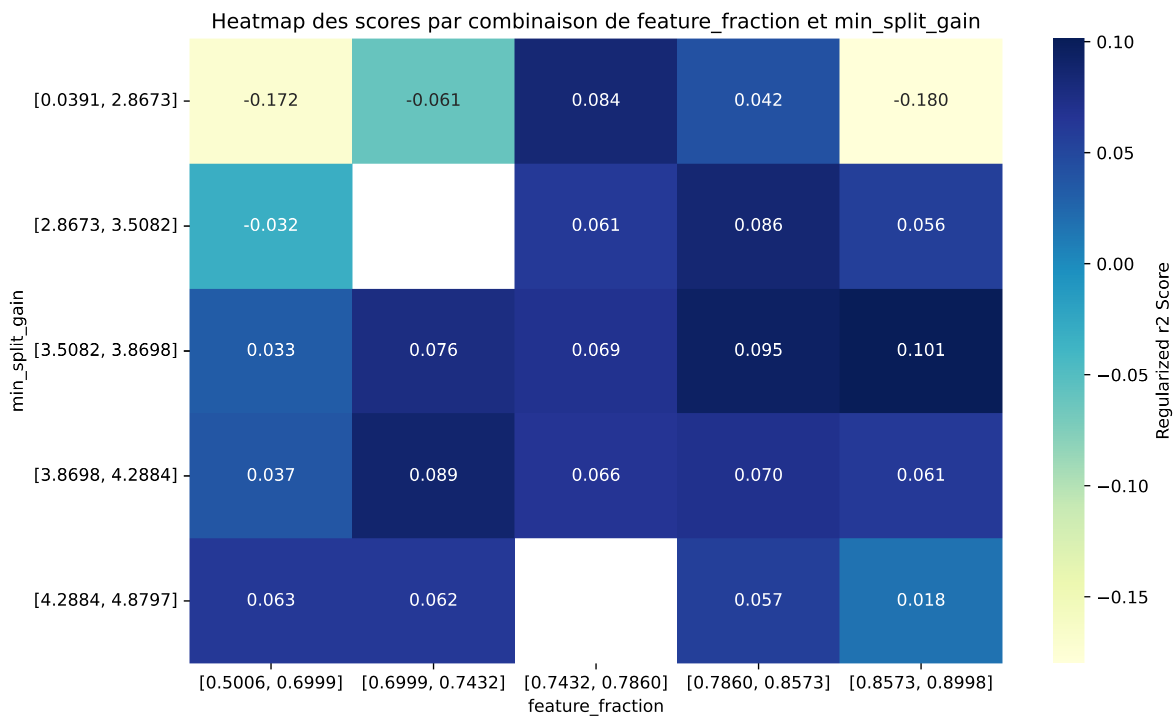



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.1672	0.158	0.0524	0.0437	0.1119	0.072	-0.086	0.4558	1.0583
2	0.1815	0.1691	0.0492	0.0323	0.1067	0.0821	-0.087	0.4854	1.0738
3	0.1652	0.155	0.0481	0.0424	0.104	0.0862	-0.0688	0.5559	1.0658
4	0.1748	0.1626	0.0497	0.0289	0.1014	0.0813	-0.0813	0.5	1.0753
5	0.1861	0.1718	0.0575	0.043	0.1003	0.1053	-0.0665	0.613	1.0833
6	0.2036	0.186	0.0679	0.0471	0.0981	0.1045	-0.0815	0.5617	1.0945
7	0.2042	0.186	0.0583	0.0594	0.0954	0.0946	-0.0914	0.5087	1.0974
8	0.1839	0.1689	0.0529	0.0469	0.0941	0.125	-0.0439	0.7401	1.0886
9	0.1974	0.1799	0.0489	0.0319	0.0923	0.1062	-0.0736	0.5906	1.0973
10	0.1885	0.1722	0.0567	0.045	0.091	0.1175	-0.0547	0.6822	1.0943
---	---	---	---	---	---	---	---	---	---
100	0.3816	0.25	0.1202	0.0314	-0.4081	0.1396	-0.1104	0.5583	1.5265

Hyperparameters:

Rank 1: {'reg___n_estimators': 600, 'reg___max_depth': 3, 'reg___learning_rate': 0.08805369319442276, 'reg___num_leaves': 18, 'reg___subsample': 0.7264538270852563, 'reg___colsample_bytree': 0.7271819850175286, 'reg___min_child_samples': 46, 'reg___reg_alpha': 3.838370105843111, 'reg___reg_lambda': 39.49366228624354, 'reg___min_split_gain': 4.048117034247705, 'reg___path_smooth': 3.767100409338574, 'reg___min_child_weight': 0.005530811153143337, 'reg___feature_fraction': 0.8359361121497547, 'reg___bagging_fraction': 0.6953247328468023, 'reg___bagging_freq': 4, 'reg___max_bin': 233, 'reg___min_data_per_group': 175}

Rank 2: {'reg___n_estimators': 800, 'reg___max_depth': 5, 'reg___learning_rate': 0.09113332547904318, 'reg___num_leaves': 23, 'reg___subsample': 0.7769606960896216, 'reg___colsample_bytree': 0.8470223187462148, 'reg___min_child_samples': 38, 'reg___reg_alpha': 5.446496728547417, 'reg___reg_lambda': 32.452264122397196, 'reg___min_split_gain': 3.8274577540099557, 'reg___path_smooth': 3.4331986410162143, 'reg___min_child_weight': 0.007650081597184888, 'reg___feature_fraction': 0.857193822033023, 'reg___bagging_fraction': 0.7862294043895404, 'reg___bagging_freq': 4, 'reg___max_bin': 185, 'reg___min_data_per_group': 169}

Rank 3: {'reg___n_estimators': 600, 'reg___max_depth': 7, 'reg___learning_rate': 0.08878870864477173, 'reg___num_leaves': 33, 'reg___subsample': 0.7324628656773144, 'reg___colsample_bytree': 0.873839598722737, 'reg___min_child_samples': 46, 'reg___reg_alpha': 3.8532461605781068, 'reg___reg_lambda': 21.10860918349348, 'reg___min_split_gain': 4.058833805619829, 'reg___path_smooth': 3.7615880556764636, 'reg___min_child_weight': 0.011649125121070374, 'reg___feature_fraction': 0.8400458332908652, 'reg___bagging_fraction': 0.6744195486353606, 'reg___bagging_freq': 4, 'reg___max_bin': 161, 'reg___min_data_per_group': 179}

Rank 4: {’reg__n_estimators’: 800, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.09501424651208508, ’reg__num_leaves’: 25, ’reg__subsample’: 0.772484403908087, ’reg__colsample_bytree’: 0.8539250470752886, ’reg__min_child_samples’: 45, ’reg__reg_alpha’: 5.399527893658744, ’reg__reg_lambda’: 33.19664258946726, ’reg__min_split_gain’: 3.8661423709095426, ’reg__path_smooth’: 3.5216430259690052, ’reg__min_child_weight’: 0.00822041313868622, ’reg__feature_fraction’: 0.8586957850630911, ’reg__bagging_fraction’: 0.7800378592607328, ’reg__bagging_freq’: 4, ’reg__max_bin’: 184, ’reg__min_data_per_group’: 170}

Rank 5: {’reg__n_estimators’: 800, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.09402081460988992, ’reg__num_leaves’: 27, ’reg__subsample’: 0.7824444879657415, ’reg__colsample_bytree’: 0.8576081772974371, ’reg__min_child_samples’: 39, ’reg__reg_alpha’: 3.195999879911531, ’reg__reg_lambda’: 46.024710881227165, ’reg__min_split_gain’: 3.8752136191129396, ’reg__path_smooth’: 3.5989298348475547, ’reg__min_child_weight’: 0.007032688267492776, ’reg__feature_fraction’: 0.7235801657563109, ’reg__bagging_fraction’: 0.7833168388413565, ’reg__bagging_freq’: 4, ’reg__max_bin’: 184, ’reg__min_data_per_group’: 168}

Rank 6: {’reg__n_estimators’: 500, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.05920539140843826, ’reg__num_leaves’: 18, ’reg__subsample’: 0.8314038134164937, ’reg__colsample_bytree’: 0.8898441881399071, ’reg__min_child_samples’: 34, ’reg__reg_alpha’: 3.884360355574424, ’reg__reg_lambda’: 41.122828992123885, ’reg__min_split_gain’: 3.526232164327048, ’reg__path_smooth’: 4.938668910945236, ’reg__min_child_weight’: 0.001592212787365959, ’reg__feature_fraction’: 0.7669391506926433, ’reg__bagging_fraction’: 0.8245601809045188, ’reg__bagging_freq’: 6, ’reg__max_bin’: 156, ’reg__min_data_per_group’: 200}

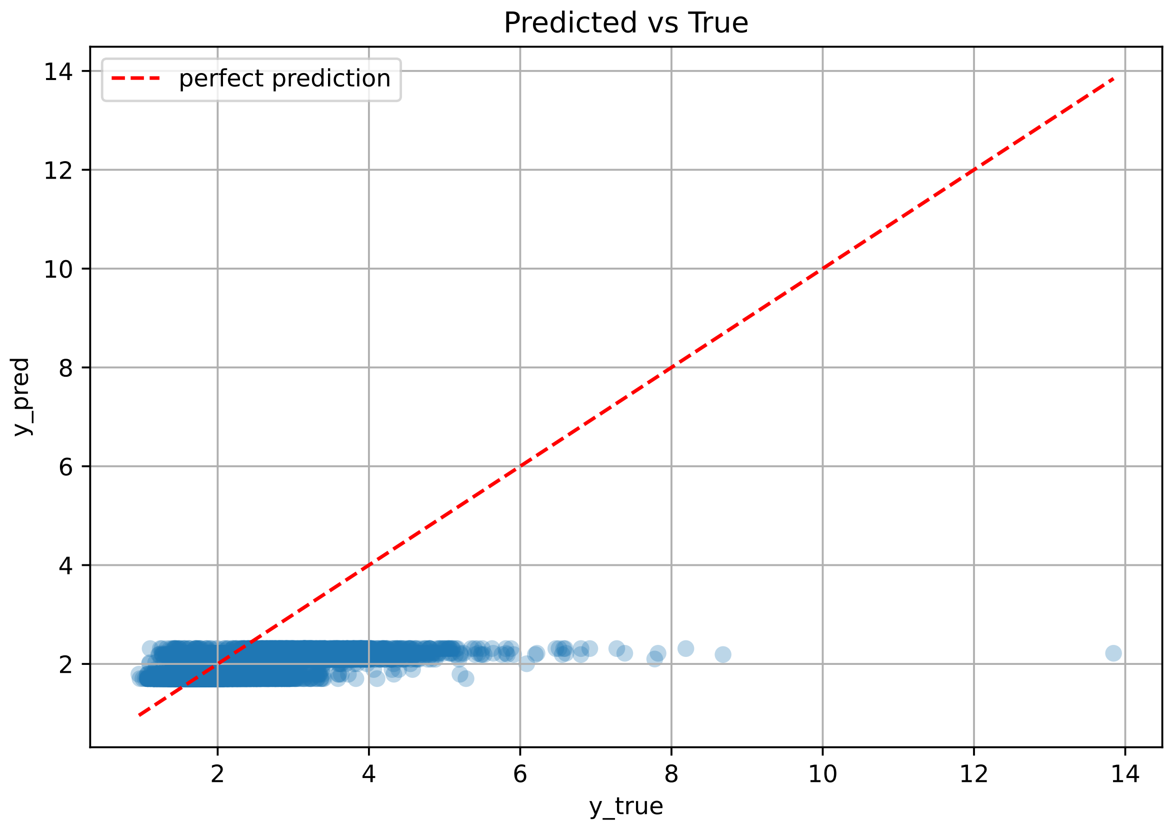
Rank 7: {’reg__n_estimators’: 400, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.07689834228685406, ’reg__num_leaves’: 15, ’reg__subsample’: 0.7924592548833778, ’reg__colsample_bytree’: 0.8280155270663115, ’reg__min_child_samples’: 45, ’reg__reg_alpha’: 4.487915542969104, ’reg__reg_lambda’: 42.023775138746046, ’reg__min_split_gain’: 3.0944850977255696, ’reg__path_smooth’: 4.5935752173731395, ’reg__min_child_weight’: 0.00397583051592986, ’reg__feature_fraction’: 0.7907193868412076, ’reg__bagging_fraction’: 0.8577675612339265, ’reg__bagging_freq’: 5, ’reg__max_bin’: 177, ’reg__min_data_per_group’: 189}

Rank 8: {’reg__n_estimators’: 700, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.0658746229347048, ’reg__num_leaves’: 42, ’reg__subsample’: 0.6960526466291886, ’reg__colsample_bytree’: 0.7758886950555154, ’reg__min_child_samples’: 40, ’reg__reg_alpha’: 3.265636344848776, ’reg__reg_lambda’: 27.761122098158918, ’reg__min_split_gain’: 3.725354278111619, ’reg__path_smooth’: 2.8046581854316837, ’reg__min_child_weight’: 0.0025574068512433383, ’reg__feature_fraction’: 0.7238644144733024, ’reg__bagging_fraction’: 0.7203022662730528, ’reg__bagging_freq’: 4, ’reg__max_bin’: 205, ’reg__min_data_per_group’: 152}

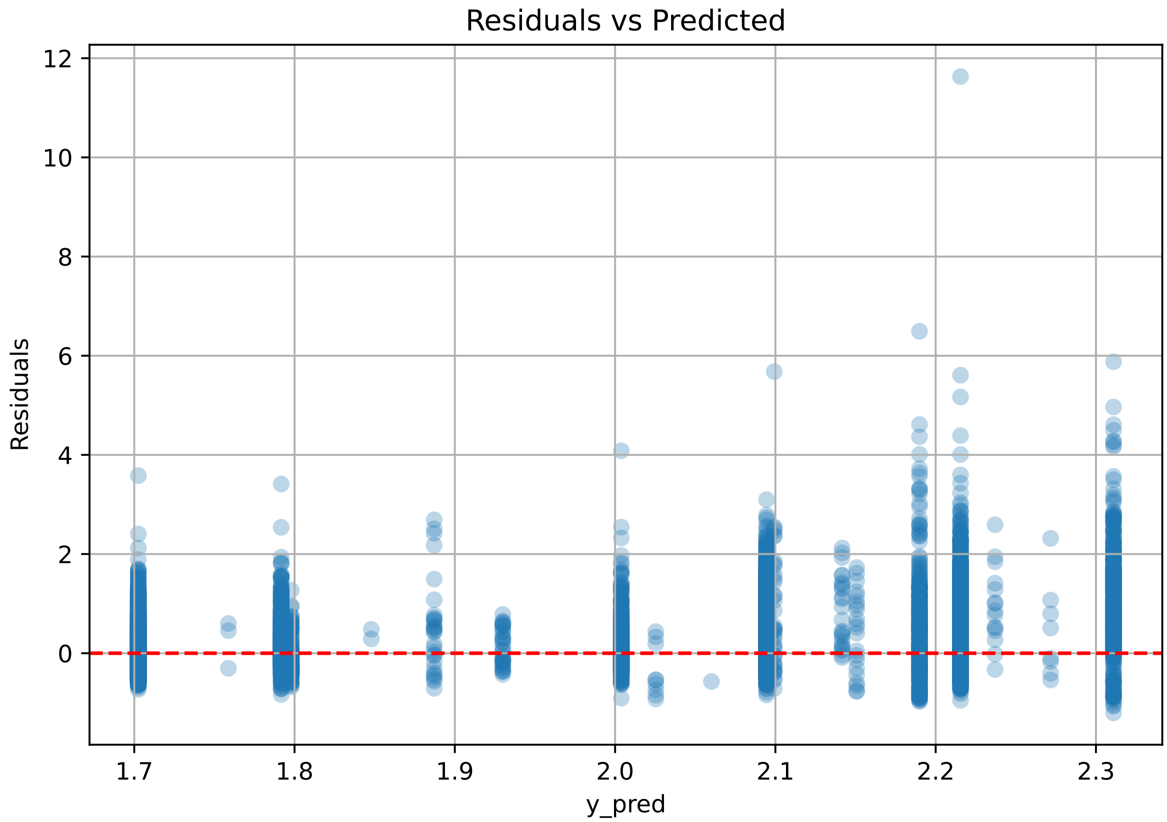
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.09946882341451506, 'reg__num_leaves': 24, 'reg__subsample': 0.772822094965967, 'reg__colsample_bytree': 0.768097399134577, 'reg__min_child_samples': 45, 'reg__reg_alpha': 3.002732005535586, 'reg__reg_lambda': 33.70760771142343, 'reg__min_split_gain': 3.693209233175715, 'reg__path_smooth': 3.5913294929895114, 'reg__min_child_weight': 0.006928265132712065, 'reg__feature_fraction': 0.7352160092616677, 'reg__bagging_fraction': 0.7825837523202377, 'reg__bagging_freq': 4, 'reg__max_bin': 193, 'reg__min_data_per_group': 171}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.047250756579606784, 'reg__num_leaves': 23, 'reg__subsample': 0.773295801165757, 'reg__colsample_bytree': 0.8423610612895109, 'reg__min_child_samples': 44, 'reg__reg_alpha': 5.214019825339021, 'reg__reg_lambda': 33.88071412493534, 'reg__min_split_gain': 2.8181364823853454, 'reg__path_smooth': 3.321412551418877, 'reg__min_child_weight': 0.002959053040600195, 'reg__feature_fraction': 0.7740189608273104, 'reg__bagging_fraction': 0.7763160976111807, 'reg__bagging_freq': 6, 'reg__max_bin': 187, 'reg__min_data_per_group': 171}

2.3.10 Scatter



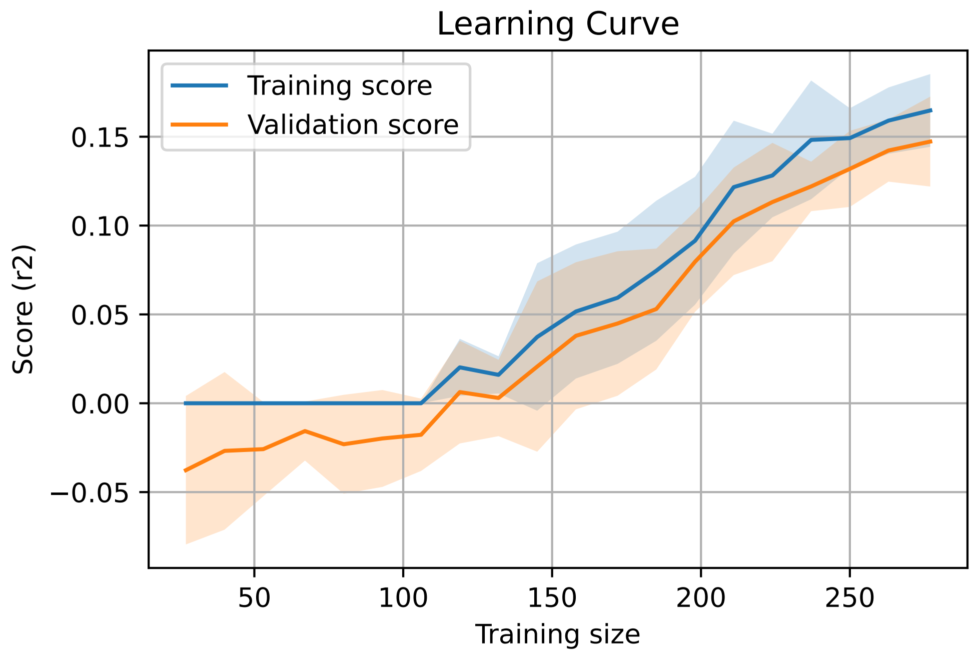
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 3

Len(group) : 568, **Len(training)** : 398,

Len(training_real) : 279, **Len(test_of_training)** : 119, **Len(test)** : 170

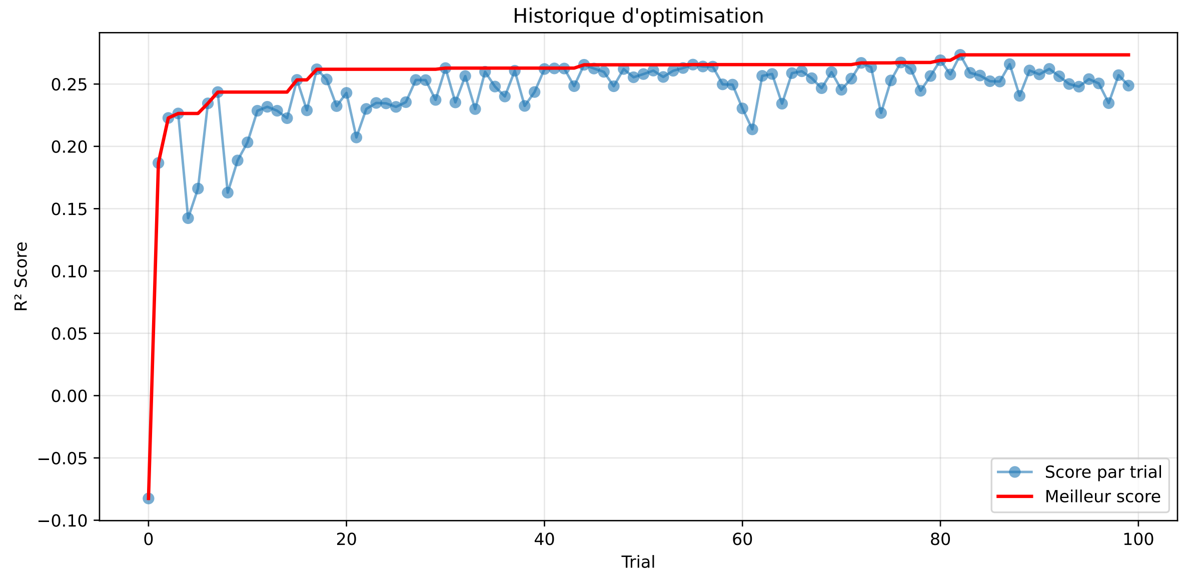


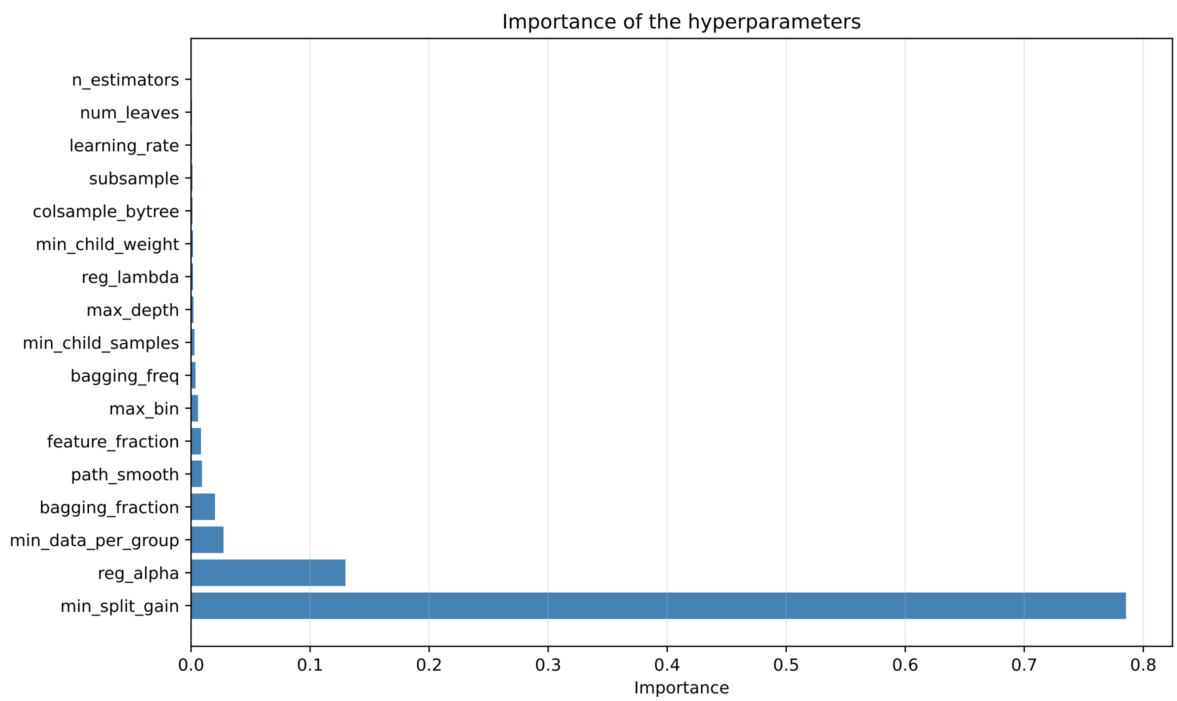
2.3.13 Gridsearch — Group: 4

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

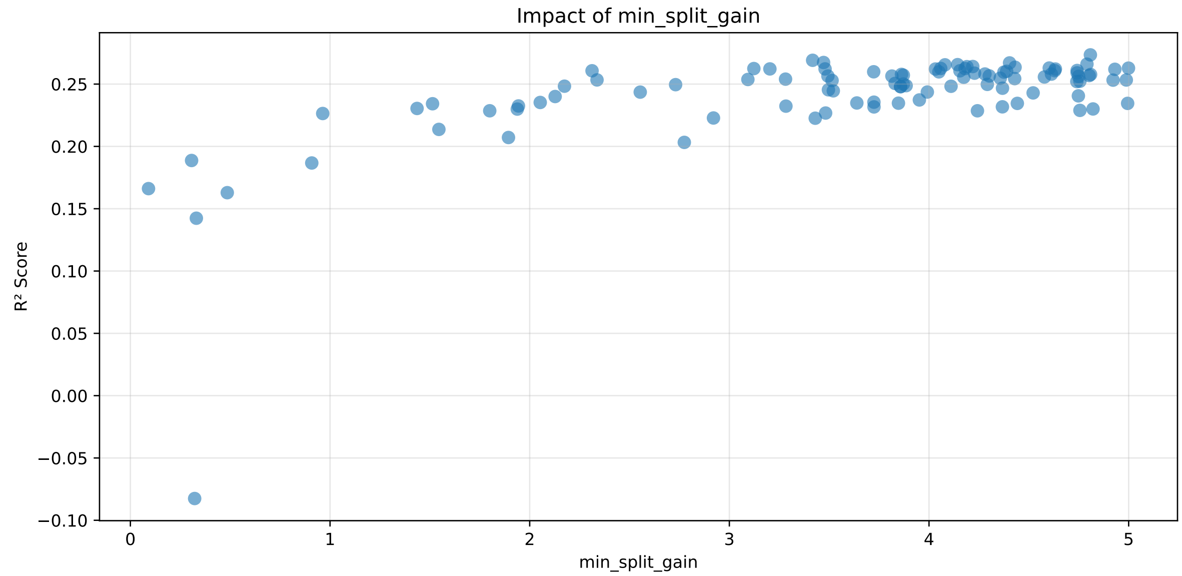
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813 y_test: mean=2.212, std=0.659, min=1.071, max=4.858

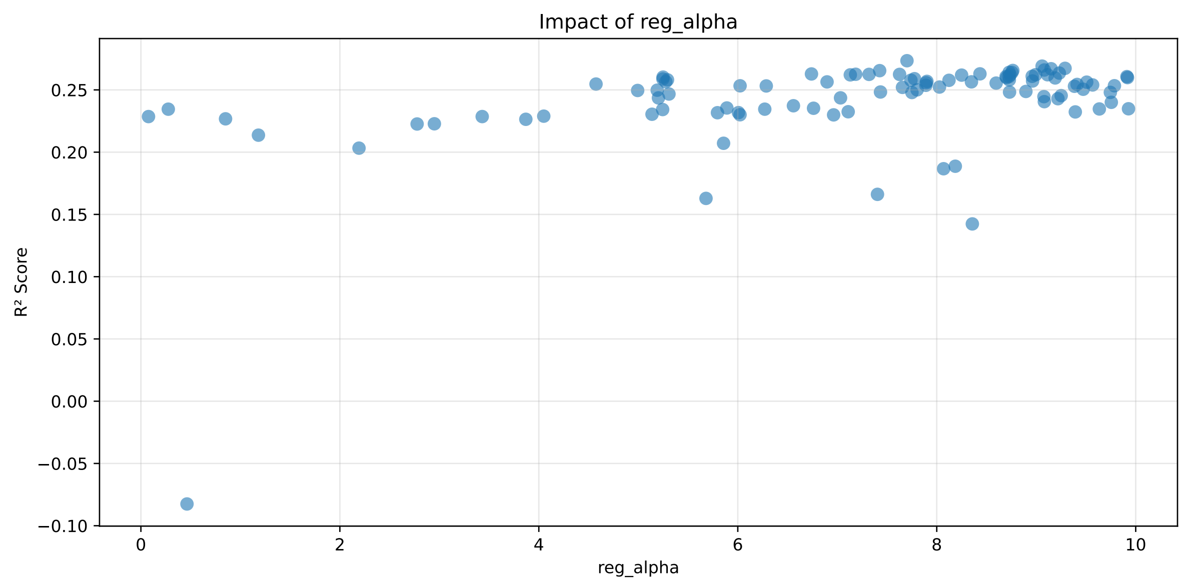
==================== TOP Importance FEATURES ==================== feature importance r_umidity 96 r_temperature 19 r_NH3 3 r_CO2 0 r_THI 0





Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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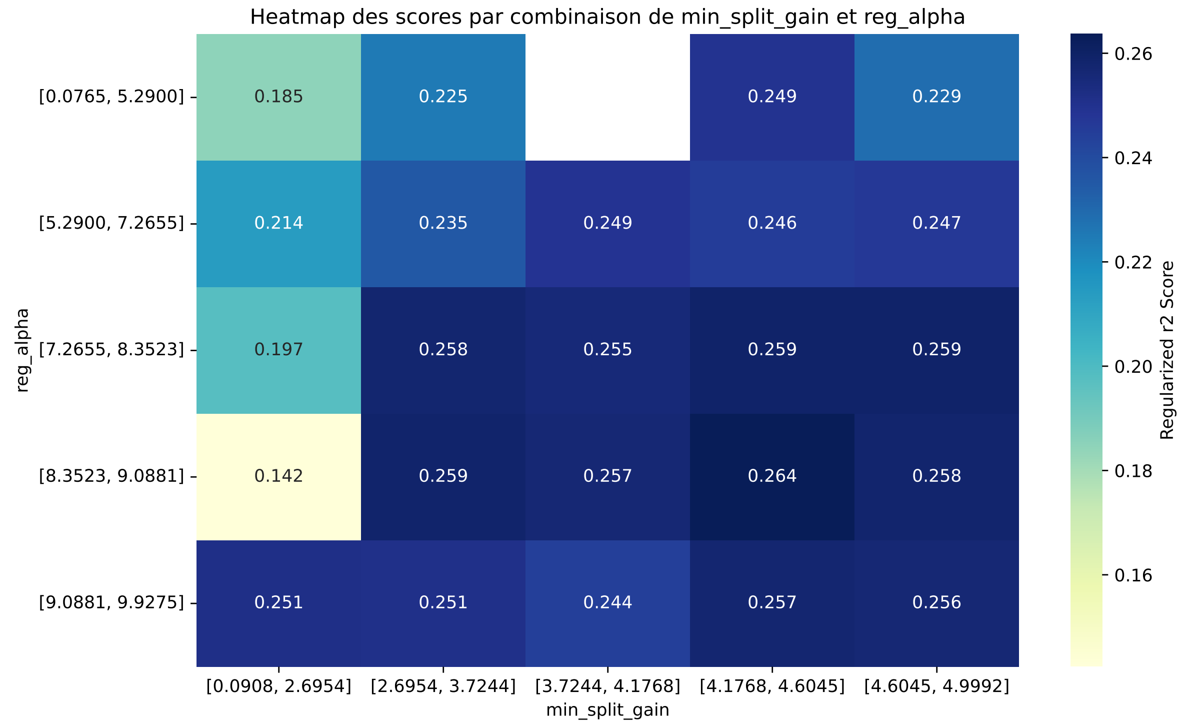
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3943	0.3741	0.4231	0.0258	0.2734	0.4357	0.0616	1.1647	1.0538
2	0.4064	0.3835	0.436	0.0263	0.2691	0.454	0.0705	1.1839	1.0597
3	0.3941	0.373	0.4244	0.0208	0.2673	0.4494	0.0765	1.205	1.0566
4	0.3857	0.3659	0.4161	0.0195	0.267	0.4336	0.0678	1.1852	1.0541
5	0.3905	0.3698	0.4216	0.019	0.266	0.4409	0.0711	1.1923	1.0561
6	0.4013	0.3787	0.4323	0.022	0.2656	0.441	0.0623	1.1645	1.0597
7	0.3988	0.3766	0.4282	0.0226	0.2655	0.442	0.0655	1.1738	1.059
8	0.3956	0.3737	0.4258	0.0232	0.2641	0.4244	0.0507	1.1356	1.0587
9	0.3979	0.3756	0.4294	0.0225	0.264	0.4342	0.0586	1.1559	1.0594
10	0.3954	0.3735	0.4268	0.0209	0.2635	0.4365	0.0631	1.1689	1.0589
---	---	---	---	---	---	---	---	---	---
100	0.5914	0.4791	0.5484	0.0227	-0.0825	0.5712	0.0921	1.1923	1.2345

Hyperparameters:

Rank 1: {'reg___n_estimators': 400, 'reg___max_depth': 6, 'reg___learning_rate': 0.02620269540944133, 'reg___num_leaves': 35, 'reg___subsample': 0.7982595786662869, 'reg___colsample_bytree': 0.60079325759206, 'reg___min_child_samples': 13, 'reg___reg_alpha': 7.700693168628662, 'reg___reg_lambda': 19.28589896067438, 'reg___min_split_gain': 4.808314934931934, 'reg___path_smooth': 0.931333436336485, 'reg___min_child_weight':

0.5490520553466952, 'reg__feature_fraction': 0.8604086639944215, 'reg__bagging_fraction': 0.5252853907756198, 'reg__bagging_freq': 3, 'reg__max_bin': 215, 'reg__min_data_per_group': 73}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.02458636450834637, 'reg__num_leaves': 35, 'reg__subsample': 0.8100243464645344, 'reg__colsample_bytree': 0.6002974783492132, 'reg__min_child_samples': 13, 'reg__reg_alpha': 9.059129467519522, 'reg__reg_lambda': 18.44319180752659, 'reg__min_split_gain': 3.417187247123839, 'reg__path_smooth': 0.3861447102423875, 'reg__min_child_weight': 8.409203883077234, 'reg__feature_fraction': 0.82725145656913, 'reg__bagging_fraction': 0.5847434485746009, 'reg__bagging_freq': 3, 'reg__max_bin': 214, 'reg__min_data_per_group': 80}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.07885040795018448, 'reg__num_leaves': 37, 'reg__subsample': 0.8005067551955194, 'reg__colsample_bytree': 0.5958629715500312, 'reg__min_child_samples': 23, 'reg__reg_alpha': 9.289689255895373, 'reg__reg_lambda': 18.268562098715293, 'reg__min_split_gain': 3.4719652490920785, 'reg__path_smooth': 0.9248228504860626, 'reg__min_child_weight': 1.8397029138149388, 'reg__feature_fraction': 0.7417091203731442, 'reg__bagging_fraction': 0.5207031398101656, 'reg__bagging_freq': 3, 'reg__max_bin': 132, 'reg__min_data_per_group': 75}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.08278574725672437, 'reg__num_leaves': 39, 'reg__subsample': 0.7323225899133994, 'reg__colsample_bytree': 0.6017845557422116, 'reg__min_child_samples': 23, 'reg__reg_alpha': 9.149727121883696, 'reg__reg_lambda': 18.548789688015802, 'reg__min_split_gain': 4.403473081400744, 'reg__path_smooth': 0.4260444996221834, 'reg__min_child_weight': 1.654990939031302, 'reg__feature_fraction': 0.7425885187902399, 'reg__bagging_fraction': 0.5262286559385787, 'reg__bagging_freq': 3, 'reg__max_bin': 75, 'reg__min_data_per_group': 79}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.024694882402844596, 'reg__num_leaves': 44, 'reg__subsample': 0.7448613368697204, 'reg__colsample_bytree': 0.6623066556897996, 'reg__min_child_samples': 29, 'reg__reg_alpha': 9.081714612778418, 'reg__reg_lambda': 13.899629875566836, 'reg__min_split_gain': 4.791107106609978, 'reg__path_smooth': 0.47965855225836007, 'reg__min_child_weight': 8.841349906086904, 'reg__feature_fraction': 0.7389965549802556, 'reg__bagging_fraction': 0.5486695666358862, 'reg__bagging_freq': 2, 'reg__max_bin': 63, 'reg__min_data_per_group': 74}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.059465882486968415, 'reg__num_leaves': 50, 'reg__subsample': 0.5815212173039832, 'reg__colsample_bytree': 0.580340375965542, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.764450215256186, 'reg__reg_lambda': 10.20835361025596, 'reg__min_split_gain': 4.143318997466703, 'reg__path_smooth': 0.7054454024683507, 'reg__min_child_weight':

1.139427549646167, ’reg___feature_fraction’: 0.7809539314668309, ’reg___bagging_fraction’: 0.5211721216034603, ’reg___bagging_freq’: 3, ’reg___max_bin’: 94, ’reg___min_data_per_group’: 81}

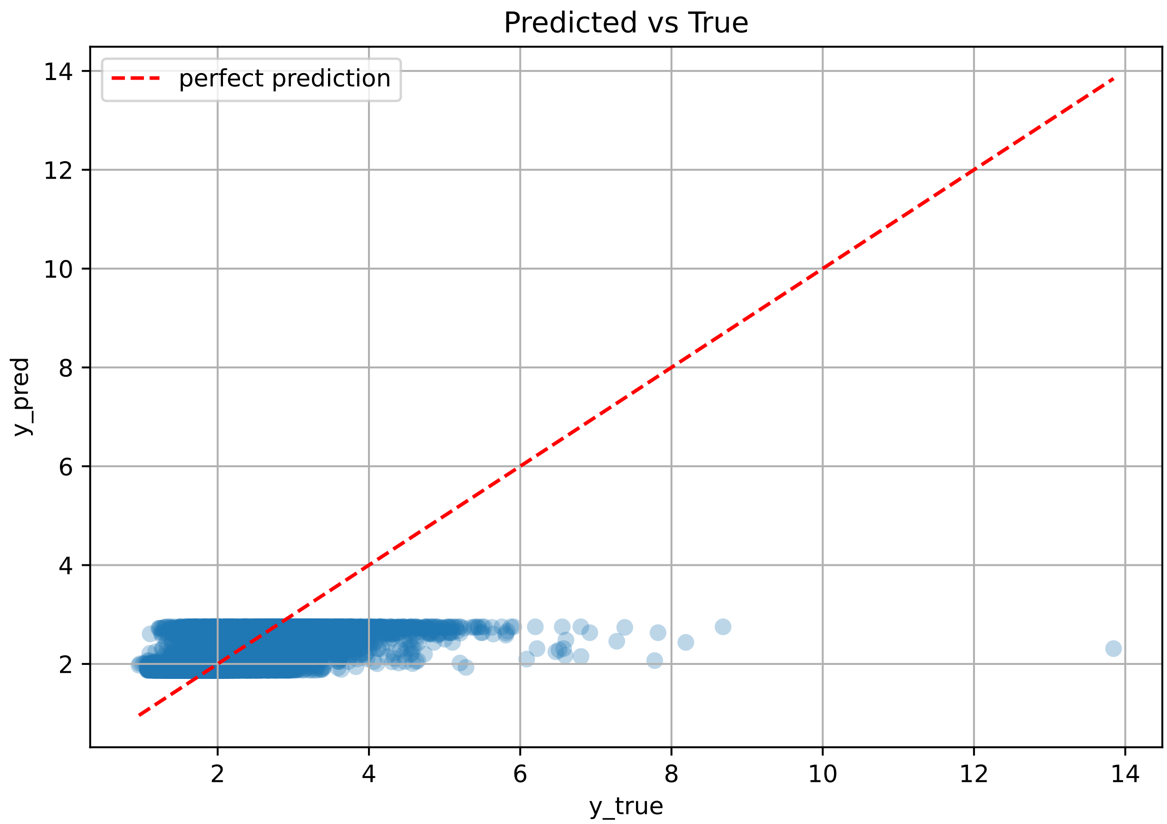
Rank 7: {’reg___n_estimators’: 500, ’reg___max_depth’: 4, ’reg___learning_rate’: 0.034850584610448904, ’reg___num_leaves’: 51, ’reg___subsample’: 0.7641497299559629, ’reg___colsample_bytree’: 0.5680787470392861, ’reg___min_child_samples’: 31, ’reg___reg_alpha’: 7.426179710771569, ’reg___reg_lambda’: 23.680879708395565, ’reg___min_split_gain’: 4.0800815917908, ’reg___path_smooth’: 0.6779512600272842, ’reg___min_child_weight’: 0.15317379228206807, ’reg___feature_fraction’: 0.8016932654901524, ’reg___bagging_fraction’: 0.539860091236712, ’reg___bagging_freq’: 3, ’reg___max_bin’: 108, ’reg___min_data_per_group’: 67}

Rank 8: {’reg___n_estimators’: 600, ’reg___max_depth’: 5, ’reg___learning_rate’: 0.026048046287632926, ’reg___num_leaves’: 41, ’reg___subsample’: 0.7640318674499655, ’reg___colsample_bytree’: 0.5880223759847052, ’reg___min_child_samples’: 15, ’reg___reg_alpha’: 8.727895542666515, ’reg___reg_lambda’: 21.834454874270552, ’reg___min_split_gain’: 4.219289170746107, ’reg___path_smooth’: 0.6498148348410563, ’reg___min_child_weight’: 1.2749644800958737, ’reg___feature_fraction’: 0.7735322687380901, ’reg___bagging_fraction’: 0.5250247742934572, ’reg___bagging_freq’: 3, ’reg___max_bin’: 94, ’reg___min_data_per_group’: 83}

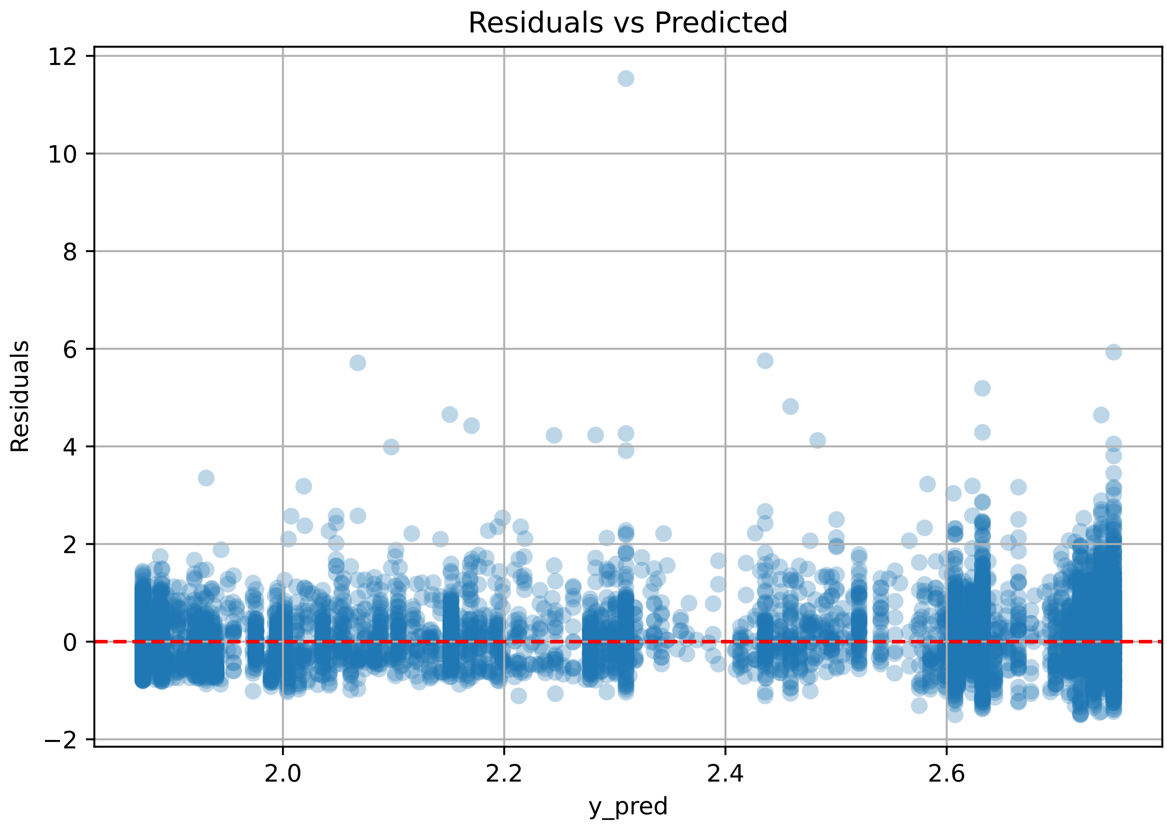
Rank 9: {’reg___n_estimators’: 600, ’reg___max_depth’: 5, ’reg___learning_rate’: 0.06373950148284548, ’reg___num_leaves’: 41, ’reg___subsample’: 0.7879380910167357, ’reg___colsample_bytree’: 0.5963864828692653, ’reg___min_child_samples’: 18, ’reg___reg_alpha’: 8.756032562924014, ’reg___reg_lambda’: 21.47598721947774, ’reg___min_split_gain’: 4.1897869776759515, ’reg___path_smooth’: 0.5155456027221629, ’reg___min_child_weight’: 0.9367285689830326, ’reg___feature_fraction’: 0.760683900684774, ’reg___bagging_fraction’: 0.5680319811719978, ’reg___bagging_freq’: 3, ’reg___max_bin’: 79, ’reg___min_data_per_group’: 85}

Rank 10: {’reg___n_estimators’: 700, ’reg___max_depth’: 6, ’reg___learning_rate’: 0.07973354499764404, ’reg___num_leaves’: 38, ’reg___subsample’: 0.8062993597454653, ’reg___colsample_bytree’: 0.601744995895052, ’reg___min_child_samples’: 22, ’reg___reg_alpha’: 9.231979661540933, ’reg___reg_lambda’: 18.798673289837133, ’reg___min_split_gain’: 4.431090152187326, ’reg___path_smooth’: 0.9431736977628659, ’reg___min_child_weight’: 1.7469628756951538, ’reg___feature_fraction’: 0.835959903248392, ’reg___bagging_fraction’: 0.5831340027345802, ’reg___bagging_freq’: 3, ’reg___max_bin’: 73, ’reg___min_data_per_group’: 77}

2.3.14 Scatter



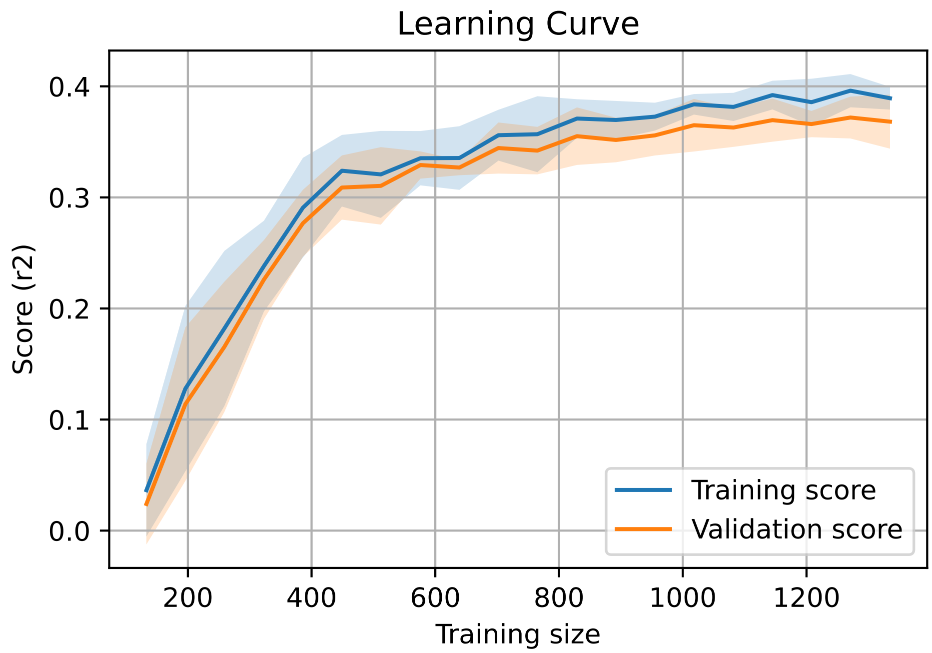
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 4

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817

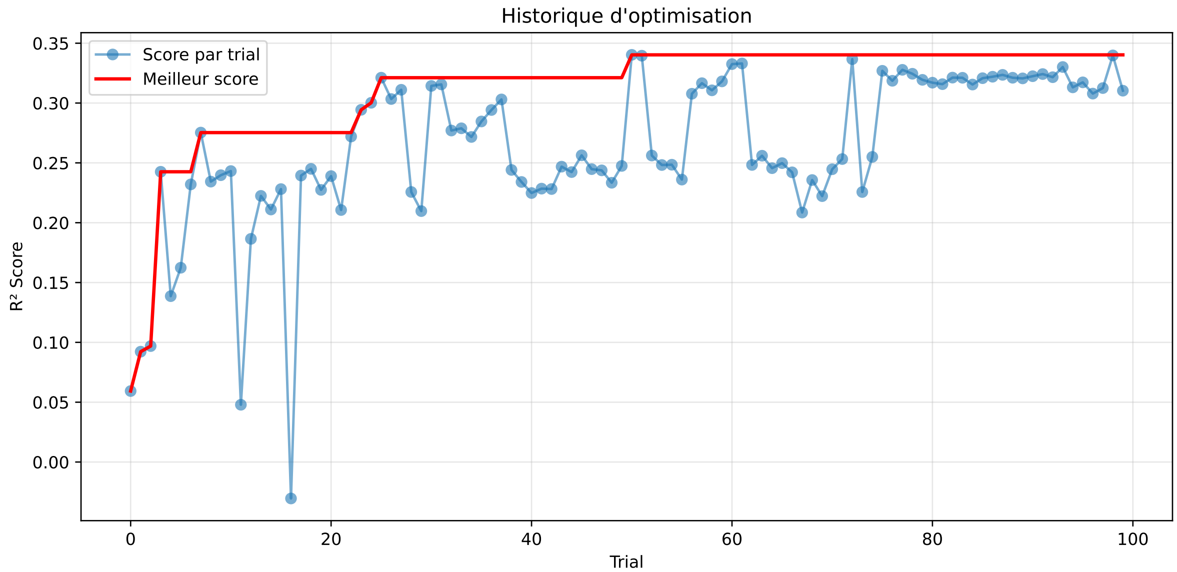


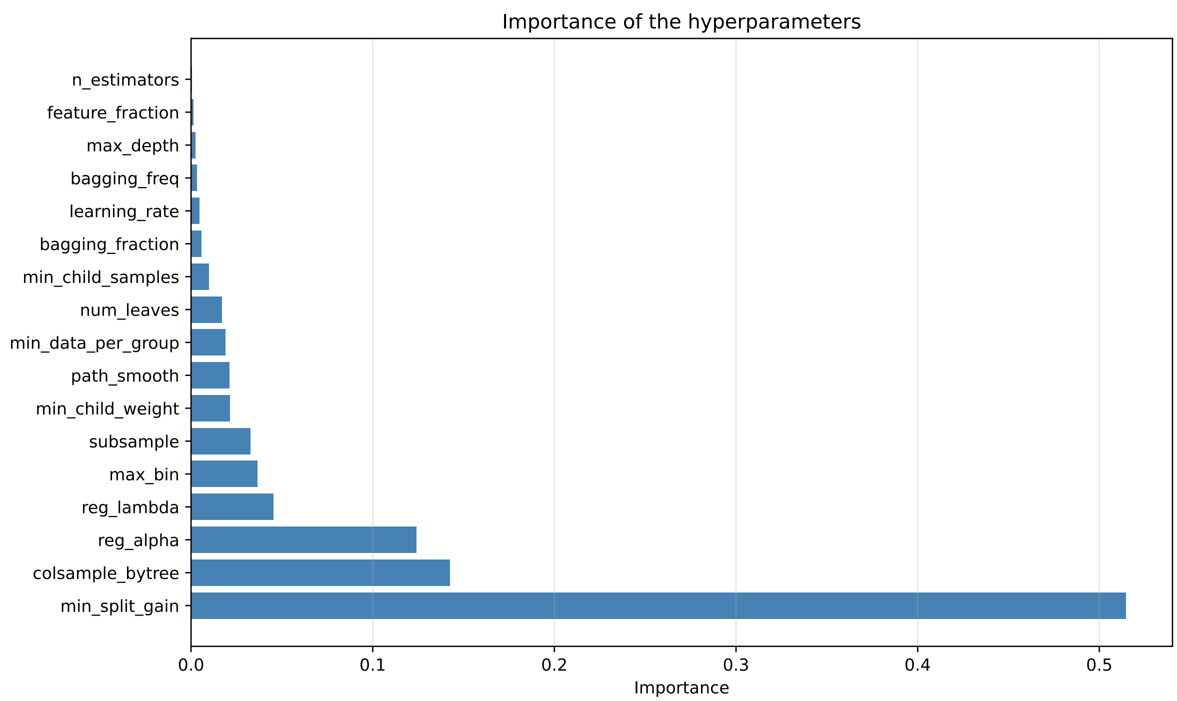
2.3.17 Gridsearch — Group: 6

Len(group) : 1353, **Len(training)** : 948, **Len(test)** : 405

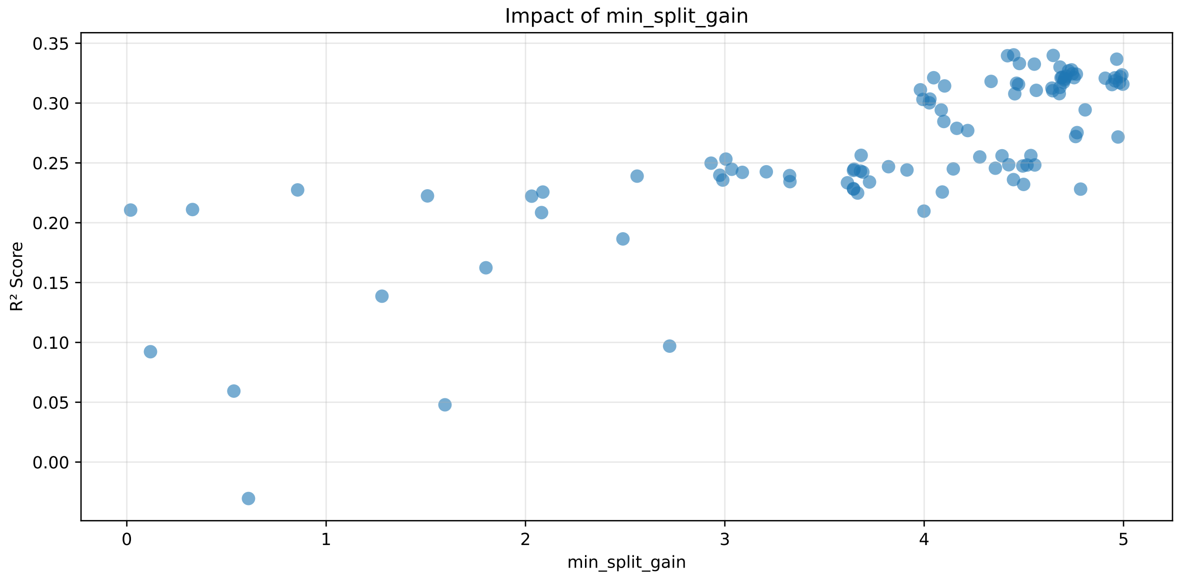
Distribution y_train vs y_test: y_train: mean=2.286, std=0.802, min=1.041, max=7.383 y_test: mean=2.187, std=0.733, min=1.078, max=5.525

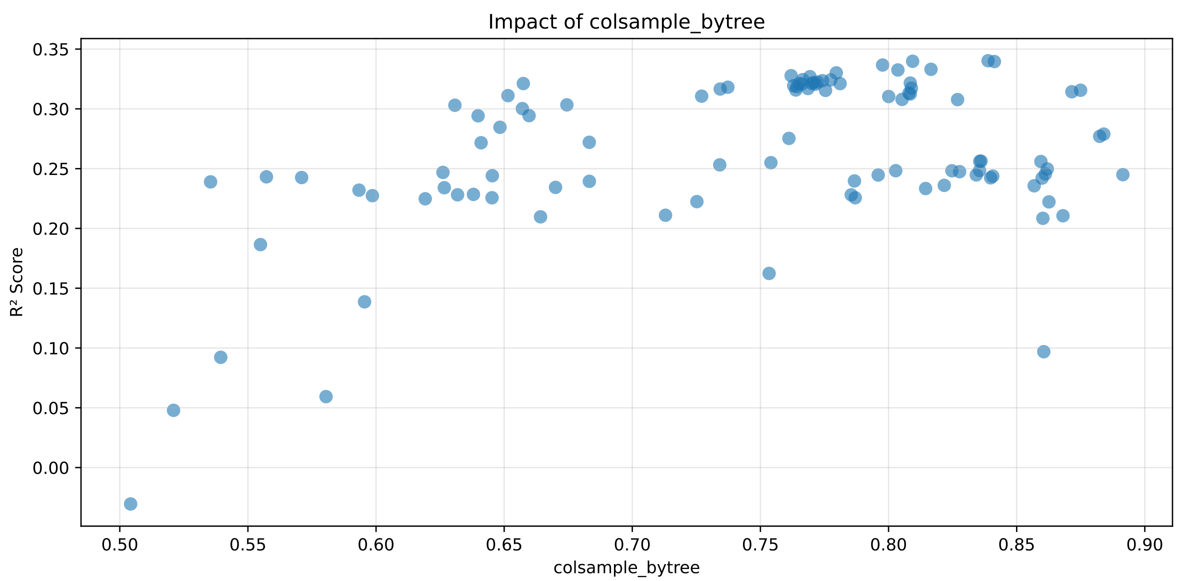
==================== TOP Importance FEATURES ==================== feature importance r_umidity 98 r_NH3 67 r_THI 49 r_temperature 42 r_CO2 0

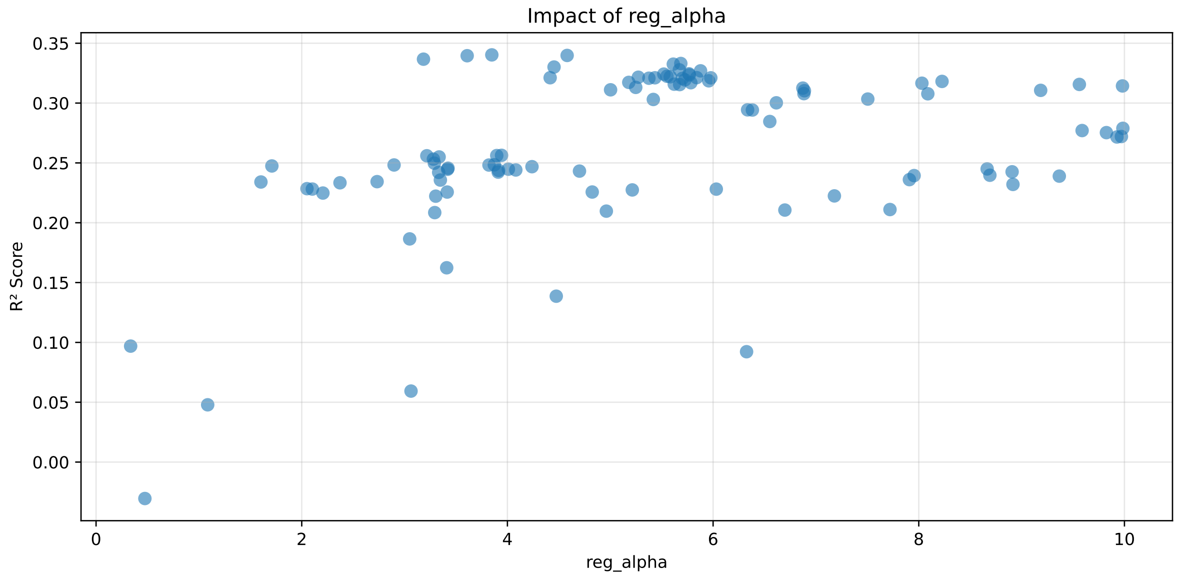




Paramètres avec >5% d'importance : ['min_split_gain', 'colsample_bytree', 'reg_alpha']


Impact of min_split_gain
R² Score
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3
4
5
min_split_gain


Impact of colsample_bytree
R² Score
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
colsample_bytree

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The default value of observed=False is deprecated and will change to observed=True in a futu
```
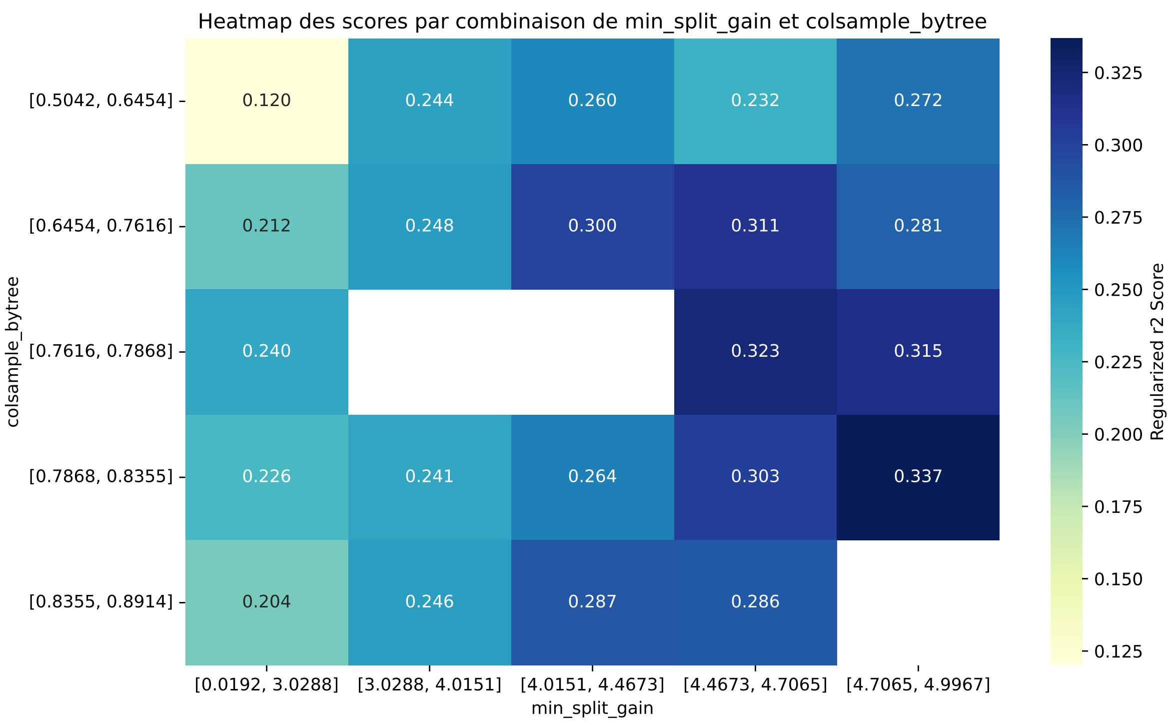



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3571	0.3403	0.4035	0.0553	0.3403	0.4286	0.0884	1.2597	1.0495
2	0.3566	0.3398	0.4063	0.0516	0.3398	0.4488	0.109	1.3207	1.0495
3	0.3563	0.3396	0.4026	0.0541	0.3396	0.4544	0.1148	1.338	1.0492
4	0.3531	0.3367	0.4001	0.0502	0.3367	0.416	0.0793	1.2355	1.0488
5	0.348	0.3331	0.395	0.0517	0.3331	0.4375	0.1044	1.3135	1.0449
6	0.3467	0.3325	0.394	0.0501	0.3325	0.4225	0.09	1.2706	1.0427
7	0.3458	0.3301	0.392	0.0455	0.3301	0.4074	0.0773	1.2341	1.0474
8	0.3417	0.3278	0.3866	0.0469	0.3278	0.4279	0.1002	1.3056	1.0424
9	0.3423	0.3269	0.3908	0.0446	0.3269	0.414	0.087	1.2661	1.0469
10	0.3385	0.3245	0.3857	0.0411	0.3245	0.3994	0.0749	1.2308	1.0432
---	---	---	---	---	---	---	---	---	---
100	0.5903	0.4868	0.6169	0.0696	-0.0305	0.616	0.1292	1.2653	1.2125

Hyperparameters:

Rank 1: {'reg___n_estimators': 700, 'reg___max_depth': 6, 'reg___learning_rate': 0.01651879267993922, 'reg___num_leaves': 26, 'reg___subsample': 0.7589802288100936, 'reg___colsample_bytree': 0.8388392528901575, 'reg___min_child_samples': 18, 'reg___reg_alpha': 3.848312474022568, 'reg___reg_lambda': 26.236667301185207, 'reg___min_split_gain': 4.449079527751974, 'reg___path_smooth': 4.3139317865993325, 'reg___min_child_weight': 0.0010538752942487338, 'reg___feature_fraction': 0.8980636990485339, 'reg___bagging_fraction': 0.6277377818212624, 'reg___bagging_freq': 3, 'reg___max_bin': 199, 'reg___min_data_per_group': 118}

Rank 2: {'reg___n_estimators': 400, 'reg___max_depth': 6, 'reg___learning_rate': 0.03272342376270277, 'reg___num_leaves': 16, 'reg___subsample': 0.6912906650954697, 'reg___colsample_bytree': 0.8094015595854205, 'reg___min_child_samples': 38, 'reg___reg_alpha': 4.580770861171073, 'reg___reg_lambda': 16.351432773032545, 'reg___min_split_gain': 4.647408855416647, 'reg___path_smooth': 3.5298574917524506, 'reg___min_child_weight': 0.001593160443924852, 'reg___feature_fraction': 0.7571446566821013, 'reg___bagging_fraction': 0.6711426111698455, 'reg___bagging_freq': 7, 'reg___max_bin': 185, 'reg___min_data_per_group': 89}

Rank 3: {'reg___n_estimators': 700, 'reg___max_depth': 6, 'reg___learning_rate': 0.015835088843680283, 'reg___num_leaves': 27, 'reg___subsample': 0.7730073819408686, 'reg___colsample_bytree': 0.8413010466270822, 'reg___min_child_samples': 25, 'reg___reg_alpha': 3.6086785871897646, 'reg___reg_lambda': 26.41017857625388, 'reg___min_split_gain': 4.418066905150074, 'reg___path_smooth': 4.2519051740897, 'reg___min_child_weight': 0.0013191764545644748, 'reg___feature_fraction': 0.8976707683183669, 'reg___bagging_fraction': 0.6181278497275235, 'reg___bagging_freq': 3, 'reg___max_bin': 175, 'reg___min_data_per_group': 118}

Rank 4: {'reg___n_estimators': 600, 'reg___max_depth': 7, 'reg___learning_rate': 0.022785513309736577, 'reg___num_leaves': 20, 'reg___subsample': 0.7492655215730543, 'reg___colsample_bytree': 0.7976855683395061, 'reg___min_child_samples': 15, 'reg___reg_alpha': 3.1838983308985487, 'reg___reg_lambda': 23.034881468262228, 'reg___min_split_gain': 4.965946868897547, 'reg___path_smooth': 3.90741566472567, 'reg___min_child_weight': 0.002385402375561533, 'reg___feature_fraction': 0.8665579378883306, 'reg___bagging_fraction': 0.5845028893782676, 'reg___bagging_freq': 6, 'reg___max_bin': 196, 'reg___min_data_per_group': 99}

Rank 5: {'reg___n_estimators': 600, 'reg___max_depth': 7, 'reg___learning_rate': 0.017366099983774204, 'reg___num_leaves': 24, 'reg___subsample': 0.7564288048132162, 'reg___colsample_bytree': 0.8165698515624095, 'reg___min_child_samples': 12, 'reg___reg_alpha': 5.686083393279615, 'reg___reg_lambda': 26.578251267763772, 'reg___min_split_gain': 4.477770352350298, 'reg___path_smooth': 4.344272508047972, 'reg___min_child_weight': 0.00138343257492917, 'reg___feature_fraction': 0.7999908206567521, 'reg___bagging_fraction': 0.6062728939442519, 'reg___bagging_freq': 4, 'reg___max_bin': 188, 'reg___min_data_per_group': 119}

Rank 6: {'reg___n_estimators': 600, 'reg___max_depth': 6, 'reg___learning_rate': 0.028034016072788003, 'reg___num_leaves': 25, 'reg___subsample': 0.7551641632017981, 'reg___colsample_bytree': 0.8036512004333282, 'reg___min_child_samples': 12, 'reg___reg_alpha': 5.613067207573976, 'reg___reg_lambda': 24.94158582908179, 'reg___min_split_gain': 4.552517654075373, 'reg___path_smooth': 4.430216260233826, 'reg___min_child_weight': 0.0013623149530093018, 'reg___feature_fraction': 0.855806017739711, 'reg___bagging_fraction': 0.6084498457177153, 'reg___bagging_freq': 6, 'reg___max_bin': 199, 'reg___min_data_per_group': 119}

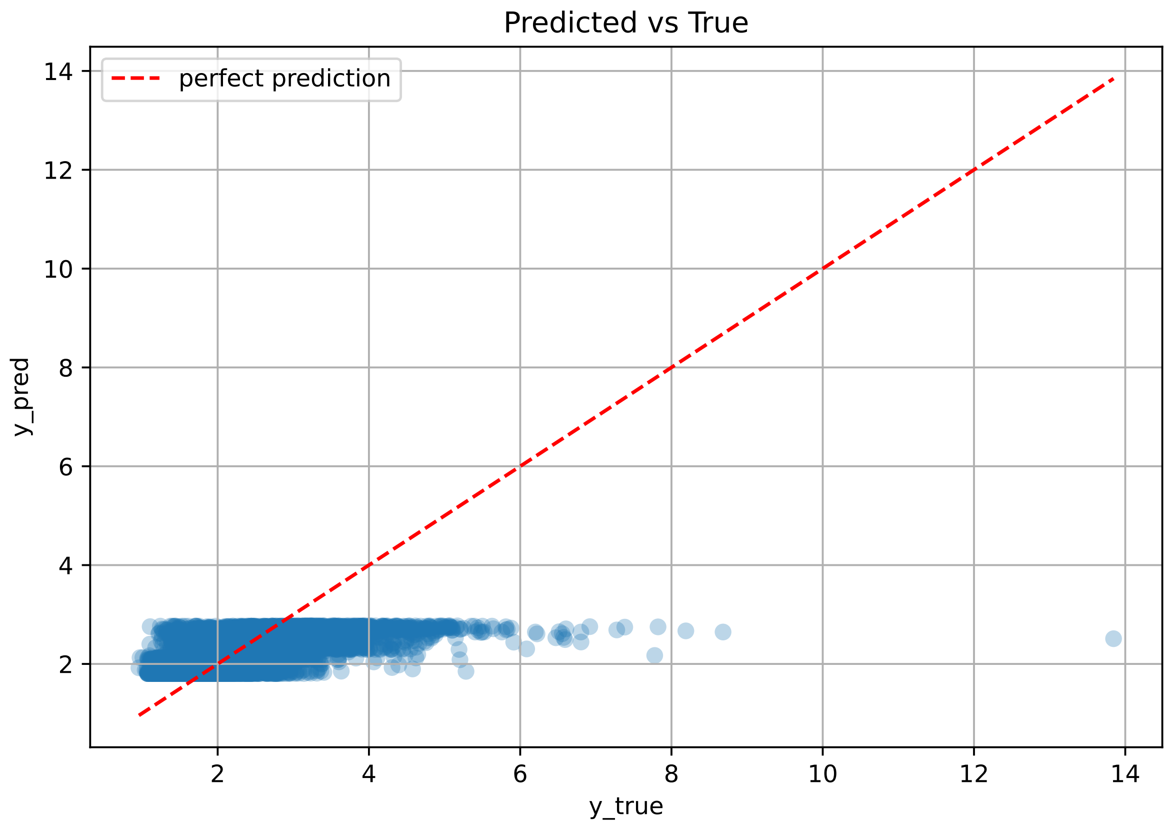
Rank 7: {'reg___n_estimators': 500, 'reg___max_depth': 6, 'reg___learning_rate': 0.01894599460495914, 'reg___num_leaves': 18, 'reg___subsample': 0.7128613621416384, 'reg___colsample_bytree': 0.7796036124566602, 'reg___min_child_samples': 16, 'reg___reg_alpha': 4.45341252861459, 'reg___reg_lambda': 16.445304158118255, 'reg___min_split_gain': 4.681282315011505, 'reg___path_smooth': 4.63927884251529, 'reg___min_child_weight': 0.0017554755251887968, 'reg___feature_fraction': 0.7419228388200726, 'reg___bagging_fraction': 0.5414697868125382, 'reg___bagging_freq': 7, 'reg___max_bin': 214, 'reg___min_data_per_group': 90}

Rank 8: {'reg___n_estimators': 500, 'reg___max_depth': 6, 'reg___learning_rate': 0.020618577631172464, 'reg___num_leaves': 20, 'reg___subsample': 0.7480375510428413, 'reg___colsample_bytree': 0.7619898452645015, 'reg___min_child_samples': 32, 'reg___reg_alpha': 5.67197960128132, 'reg___reg_lambda': 20.407195683116345, 'reg___min_split_gain': 4.739441735947523, 'reg___path_smooth': 3.6506274973878314, 'reg___min_child_weight': 0.0022532089761853535, 'reg___feature_fraction': 0.8767660287030529, 'reg___bagging_fraction': 0.5888932008000822, 'reg___bagging_freq': 6, 'reg___max_bin': 208, 'reg___min_data_per_group': 132}

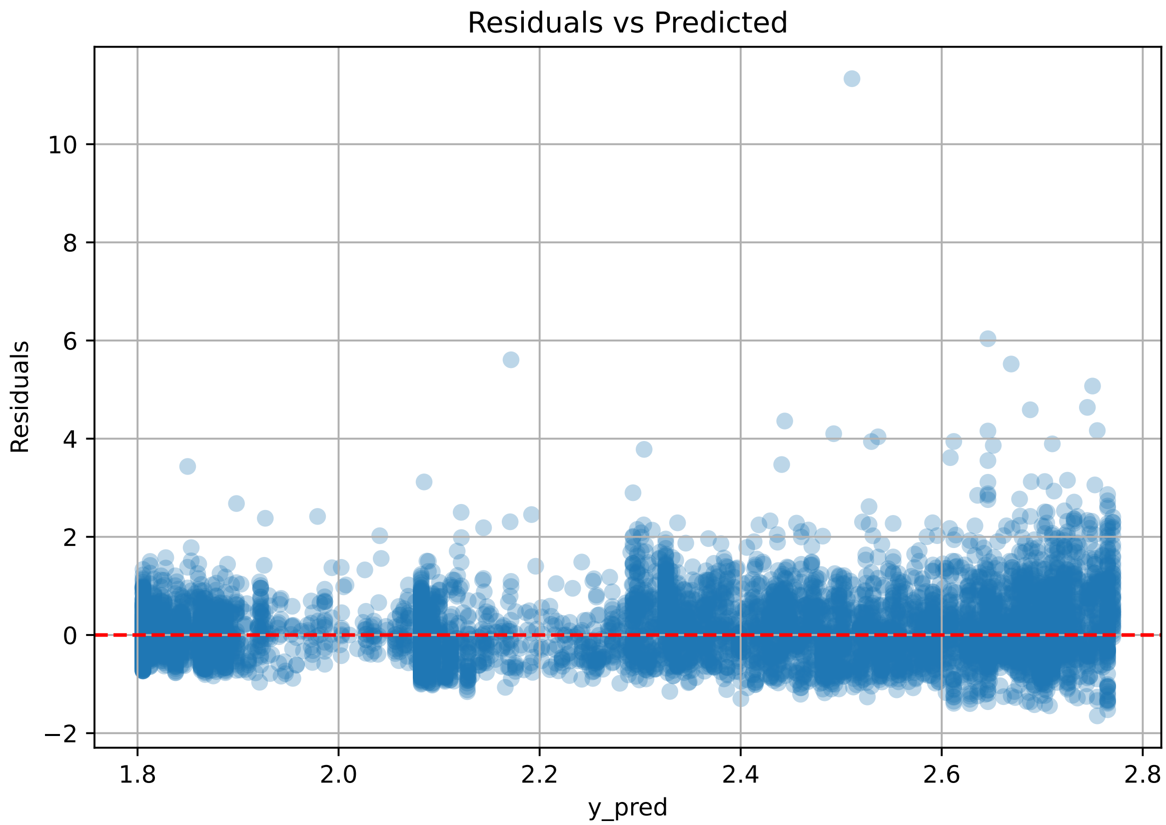
Rank 9: {'reg___n_estimators': 600, 'reg___max_depth': 6, 'reg___learning_rate': 0.024947422042753693, 'reg___num_leaves': 20, 'reg___subsample': 0.7353402874995331, 'reg___colsample_bytree': 0.769345477857414, 'reg___min_child_samples': 33, 'reg___reg_alpha': 5.878103409308824, 'reg___reg_lambda': 21.77165921255304, 'reg___min_split_gain': 4.723843707532525, 'reg___path_smooth': 4.0546136586884645, 'reg___min_child_weight': 0.001003193967829553, 'reg___feature_fraction': 0.8793609646814218, 'reg___bagging_fraction': 0.584646168455344, 'reg___bagging_freq': 6, 'reg___max_bin': 193, 'reg___min_data_per_group': 100}

Rank 10: {'reg___n_estimators': 500, 'reg___max_depth': 6, 'reg___learning_rate': 0.03996033933356695, 'reg___num_leaves': 20, 'reg___subsample': 0.7314962421276996, 'reg___colsample_bytree': 0.766531712315085, 'reg___min_child_samples': 32, 'reg___reg_alpha': 5.76535786587341, 'reg___reg_lambda': 15.207883915086597, 'reg___min_split_gain': 4.743599101989727, 'reg___path_smooth': 3.812234984859164, 'reg___min_child_weight': 0.0024780364412319436, 'reg___feature_fraction': 0.7796117752362679, 'reg___bagging_fraction': 0.5764977623132611, 'reg___bagging_freq': 6, 'reg___max_bin': 187, 'reg___min_data_per_group': 133}

2.3.18 Scatter



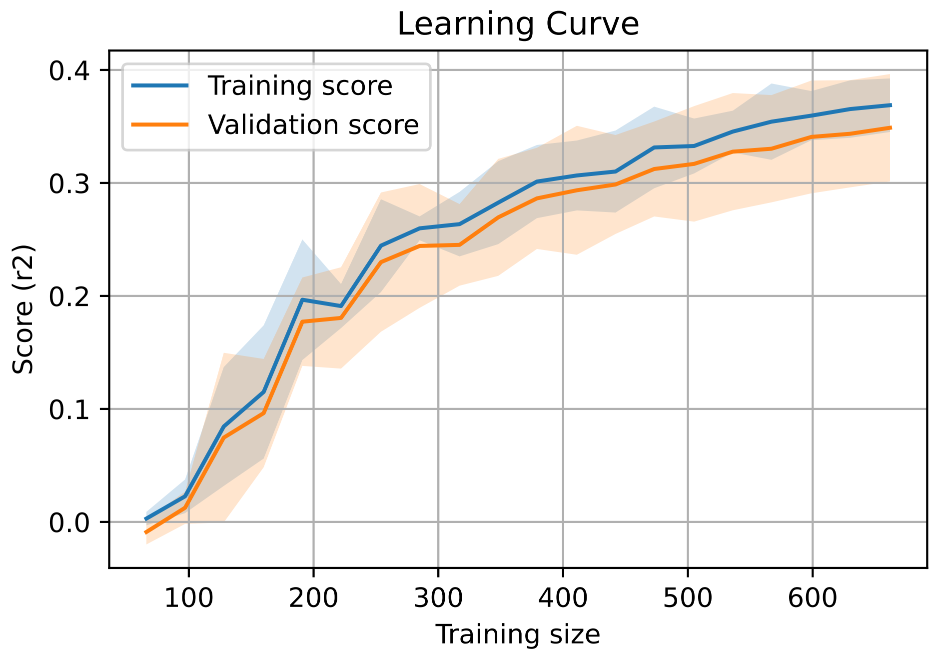
2.3.19 Scatter residuals



2.3.20 Learning curve — Group: 6

Len(group) : 1353, **Len(training)** : 948,

Len(training_real) : 664, **Len(test_of_training)** : 284, **Len(test)** : 405



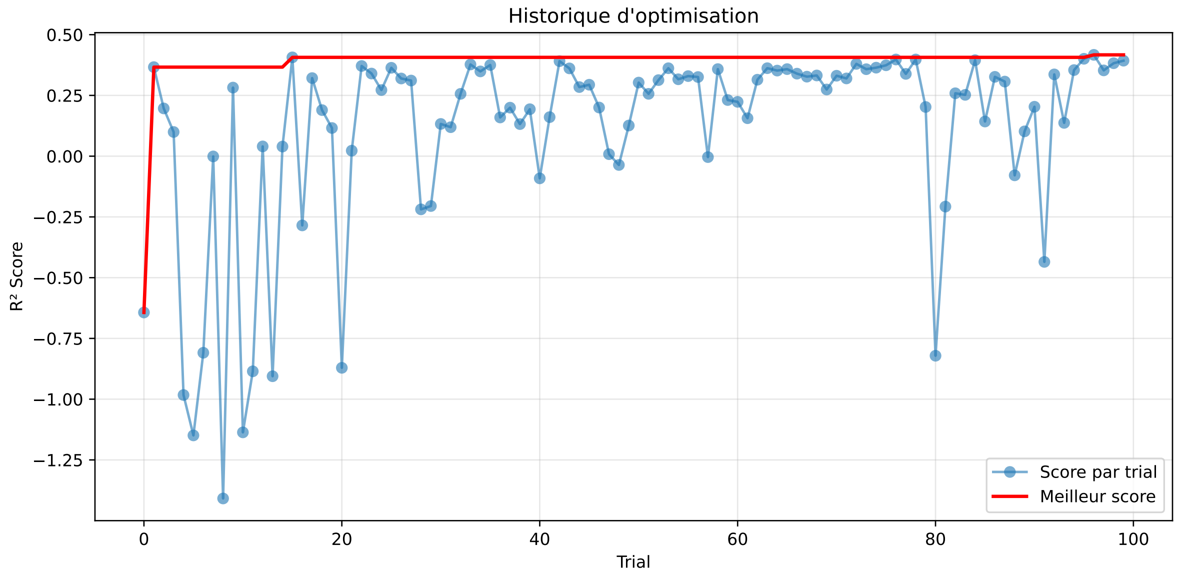
2.4 Model : CatBoost

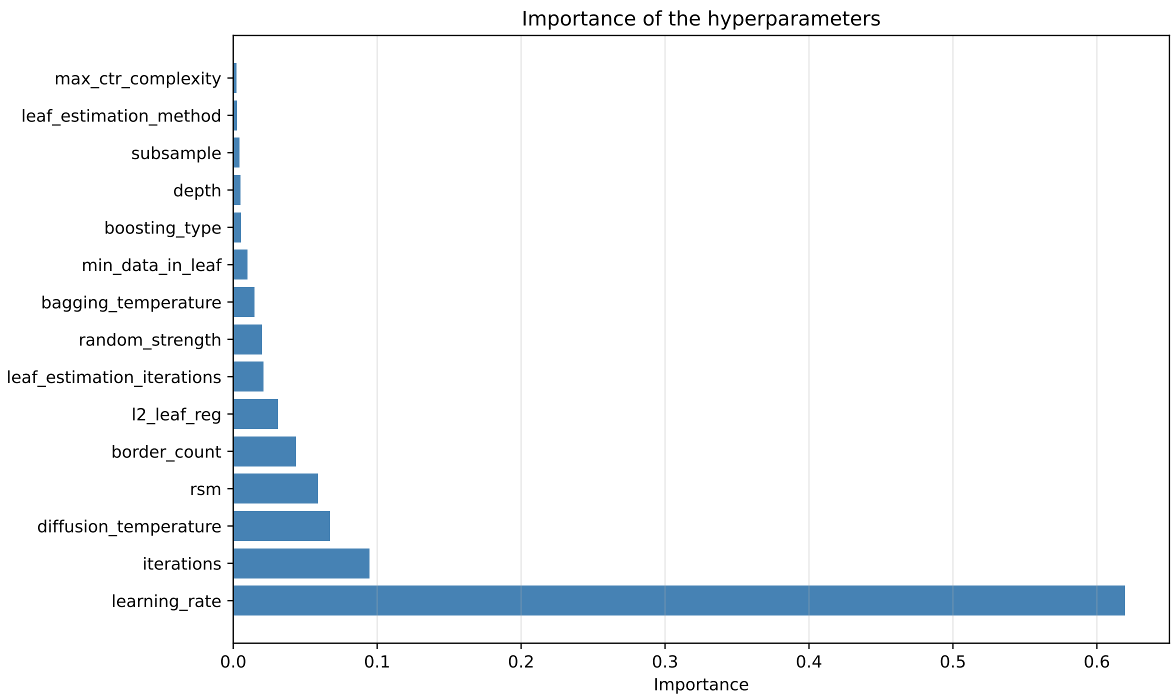
2.4.1 Gridsearch — Group: 1

Len(group) : 795, **Len(training)** : 557, **Len(test)** : 238

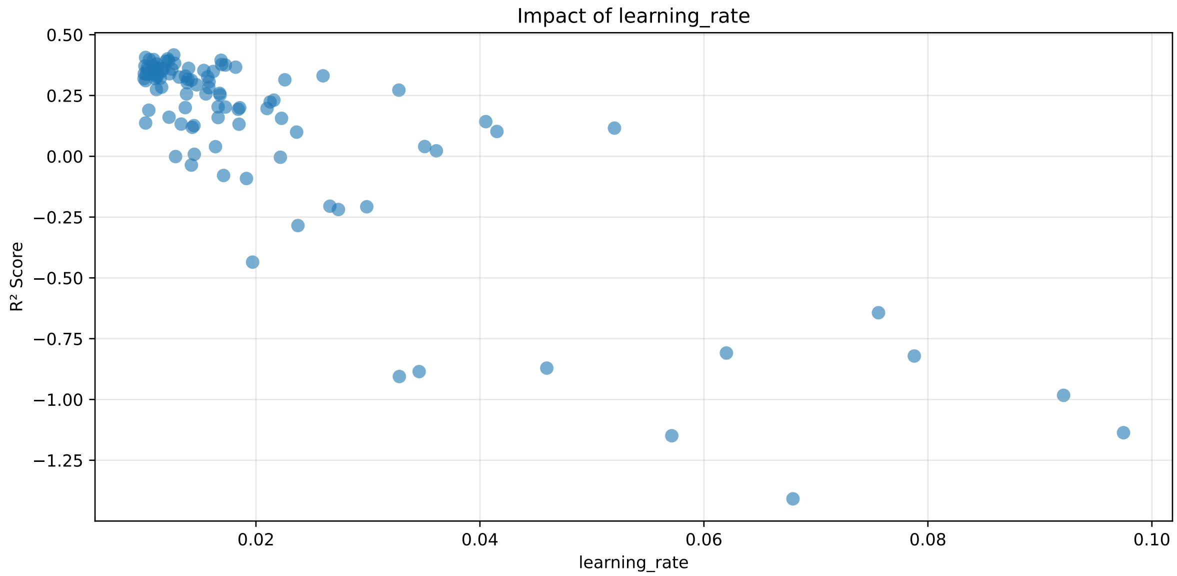
Distribution y_train vs y_test: y_train: mean=2.597, std=0.579, min=1.437, max=4.986 y_test: mean=2.492, std=0.625, min=1.466, max=5.447

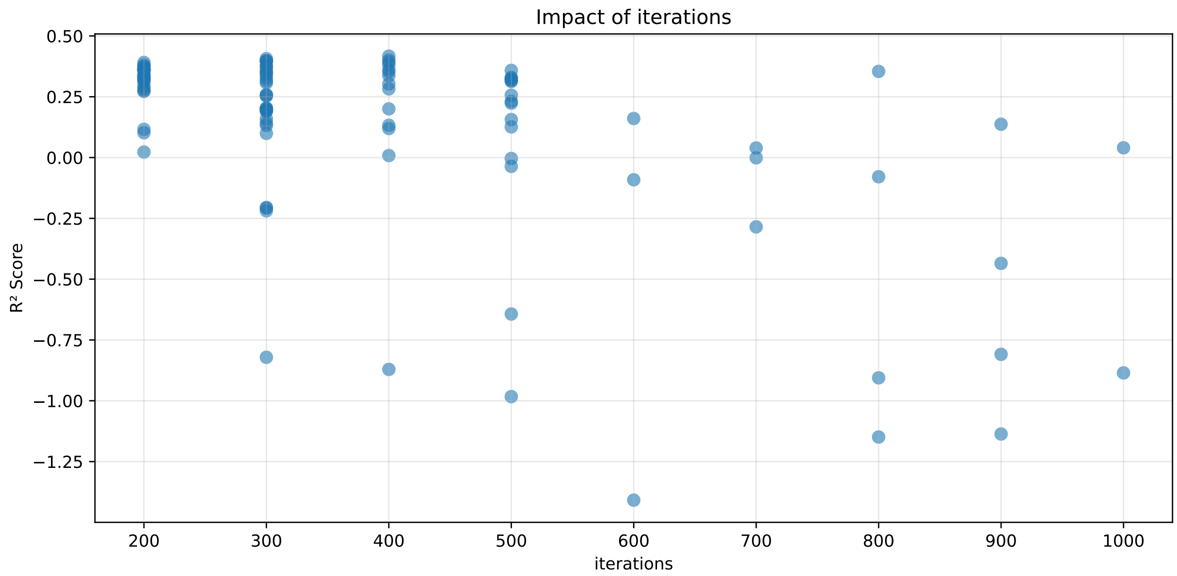
==================== TOP Importance FEATURES ==================== feature importance r_temperature 30.461256 r_THI 24.740957 r_umidity 22.149818 r_CO2 12.716653 r_NH3 9.931315

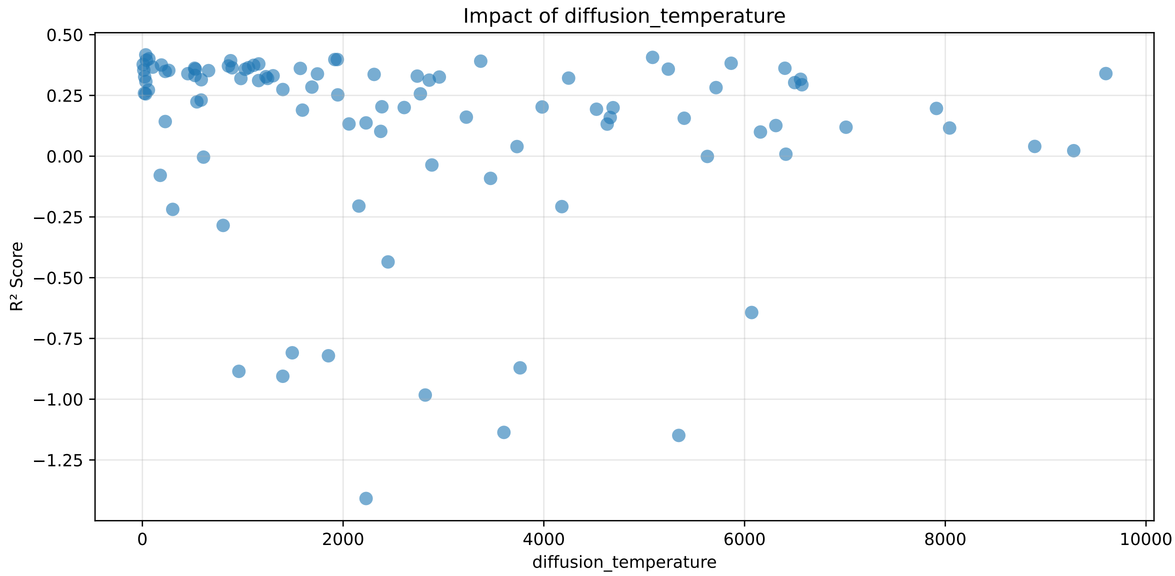


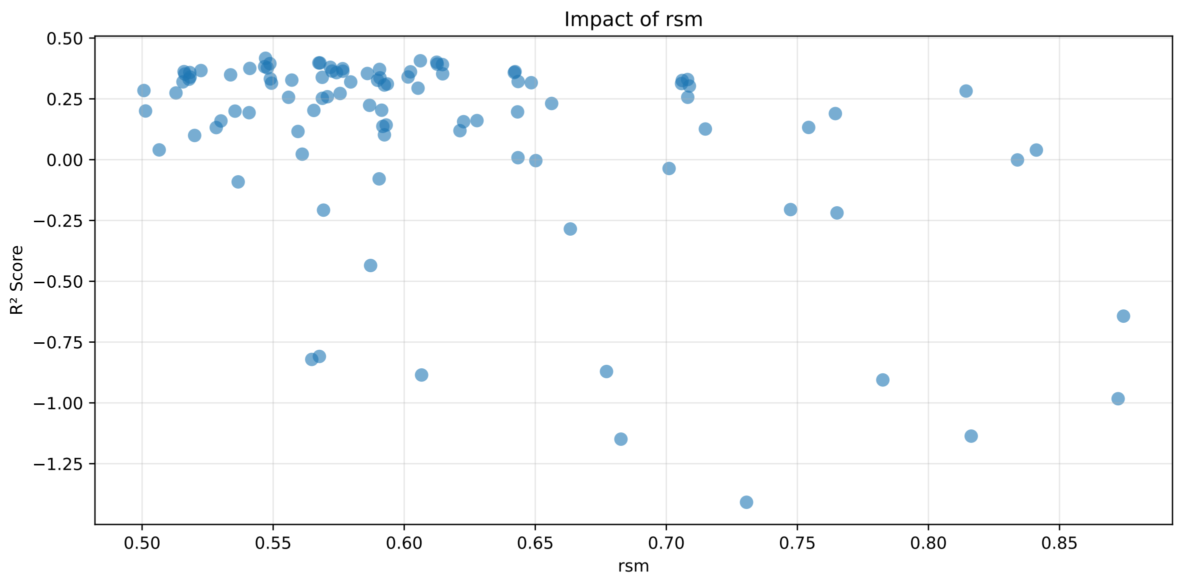


Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'diffusion_temperature', 'rsm']

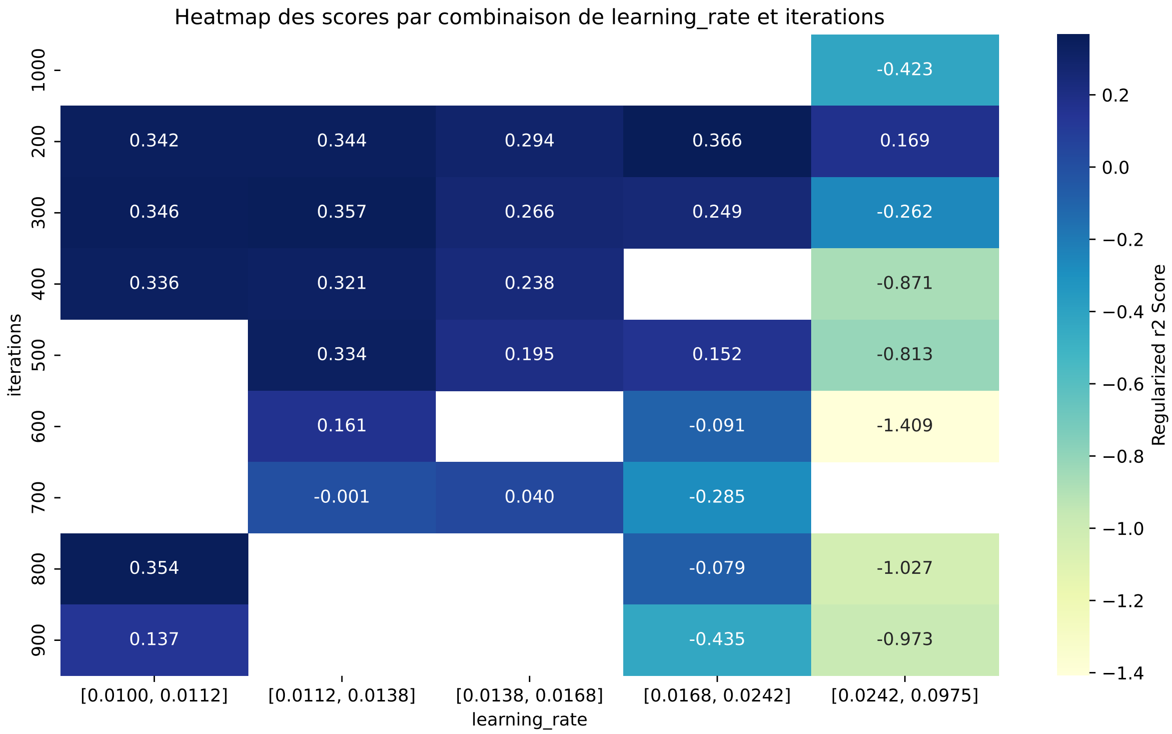

Impact of learning_rate
R² Score
learning_rate
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
0.02
0.04
0.06
0.08
0.10


Impact of iterations
R² Score
iterations
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
200
300
400
500
600
700
800
900
1000






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6362	0.5996	0.5601	0.0431	0.4169	0.5815	-0.0182	0.9697	1.0609
2	0.6397	0.6008	0.5676	0.045	0.4066	0.5834	-0.0174	0.971	1.0647
3	0.6351	0.596	0.5597	0.0427	0.4007	0.5778	-0.0182	0.9694	1.0655
4	0.611	0.5755	0.5363	0.0434	0.3978	0.5537	-0.0218	0.9622	1.0618
5	0.6042	0.5698	0.5231	0.0432	0.3977	0.5435	-0.0263	0.9539	1.0604
6	0.6396	0.5988	0.5616	0.0438	0.3949	0.5874	-0.0114	0.9809	1.0681
7	0.6465	0.6043	0.5682	0.0449	0.3929	0.5897	-0.0146	0.9758	1.07
8	0.6101	0.5736	0.5282	0.0429	0.3909	0.5532	-0.0203	0.9645	1.0637
9	0.653	0.6079	0.5699	0.0445	0.3824	0.5963	-0.0116	0.9808	1.0742
10	0.5422	0.5151	0.4658	0.0376	0.3798	0.4895	-0.0256	0.9503	1.0526
---	---	---	---	---	---	---	---	---	---
100	0.99	0.5902	0.6553	0.0443	-1.4086	0.6613	0.0711	1.1205	1.6773

Hyperparameters:

Rank 1: {'reg___iterations': 400, 'reg___depth': 6, 'reg___learning_rate': 0.01266028480262644, 'reg___l2_leaf_reg': 29.10727941944802, 'reg___bagging_temperature': 3.099755399602925, 'reg___random_strength': 3.1829874003371077, 'reg___subsample': 0.8777588270761061, 'reg___min_data_in_leaf': 33, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.5471015043236325, 'reg___border_count': 48, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 33.586173279262}

Rank 2: {'reg___iterations': 300, 'reg___depth': 5, 'reg___learning_rate': 0.010145677429373563, 'reg___l2_leaf_reg': 31.58167301678971, 'reg___bagging_temperature': 0.6792547463779819, 'reg___random_strength': 1.485791004123043, 'reg___subsample': 0.5955373204716761, 'reg___min_data_in_leaf': 21, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.6061656040293726, 'reg___border_count': 40, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 5083.7197530479925}

Rank 3: {'reg___iterations': 400, 'reg___depth': 6, 'reg___learning_rate': 0.01210959295462592, 'reg___l2_leaf_reg': 32.45021167738264, 'reg___bagging_temperature': 2.872974050300316, 'reg___random_strength': 3.171692289422789, 'reg___subsample': 0.8850945514676314, 'reg___min_data_in_leaf': 35, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.6123997448848119, 'reg___border_count': 128, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 65.06441276557122}

Rank 4: {'reg___iterations': 300, 'reg___depth': 6, 'reg___learning_rate': 0.010487251129626425, 'reg___l2_leaf_reg': 28.352931213174458, 'reg___bagging_temperature': 3.4235625452210883, 'reg___random_strength': 1.8035195675123041, 'reg___subsample': 0.8605610207380321, 'reg___min_data_in_leaf': 25, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.5674396161061595, 'reg___border_count': 127, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 1942.7201561190661}

Rank 5: {'reg___iterations': 300, 'reg___depth': 6, 'reg___learning_rate': 0.010864311297887412, 'reg___l2_leaf_reg': 43.612129560226876, 'reg___bagging_temperature': 0.6238861303147736, 'reg___random_strength': 0.7051610952032537, 'reg___subsample': 0.8645674086029245, 'reg___min_data_in_leaf': 25, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.5678340848880022, 'reg___border_count': 45, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 1917.4456148402385}

Rank 6: {'reg___iterations': 300, 'reg___depth': 6, 'reg___learning_rate': 0.01688673505846059, 'reg___l2_leaf_reg': 25.409784698920028, 'reg___bagging_temperature': 0.6795852440081752, 'reg___random_strength': 3.210125010806593, 'reg___subsample': 0.8360137911789354, 'reg___min_data_in_leaf': 34, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.5487596567693256, 'reg___border_count': 53, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 40.17280983640808}

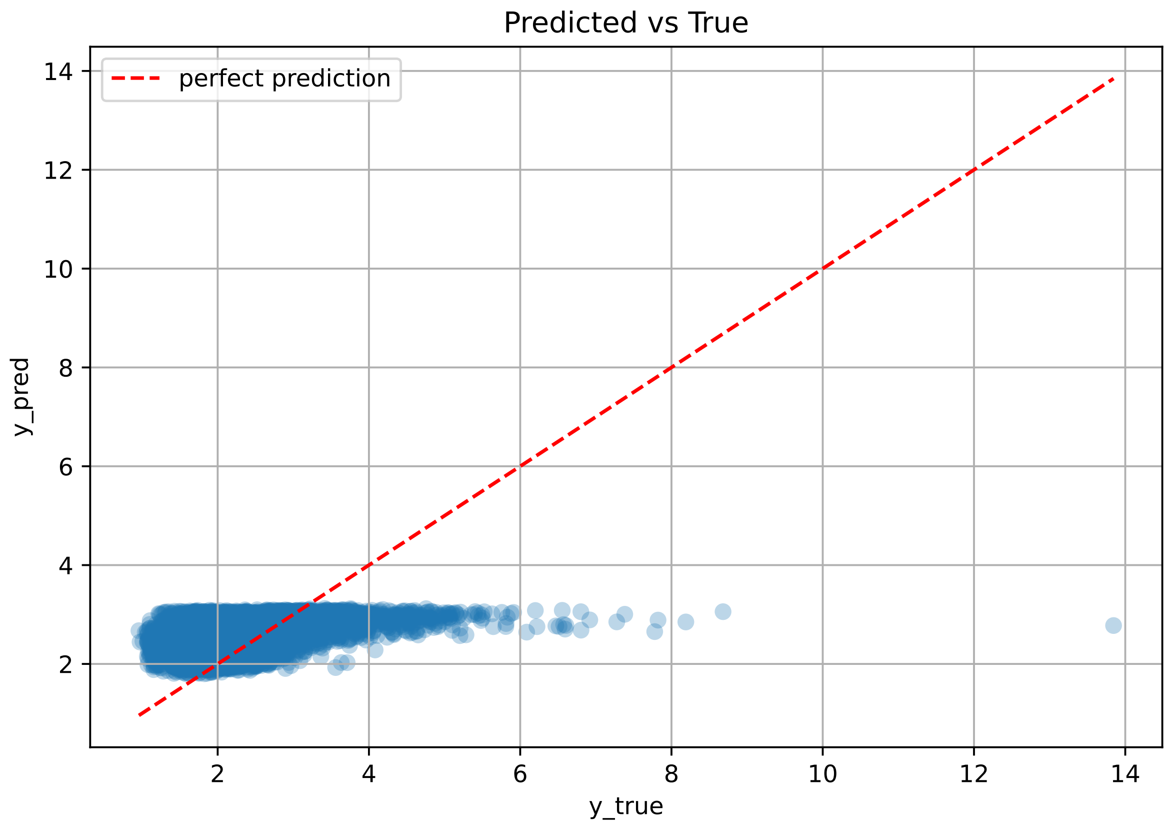
Rank 7: {'reg___iterations': 400, 'reg___depth': 6, 'reg___learning_rate': 0.012216114968743292, 'reg___l2_leaf_reg': 25.99074796449287, 'reg___bagging_temperature': 4.316200459762689, 'reg___random_strength': 2.19498563618052, 'reg___subsample': 0.8158235421582242, 'reg___min_data_in_leaf': 21, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.6125965229325468, 'reg___border_count': 38, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 879.3068973917873}

Rank 8: {'reg___iterations': 200, 'reg___depth': 6, 'reg___learning_rate': 0.01195978701740966, 'reg___l2_leaf_reg': 45.06686914317591, 'reg___bagging_temperature': 0.5932848550956709, 'reg___random_strength': 0.8135192983734509, 'reg___subsample': 0.8009153781337881, 'reg___min_data_in_leaf': 22, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.6145931827830671, 'reg___border_count': 98, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 3371.2317762453126}

Rank 9: {'reg___iterations': 400, 'reg___depth': 6, 'reg___learning_rate': 0.012739504673355237, 'reg___l2_leaf_reg': 22.5545531761347, 'reg___bagging_temperature': 0.8332271207961356, 'reg___random_strength': 1.834800014626587, 'reg___subsample': 0.8760908764887874, 'reg___min_data_in_leaf': 21, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.5467475124392671, 'reg___border_count': 50, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 5866.446260909549}

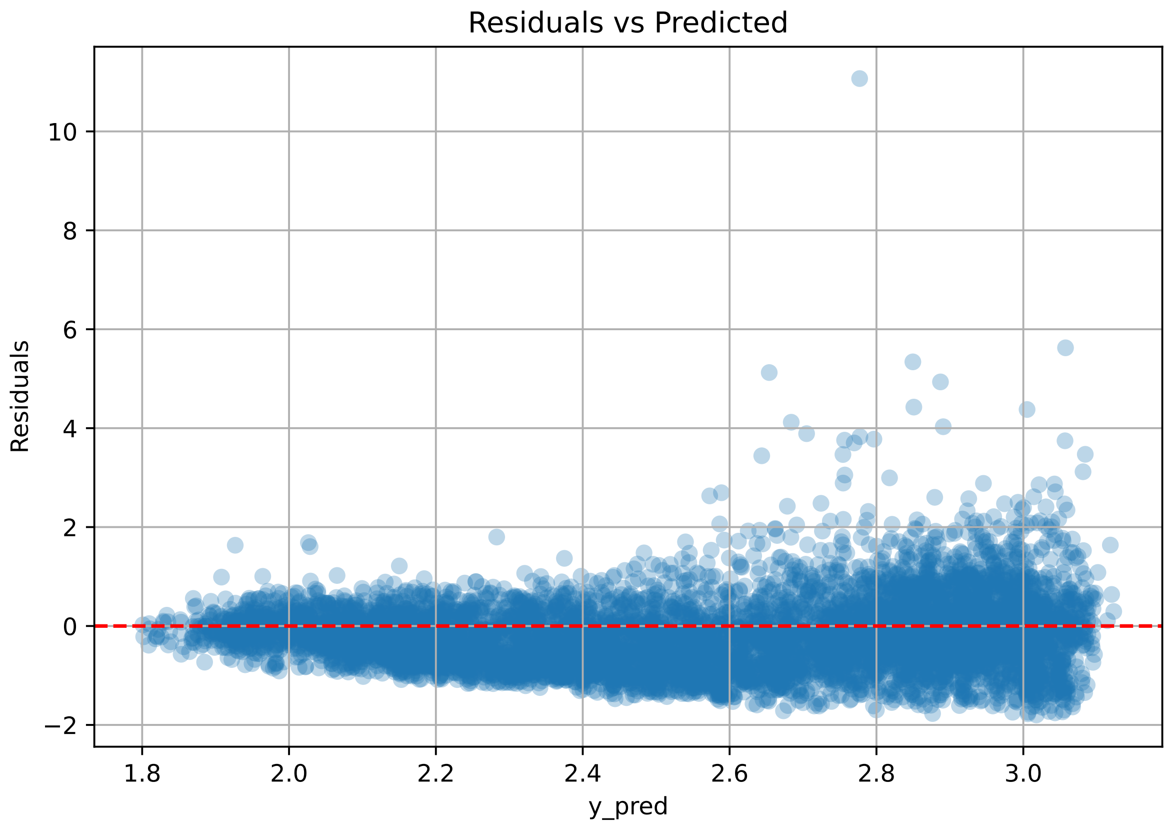
Rank 10: {'reg___iterations': 200, 'reg___depth': 6, 'reg___learning_rate': 0.011112063337307963, 'reg___l2_leaf_reg': 47.86407313736675, 'reg___bagging_temperature': 0.251202188077286, 'reg___random_strength': 1.1276514789054437, 'reg___subsample': 0.8632836391472216, 'reg___min_data_in_leaf': 25, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.5718121161824584, 'reg___border_count': 44, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 1159.9357244334062}

2.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

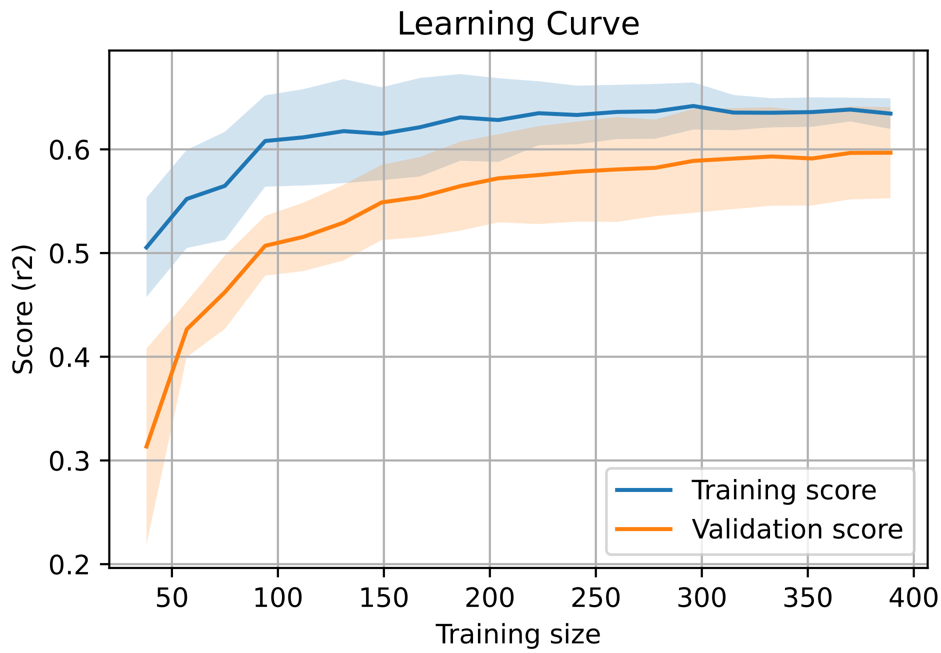
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 1

Len(group) : 795, **Len(training)** : 557,

Len(training_real) : 390, **Len(test_of_training)** : 167, **Len(test)** : 238

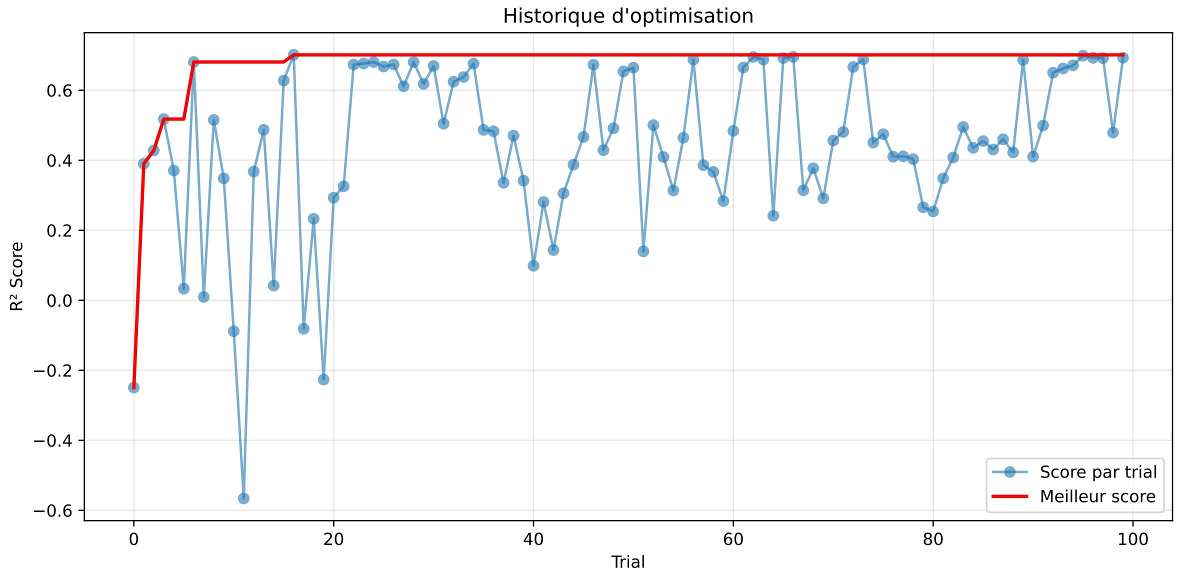


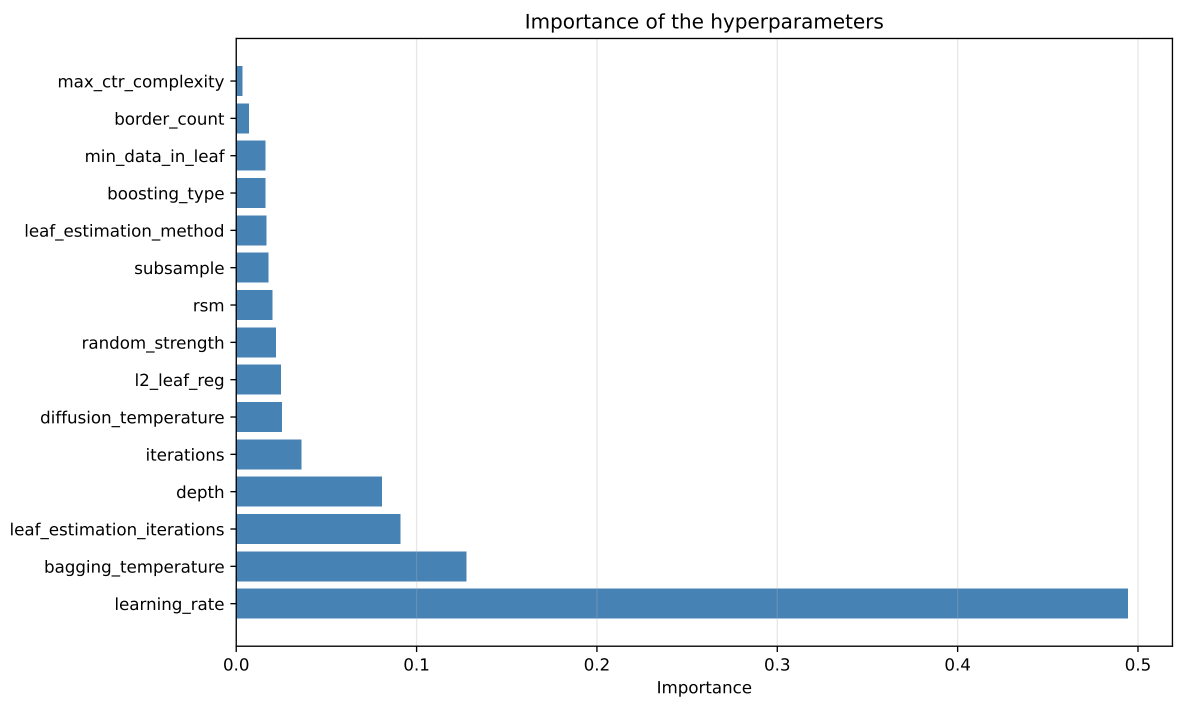
2.4.5 Gridsearch — Group: 2

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

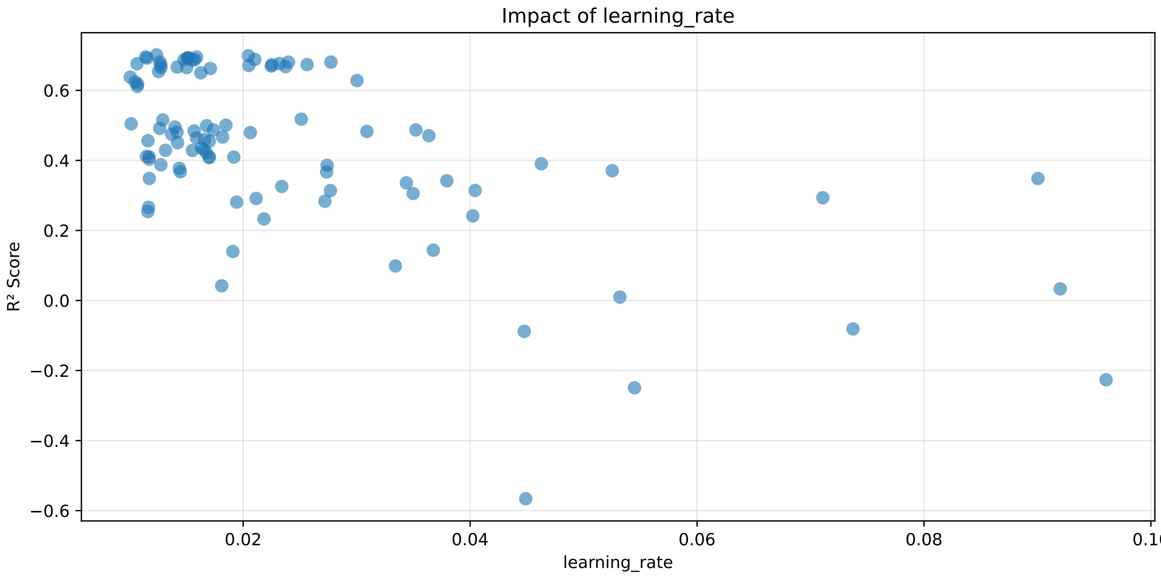
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682 y_test: mean=2.561, std=0.947, min=1.120, max=13.845

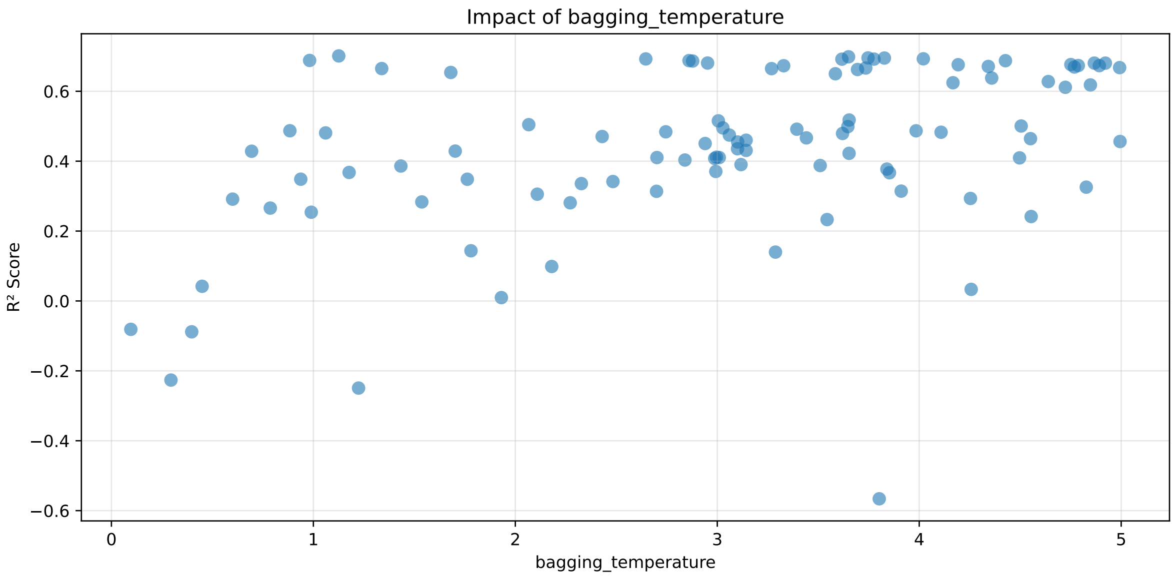
==================== TOP Importance FEATURES ==================== feature importance r_NH3 32.586093 r_umidity 29.570571 r_CO2 15.800026 r_temperature 13.179639 r_THI 8.863671

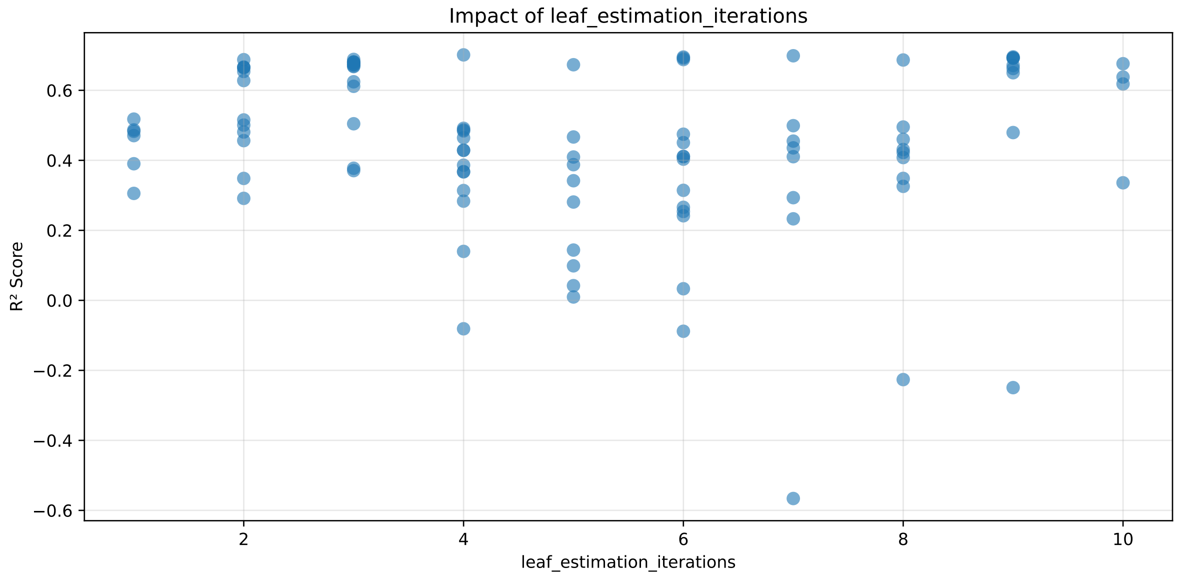


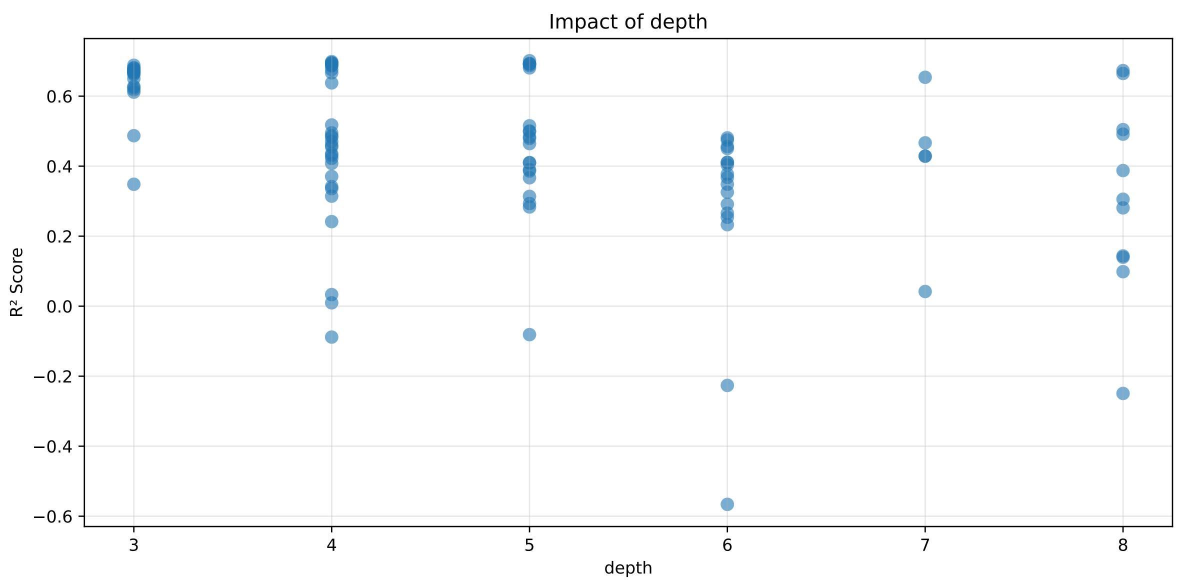


Paramètres avec >5% d'importance : ['learning_rate', 'bagging_temperature', 'leaf_estimation_iterations', 'depth']

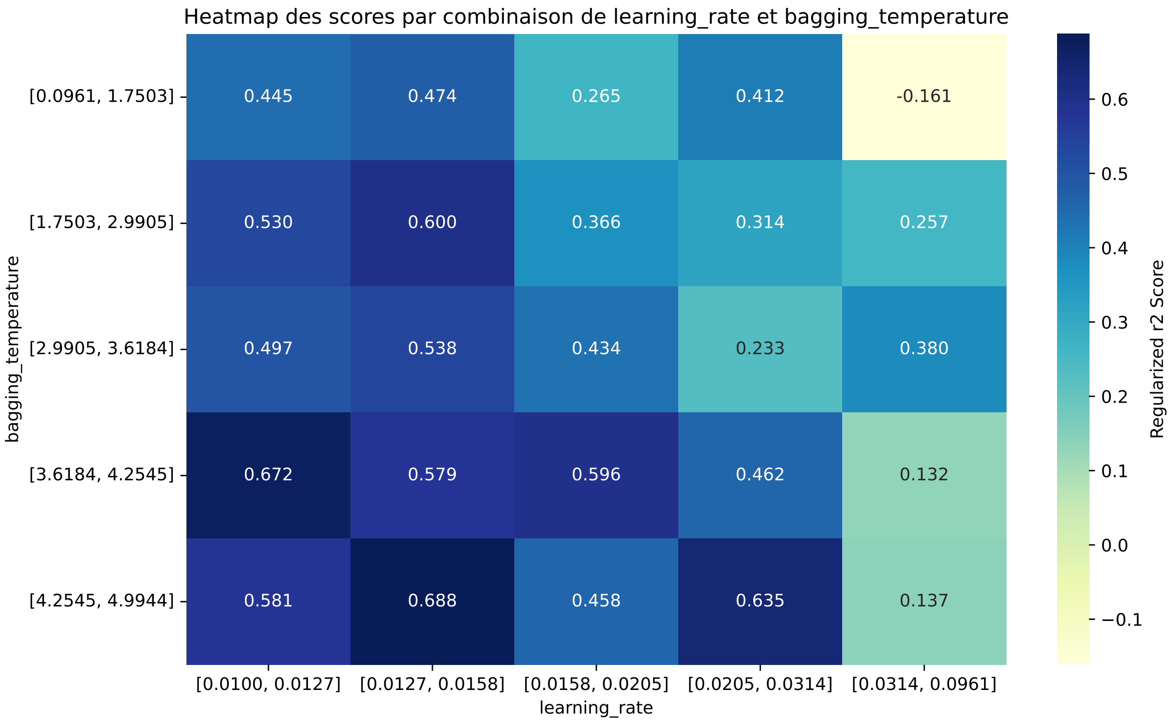

Impact of learning_rate
R² Score
learning_rate
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.02
0.04
0.06
0.08
0.10


Impact of bagging_temperature
R² Score
bagging_temperature
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
1
2
3
4
5






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.735	0.7012	0.6023	0.0168	0.7012	0.6089	-0.0923	0.8684	1.0482
2	0.73	0.6987	0.6008	0.0176	0.6987	0.6035	-0.0951	0.8638	1.0448
3	0.7247	0.6955	0.599	0.0205	0.6955	0.6021	-0.0934	0.8657	1.042
4	0.7261	0.6951	0.597	0.02	0.6951	0.5992	-0.096	0.862	1.0445
5	0.7232	0.6929	0.5961	0.0146	0.6929	0.6027	-0.0902	0.8699	1.0437
6	0.7225	0.6926	0.598	0.015	0.6926	0.6034	-0.0892	0.8712	1.0432
7	0.7195	0.6921	0.5977	0.0174	0.6921	0.6002	-0.0919	0.8672	1.0396
8	0.7225	0.6919	0.5972	0.0153	0.6919	0.6049	-0.087	0.8742	1.0442
9	0.7148	0.6883	0.5923	0.0221	0.6883	0.5876	-0.1006	0.8538	1.0386
10	0.7112	0.6878	0.5946	0.0165	0.6878	0.5968	-0.091	0.8677	1.0339
---	---	---	---	---	---	---	---	---	---
100	0.9574	0.7035	0.5841	0.0232	-0.5661	0.6014	-0.1022	0.8548	1.3609

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.012363552656751986, 'reg__l2_leaf_reg': 12.680825129330584, 'reg__bagging_temperature': 1.1256414858632513, 'reg__random_strength': 4.140105528890884, 'reg__subsample': 0.6489362515032807, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5623479623045151, 'reg__border_count': 105, 'reg__leaf_estimation_method': 'Gradient',

’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 9463.261988939239}

Rank 2: {’reg__iterations’: 600, ’reg__depth’: 4, ’reg__learning_rate’: 0.020455098777038203, ’reg__l2_leaf_reg’: 48.75540809282379, ’reg__bagging_temperature’: 3.6499581002386643, ’reg__random_strength’: 3.722153334051535, ’reg__subsample’: 0.6047304416571557, ’reg__min_data_in_leaf’: 30, ’reg__leaf_estimation_iterations’: 7, ’reg__rsm’: 0.7499000690362392, ’reg__border_count’: 112, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 9783.820078188413}

Rank 3: {’reg__iterations’: 900, ’reg__depth’: 4, ’reg__learning_rate’: 0.01140418139502721, ’reg__l2_leaf_reg’: 16.66050125910854, ’reg__bagging_temperature’: 3.7463555989625545, ’reg__random_strength’: 4.737415168915815, ’reg__subsample’: 0.7633916958197091, ’reg__min_data_in_leaf’: 26, ’reg__leaf_estimation_iterations’: 9, ’reg__rsm’: 0.7443053788496107, ’reg__border_count’: 117, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 7645.203324793038}

Rank 4: {’reg__iterations’: 600, ’reg__depth’: 4, ’reg__learning_rate’: 0.015890931407645543, ’reg__l2_leaf_reg’: 5.412421443200735, ’reg__bagging_temperature’: 3.82753022349993, ’reg__random_strength’: 3.1851427362853912, ’reg__subsample’: 0.7051160490057721, ’reg__min_data_in_leaf’: 27, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.6579991372450006, ’reg__border_count’: 116, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 3524.1932889566356}

Rank 5: {’reg__iterations’: 600, ’reg__depth’: 5, ’reg__learning_rate’: 0.015093693817011993, ’reg__l2_leaf_reg’: 43.38789303348035, ’reg__bagging_temperature’: 4.02038569540861, ’reg__random_strength’: 3.740895605683446, ’reg__subsample’: 0.5819703455262596, ’reg__min_data_in_leaf’: 30, ’reg__leaf_estimation_iterations’: 9, ’reg__rsm’: 0.7587276493740841, ’reg__border_count’: 125, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 8801.854075322008}

Rank 6: {’reg__iterations’: 600, ’reg__depth’: 5, ’reg__learning_rate’: 0.015186869482373766, ’reg__l2_leaf_reg’: 48.28230345287401, ’reg__bagging_temperature’: 2.6460151174403603, ’reg__random_strength’: 3.7967141154328097, ’reg__subsample’: 0.7127936300973757, ’reg__min_data_in_leaf’: 30, ’reg__leaf_estimation_iterations’: 9, ’reg__rsm’: 0.7547542392356247, ’reg__border_count’: 125, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 9686.79474482312}

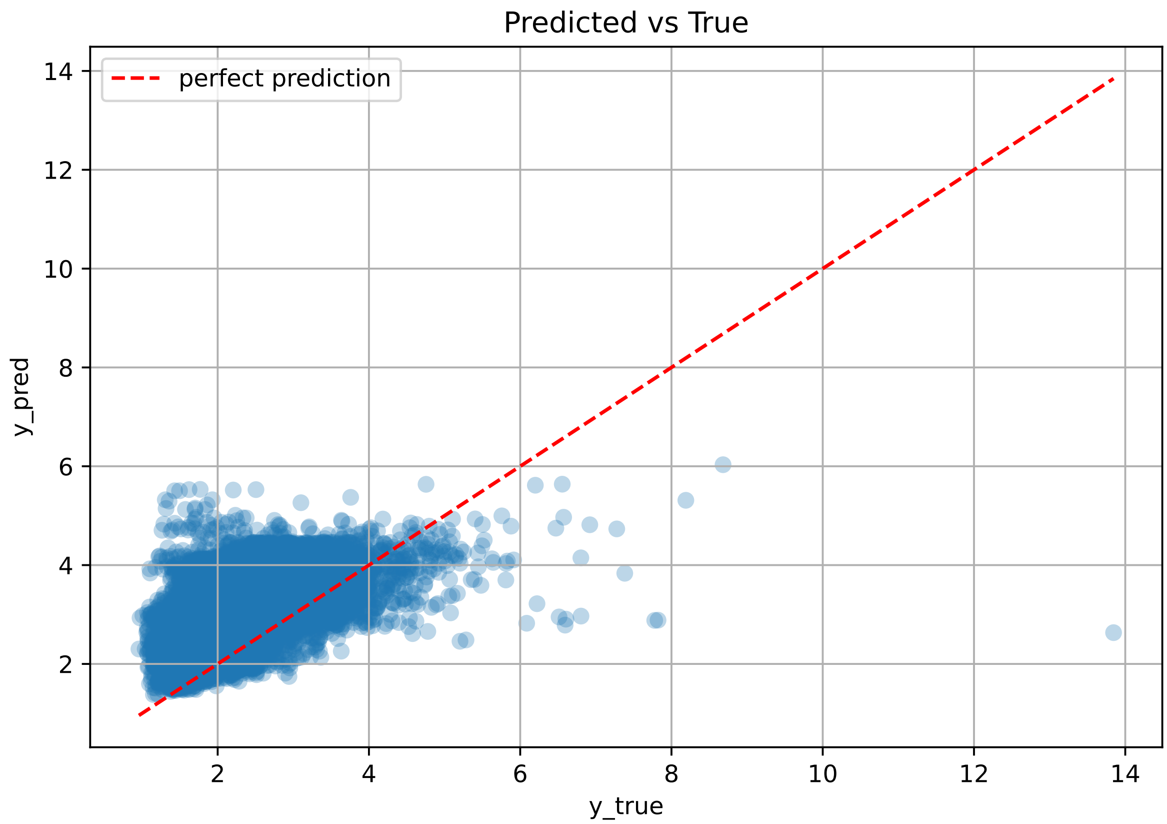
Rank 7: {’reg__iterations’: 900, ’reg__depth’: 4, ’reg__learning_rate’: 0.01153567414623452, ’reg__l2_leaf_reg’: 17.690195160511948, ’reg__bagging_temperature’: 3.776020918737306, ’reg__random_strength’: 4.69020642117927, ’reg__subsample’: 0.6562766674421058, 'reg___min_data_in_leaf': 27, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.6246349409089897, 'reg___border_count': 121, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 9921.135341244171}

Rank 8: {'reg___iterations': 600, 'reg___depth': 5, 'reg___learning_rate': 0.015117851409682425, 'reg___l2_leaf_reg': 42.30794588899658, 'reg___bagging_temperature': 3.61706375004374, 'reg___random_strength': 3.680842111692538, 'reg___subsample': 0.7165235412176563, 'reg___min_data_in_leaf': 23, 'reg___leaf_estimation_iterations': 9, 'reg___rsm': 0.7673776327882939, 'reg___border_count': 125, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 8778.049848278035}

Rank 9: {'reg___iterations': 600, 'reg___depth': 3, 'reg___learning_rate': 0.021023991135192287, 'reg___l2_leaf_reg': 5.966620773010796, 'reg___bagging_temperature': 0.9813658587633769, 'reg___random_strength': 3.1172397807389833, 'reg___subsample': 0.6593059118340096, 'reg___min_data_in_leaf': 28, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.5127832194703396, 'reg___border_count': 116, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 9992.94568838636}

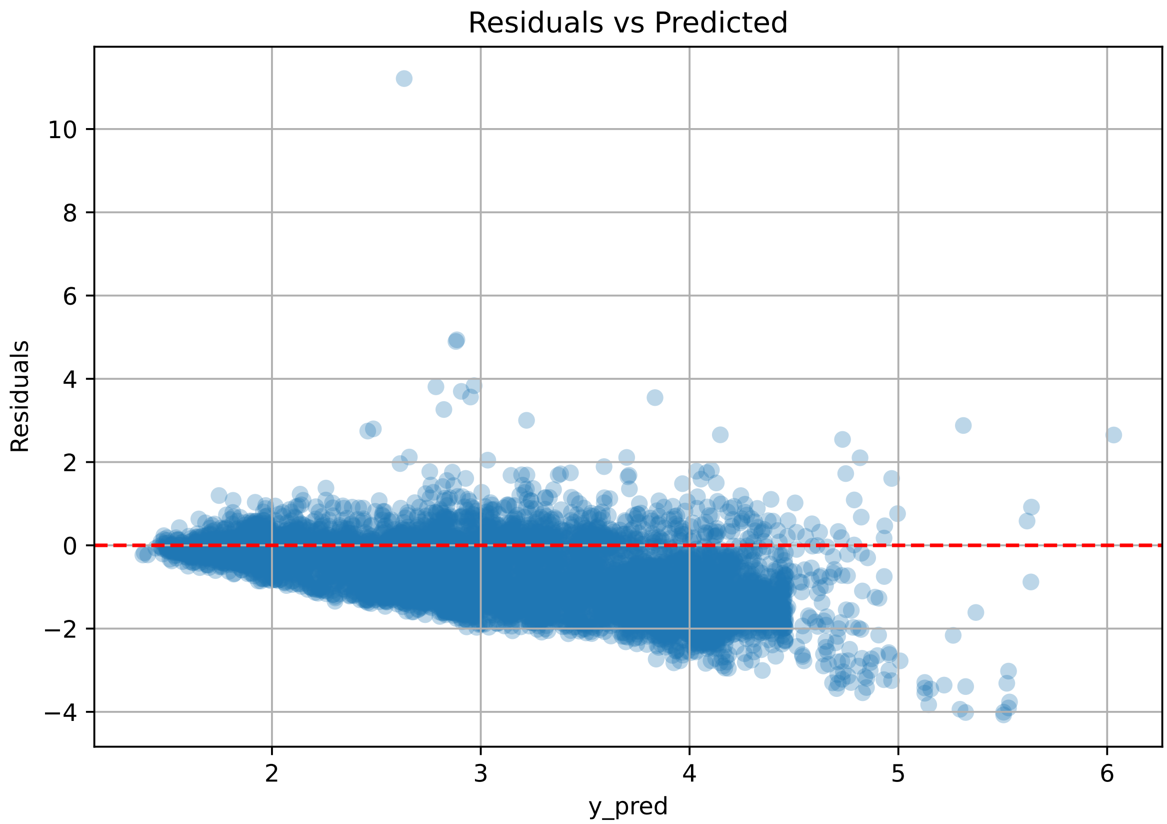
Rank 10: {'reg___iterations': 600, 'reg___depth': 4, 'reg___learning_rate': 0.015737050016449147, 'reg___l2_leaf_reg': 17.48649763739488, 'reg___bagging_temperature': 2.860109380490647, 'reg___random_strength': 4.269666209913589, 'reg___subsample': 0.799634936378419, 'reg___min_data_in_leaf': 27, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.63330980867431, 'reg___border_count': 114, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 3558.04211119278}

2.4.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

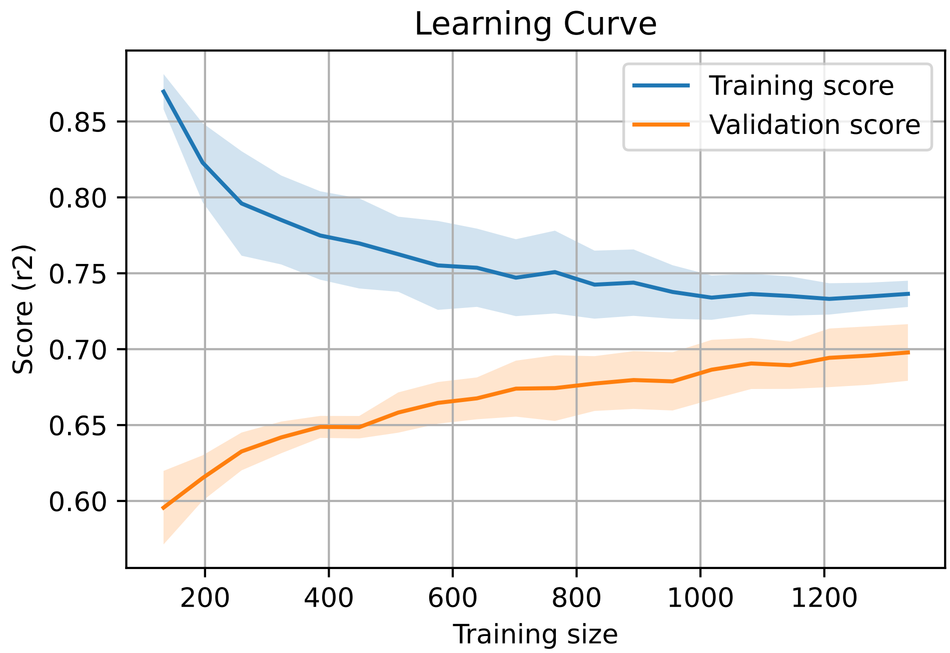
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2

$\textbf{Len(group)} : 2726$, $\textbf{Len(training)} : 1909$,

$\textbf{Len(training\_real)} : 1337$, $\textbf{Len(test\_of\_training)} : 572$, $\textbf{Len(test)} : 817$

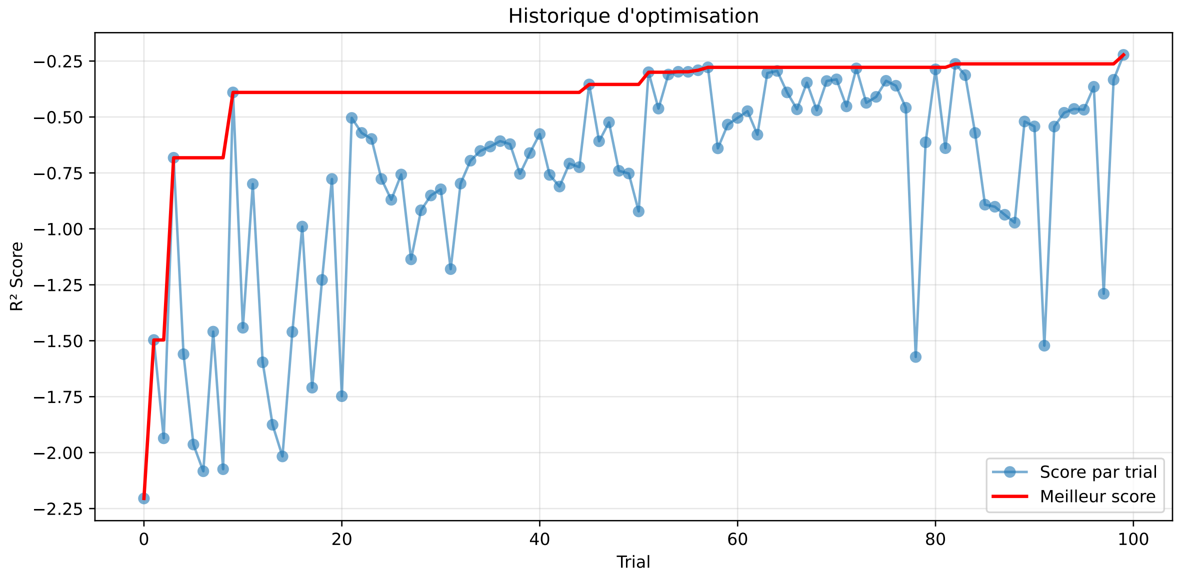


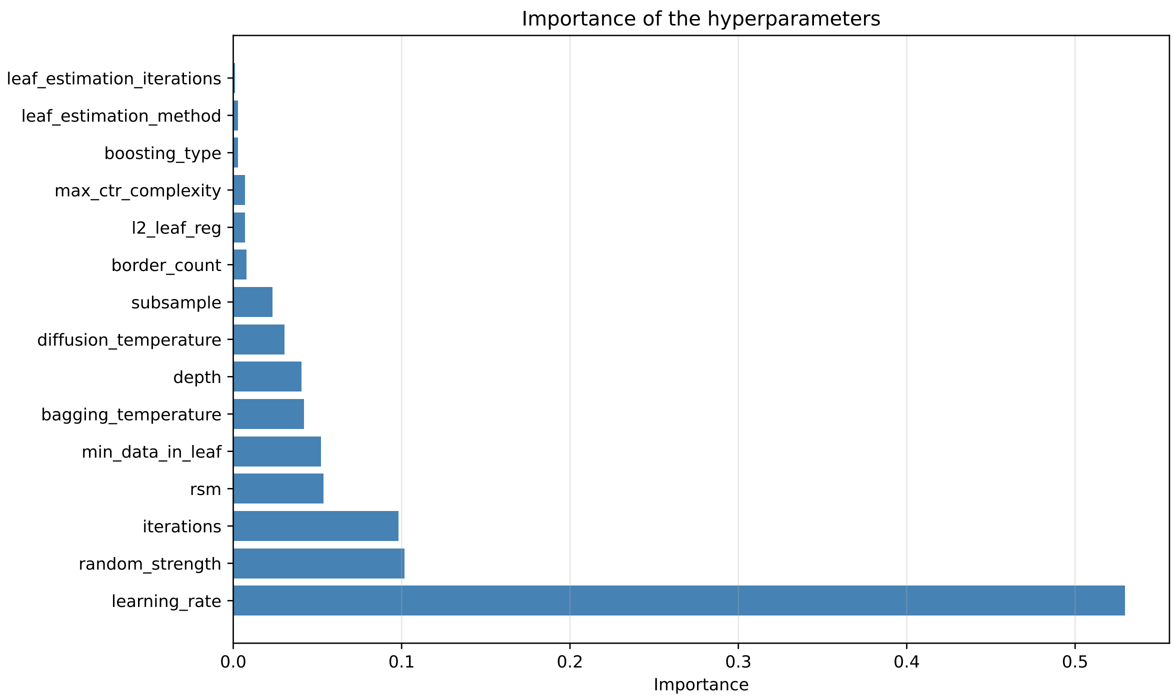
2.4.9 Gridsearch — Group: 3

Len(group) : 568, **Len(training)** : 398, **Len(test)** : 170

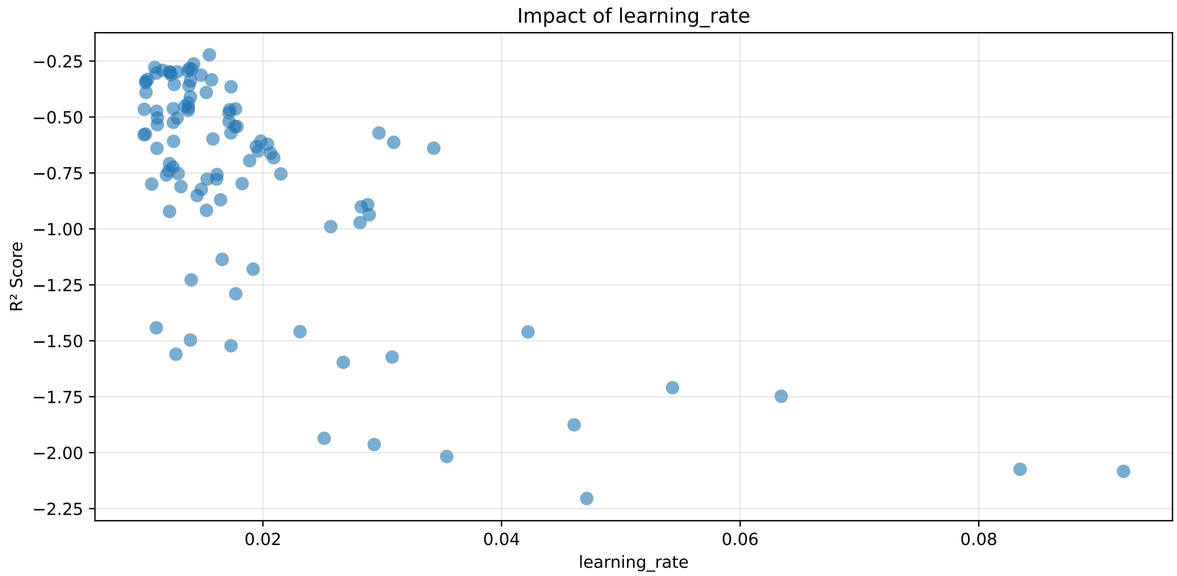
Distribution y_train vs y_test: y_train: mean=1.928, std=0.816, min=0.959, max=6.804 y_test: mean=1.833, std=0.729, min=1.098, max=7.822

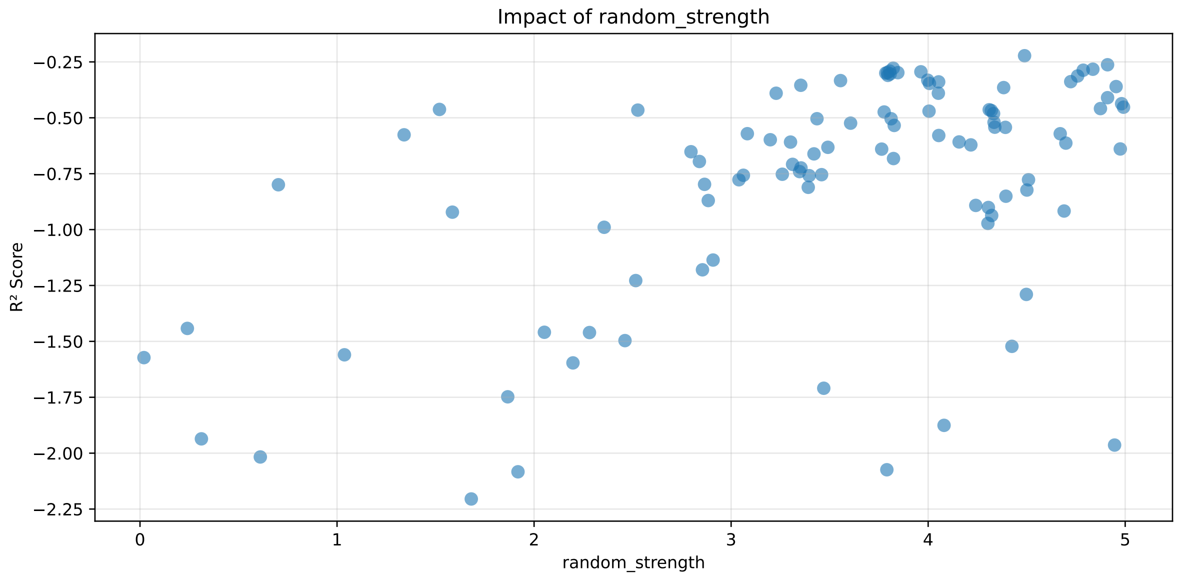
==================== TOP Importance FEATURES ==================== feature importance r_umidity 29.677598 r_NH3 26.781922 r_THI 26.107937 r_temperature 9.880096 r_CO2 7.552447

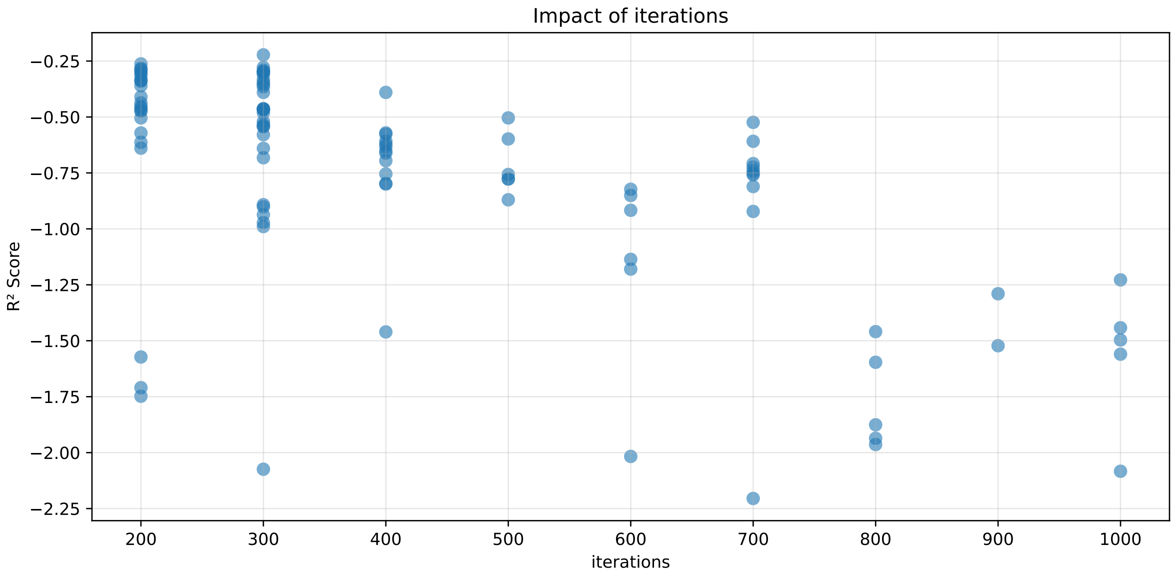


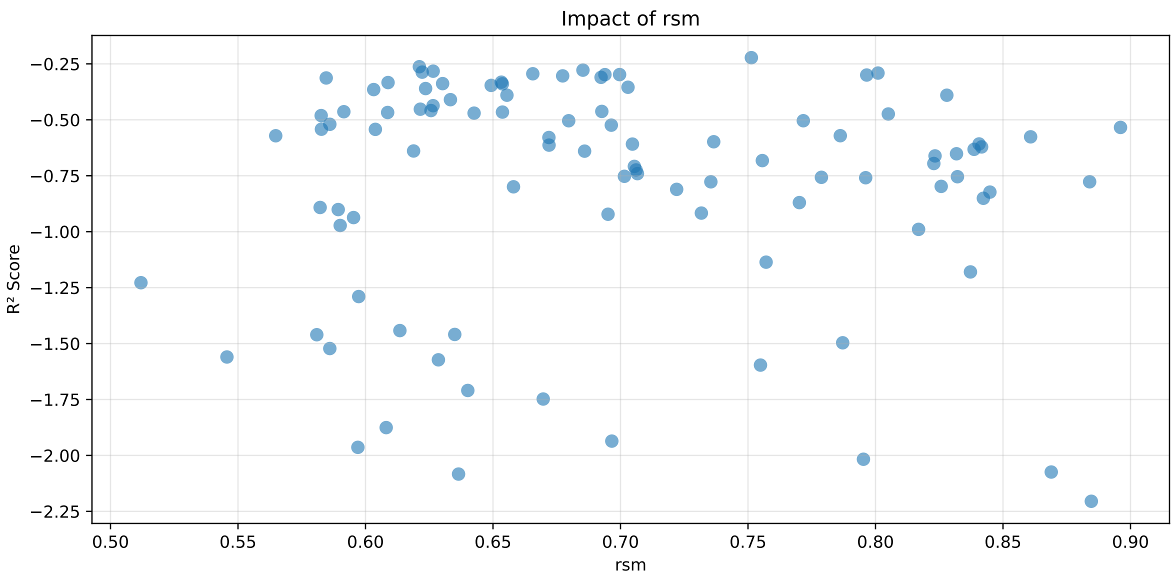


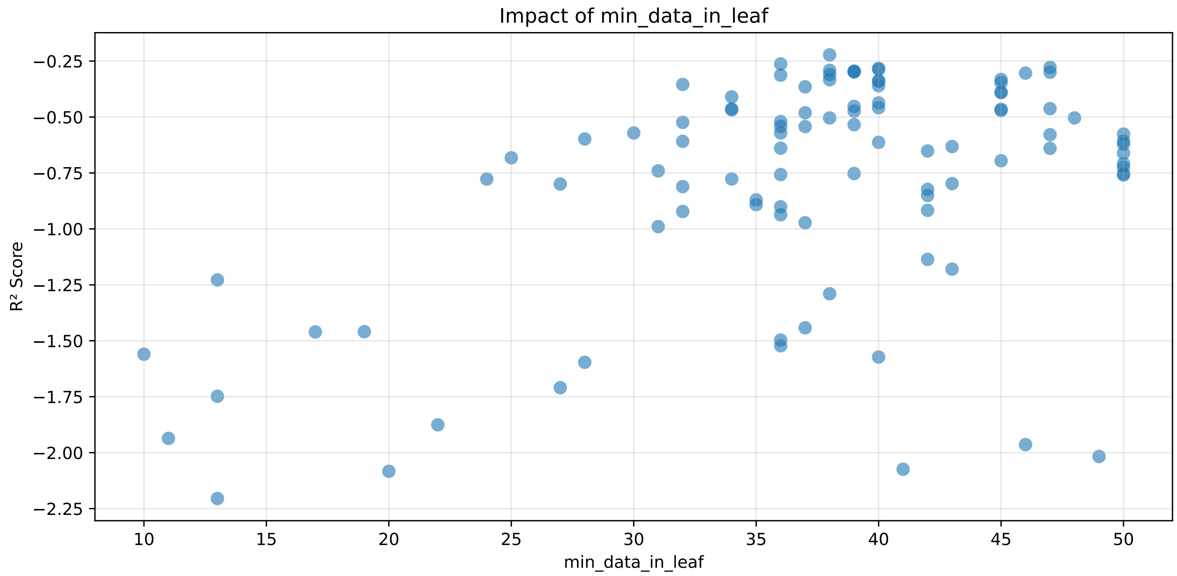
Paramètres avec >5% d'importance : ['learning_rate', 'random_strength', 'iterations', 'rsm', 'min_data_in_leaf']


Impact of learning_rate
R² Score
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
−2.25
0.02
0.04
0.06
0.08
learning_rate


Impact of random_strength
R² Score
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
−2.25
0
1
2
3
4
5
random_strength


Impact of iterations
R² Score
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
−2.25
200
300
400
500
600
700
800
900
1000
iterations


Impact of rsm
R² Score
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
−2.25
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
rsm



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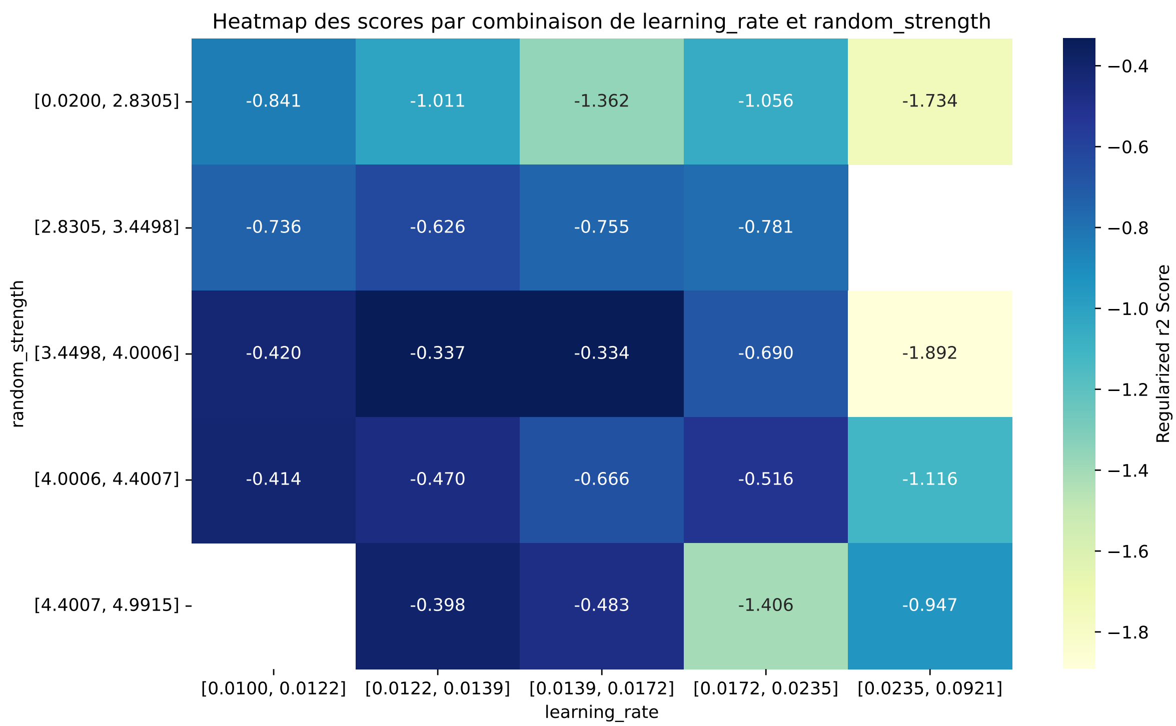
The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.424	0.3163	0.1786	0.0426	-0.2223	0.1869	-0.1294	0.5909	1.3406
2	0.4499	0.3311	0.1888	0.0472	-0.2627	0.1994	-0.1317	0.6023	1.3587
3	0.4384	0.319	0.1811	0.0446	-0.2782	0.1958	-0.1233	0.6136	1.3744
4	0.44	0.3195	0.1869	0.045	-0.2828	0.2004	-0.1191	0.6272	1.377
5	0.4536	0.3301	0.1862	0.0492	-0.2871	0.2052	-0.1249	0.6216	1.374
6	0.4504	0.3268	0.1878	0.0419	-0.2911	0.226	-0.1008	0.6916	1.3782
7	0.4471	0.3235	0.1772	0.0449	-0.2943	0.196	-0.1275	0.6059	1.3819
8	0.4495	0.3249	0.1846	0.0392	-0.2979	0.2134	-0.1115	0.6568	1.3834
9	0.4515	0.3266	0.1825	0.0413	-0.2981	0.2025	-0.1241	0.6201	1.3826
10	0.4651	0.3376	0.1808	0.0447	-0.3	0.1997	-0.1379	0.5915	1.3777
---	---	---	---	---	---	---	---	---	---
100	0.9886	0.4564	0.2421	0.0956	-2.2049	0.331	-0.1253	0.7254	2.1663

Hyperparameters:

Rank 1: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.01550838527646837, 'reg___l2_leaf_reg': 30.65895413620455, 'reg___bagging_temperature': 3.2251932952815774, 'reg___random_strength': 4.489356865351843, 'reg___subsample': 0.5304233071196921, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.7513200695828289, 'reg___border_count': 90, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 108.11943035600143}

Rank 2: {'reg___iterations': 200, 'reg___depth': 4, 'reg___learning_rate': 0.01418781909748735, 'reg___l2_leaf_reg': 25.881144012418325, 'reg___bagging_temperature': 2.4281927845628126, 'reg___random_strength': 4.91056013073154, 'reg___subsample': 0.5835083990816157, 'reg___min_data_in_leaf': 36, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.6211187691171614, 'reg___border_count': 67, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 3399.575774957205}

Rank 3: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.0109291443628931, 'reg___l2_leaf_reg': 18.995115567872922, 'reg___bagging_temperature': 1.962021218587222, 'reg___random_strength': 3.822649329736768, 'reg___subsample': 0.6268443095819676, 'reg___min_data_in_leaf': 47, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.6852724957420983, 'reg___border_count': 72, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 3770.663824040415}

Rank 4: {'reg___iterations': 200, 'reg___depth': 4, 'reg___learning_rate': 0.013832001187261807, 'reg___l2_leaf_reg': 21.345804458142698, 'reg___bagging_temperature': 2.4201493622720522, 'reg___random_strength': 4.836000168540766, 'reg___subsample': 0.6477298499868201, 'reg___min_data_in_leaf': 40, 'reg___leaf_estimation_iterations': 5, 'reg___rsm':

0.6265366956052286, 'reg___border_count': 72, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 2750.302949938545}

Rank 5: {'reg___iterations': 200, 'reg___depth': 4, 'reg___learning_rate': 0.014030344914246231, 'reg___l2_leaf_reg': 26.41257778428224, 'reg___bagging_temperature': 2.7212836774828597, 'reg___random_strength': 4.787037055324994, 'reg___subsample': 0.5454326999858536, 'reg___min_data_in_leaf': 40, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.6222380329246764, 'reg___border_count': 69, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 3433.106809968849}

Rank 6: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.011587311084556521, 'reg___l2_leaf_reg': 18.201895037841293, 'reg___bagging_temperature': 2.0149847937262146, 'reg___random_strength': 3.8049020734783783, 'reg___subsample': 0.6212971522925266, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.8009803270162842, 'reg___border_count': 48, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 3777.8792453023857}

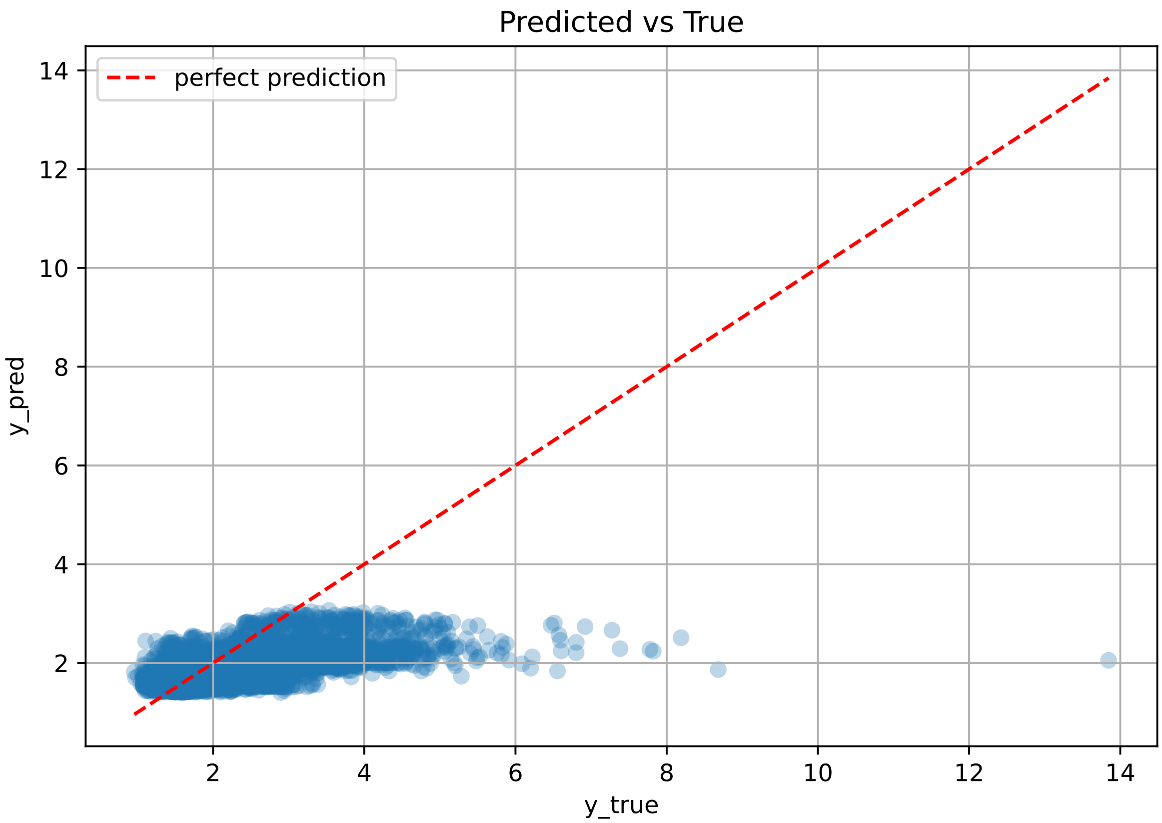
Rank 7: {'reg___iterations': 200, 'reg___depth': 4, 'reg___learning_rate': 0.013702709799100968, 'reg___l2_leaf_reg': 22.630080714549788, 'reg___bagging_temperature': 1.7331785091462986, 'reg___random_strength': 3.9631369957473352, 'reg___subsample': 0.6469771925408075, 'reg___min_data_in_leaf': 39, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.6656070401652548, 'reg___border_count': 75, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 3433.9344166721494}

Rank 8: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.012177573861114756, 'reg___l2_leaf_reg': 17.842034773252816, 'reg___bagging_temperature': 1.9554618004621618, 'reg___random_strength': 3.79346312477882, 'reg___subsample': 0.6002682354058975, 'reg___min_data_in_leaf': 39, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.6996438799833471, 'reg___border_count': 126, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 4065.7310219678934}

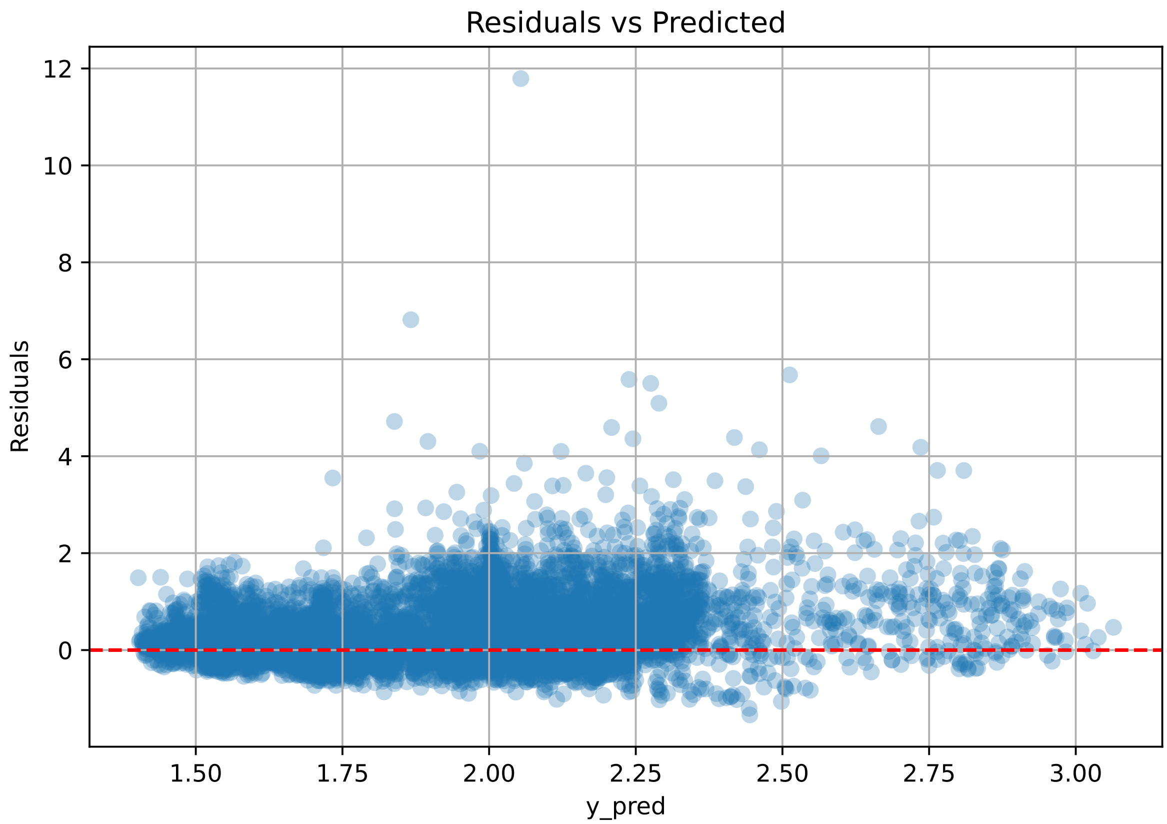
Rank 9: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.012808051542976493, 'reg___l2_leaf_reg': 19.6070472578647, 'reg___bagging_temperature': 2.095097495150715, 'reg___random_strength': 3.8462001927223444, 'reg___subsample': 0.6073893070845918, 'reg___min_data_in_leaf': 39, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.6938924032213257, 'reg___border_count': 128, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 4017.514899630612}

Rank 10: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.012201615445797998, 'reg___l2_leaf_reg': 12.424323459083723, 'reg___bagging_temperature': 1.8323296868952026, 'reg___random_strength': 3.784797506389579, 'reg___subsample': 0.6024284414597982, 'reg___min_data_in_leaf': 47, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.7965530405131153, 'reg___border_count': 128, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 4015.110678145863}

2.4.10 Scatter



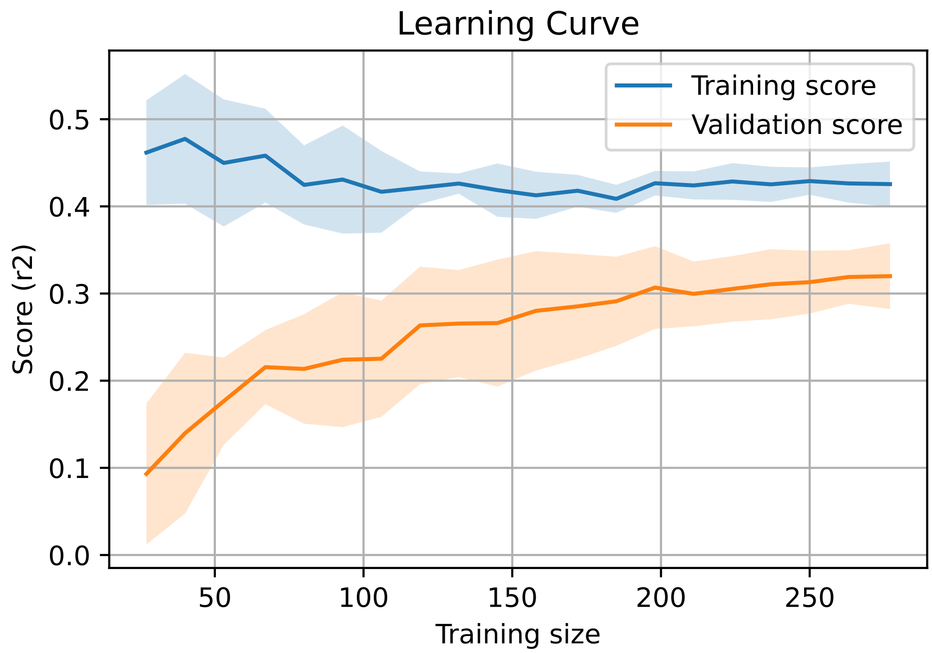
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 3

Len(group) : 568, **Len(training)** : 398,

Len(training_real) : 279, **Len(test_of_training)** : 119, **Len(test)** : 170

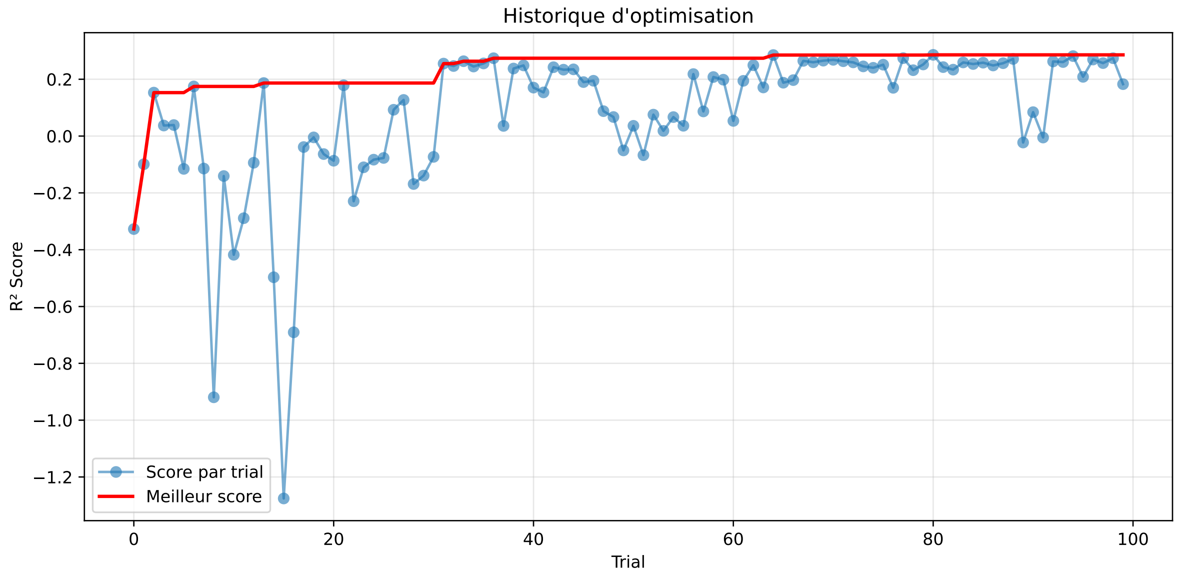


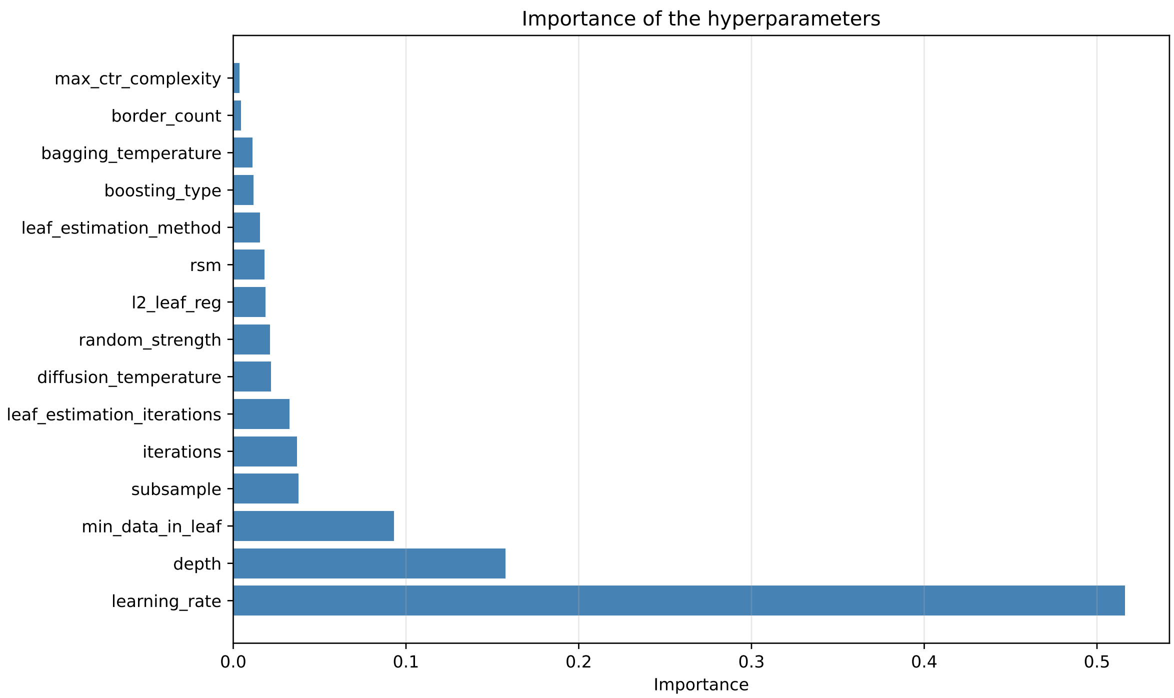
2.4.13 Gridsearch — Group: 4

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

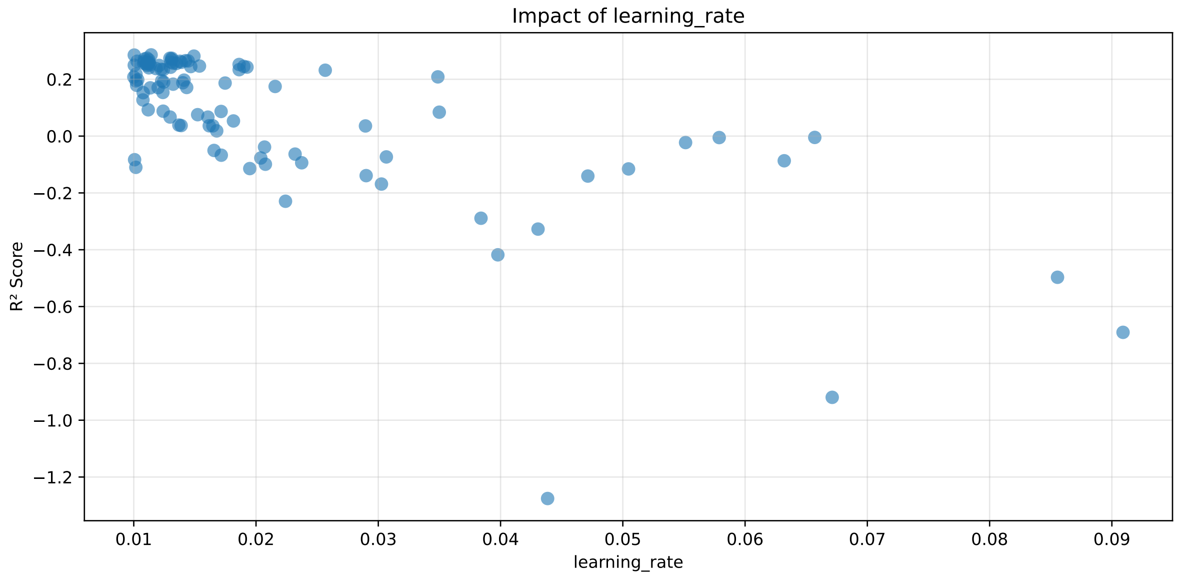
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813 y_test: mean=2.212, std=0.659, min=1.071, max=4.858

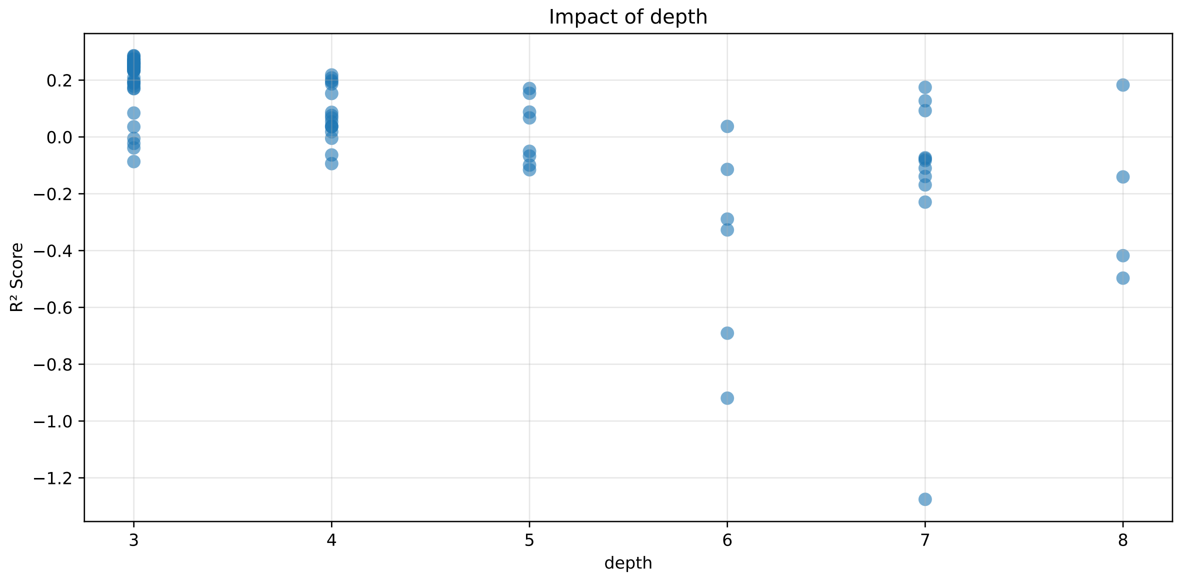
==================== TOP Importance FEATURES ==================== feature importance r_umidity 67.950083 r_temperature 15.238510 r_CO2 6.613158 r_NH3 5.645806 r_THI 4.552443

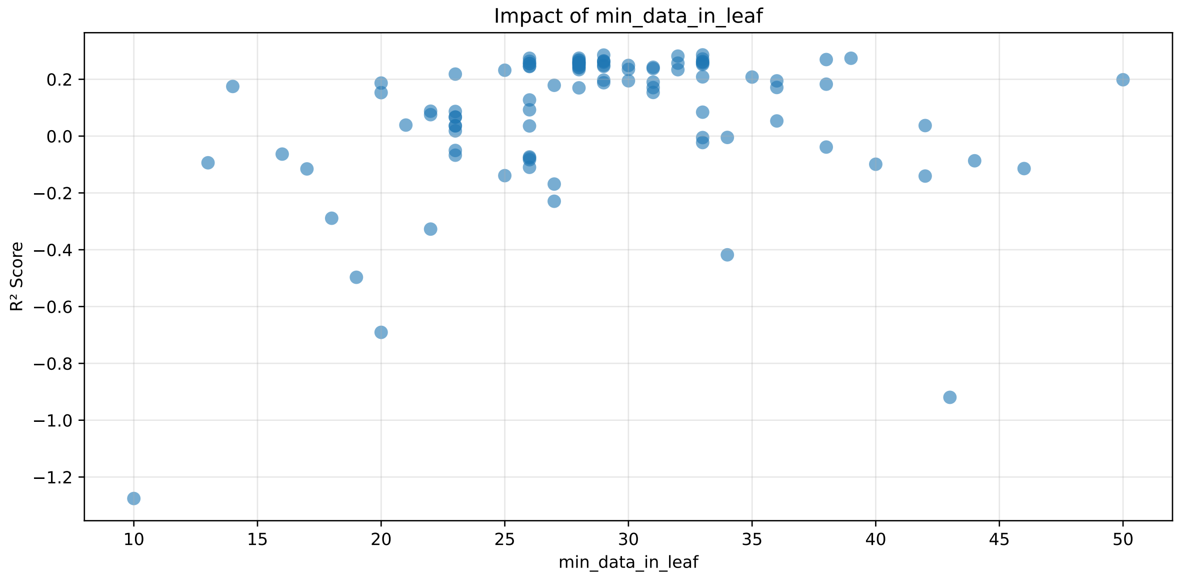




Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'min_data_in_leaf']


Impact of learning_rate
R² Score
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
learning_rate


Impact of depth
R² Score
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
3
4
5
6
7
8
depth

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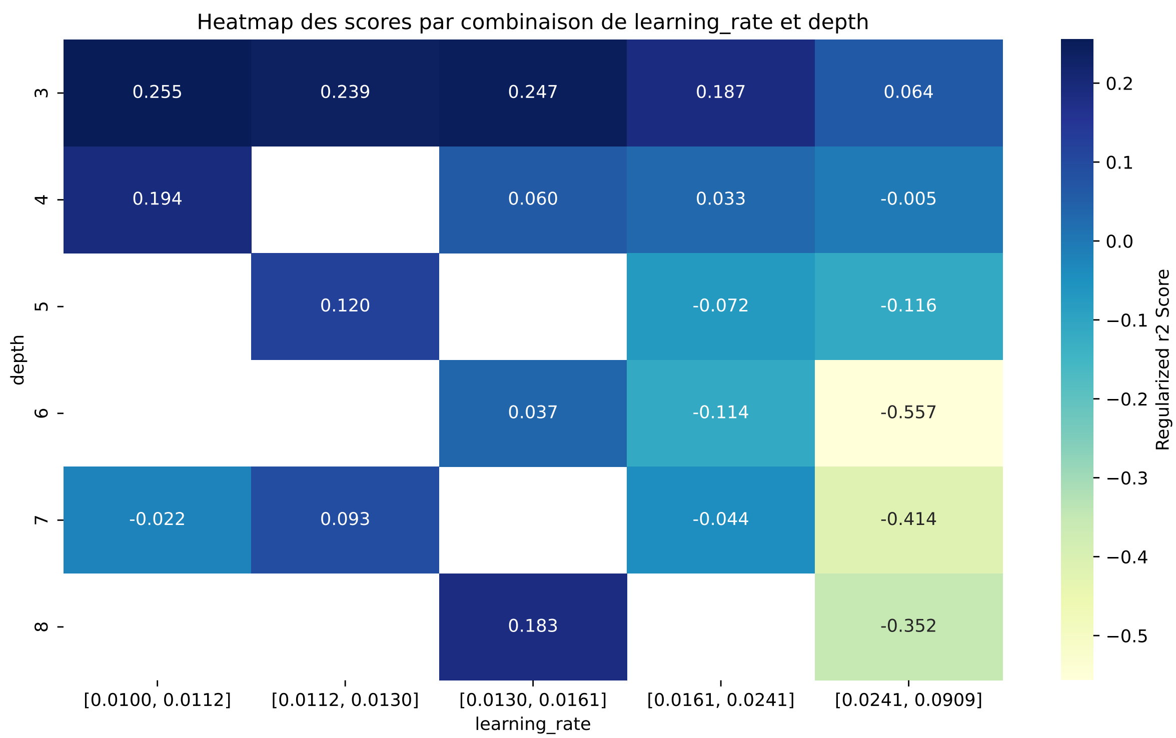
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4319	0.4076	0.4672	0.0202	0.2859	0.4684	0.0608	1.1492	1.0597
2	0.4725	0.4413	0.5027	0.0203	0.2852	0.5064	0.0651	1.1474	1.0707
3	0.4574	0.4281	0.4921	0.0225	0.2815	0.4922	0.0641	1.1496	1.0685
4	0.4355	0.4086	0.469	0.0198	0.2743	0.4708	0.0622	1.1521	1.0657
5	0.4709	0.4381	0.5025	0.0227	0.274	0.5025	0.0644	1.147	1.0749
6	0.4395	0.4119	0.4745	0.02	0.274	0.4748	0.0629	1.1527	1.067
7	0.4316	0.4049	0.4649	0.0198	0.2714	0.4621	0.0572	1.1412	1.0659
8	0.4368	0.4089	0.4707	0.021	0.2694	0.4699	0.0611	1.1494	1.0682
9	0.4513	0.4208	0.4854	0.0214	0.2683	0.4865	0.0657	1.1561	1.0725
10	0.4567	0.4248	0.4896	0.0225	0.2655	0.4947	0.0699	1.1646	1.075
---	---	---	---	---	---	---	---	---	---
100	0.9277	0.5605	0.6017	0.0176	-1.2757	0.6146	0.0542	1.0966	1.6552

Hyperparameters:

Rank 1: {’reg___iterations’: 200, ’reg___depth’: 3, ’reg___learning_rate’: 0.01141962096251727, ’reg___l2_leaf_reg’: 20.276390016920367, ’reg___bagging_temperature’: 4.431352183989506, ’reg___random_strength’: 2.859629375369349, ’reg___subsample’: 0.5217134405064868, ’reg___min_data_in_leaf’: 33, ’reg___leaf_estimation_iterations’: 10, ’reg___rsm’: 0.742360158878449, ’reg___border_count’: 51, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Plain’, ’reg___max_ctr_complexity’: 4, ’reg___diffusion_temperature’: 503.13287510905235}

Rank 2: {’reg___iterations’: 600, ’reg___depth’: 3, ’reg___learning_rate’: 0.010040878294186786, ’reg___l2_leaf_reg’: 2.599715352463864, ’reg___bagging_temperature’: 4.030030807987661, ’reg___random_strength’: 2.523347380765334, ’reg___subsample’: 0.6898082005078888, ’reg___min_data_in_leaf’: 29, ’reg___leaf_estimation_iterations’: 9, ’reg___rsm’: 0.5556840385307261, ’reg___border_count’: 61, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Plain’, ’reg___max_ctr_complexity’: 4, ’reg___diffusion_temperature’: 9.659671328620874}

Rank 3: {’reg___iterations’: 200, ’reg___depth’: 3, ’reg___learning_rate’: 0.014921766887963393, ’reg___l2_leaf_reg’: 5.919223699714605, ’reg___bagging_temperature’: 4.2569148504599585, ’reg___random_strength’: 3.09879323877042, ’reg___subsample’: 0.8950664587745116, ’reg___min_data_in_leaf’: 32, ’reg___leaf_estimation_iterations’: 10, ’reg___rsm’: 0.750804680954421, ’reg___border_count’: 58, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Plain’, ’reg___max_ctr_complexity’: 4, ’reg___diffusion_temperature’: 274.75811294316776}

Rank 4: {’reg___iterations’: 200, ’reg___depth’: 3, ’reg___learning_rate’: 0.011077384632443286, ’reg___l2_leaf_reg’: 19.87489580756254, ’reg___bagging_temperature’: 4.700879590345391, ’reg___random_strength’: 2.8448712696696004, ’reg___subsample’: 0.6055578343467263, ’reg___min_data_in_leaf’: 28, ’reg___leaf_estimation_iterations’: 10, ’reg___rsm’:

0.6518164944140783, 'reg___border_count': 54, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 527.731070184863}

Rank 5: {'reg___iterations': 500, 'reg___depth': 3, 'reg___learning_rate': 0.012955126169200004, 'reg___l2_leaf_reg': 14.746212339220891, 'reg___bagging_temperature': 3.2889650091733182, 'reg___random_strength': 3.550864065425267, 'reg___subsample': 0.7492019985283952, 'reg___min_data_in_leaf': 26, 'reg___leaf_estimation_iterations': 8, 'reg___rsm': 0.593190774577761, 'reg___border_count': 72, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 14.204951029902077}

Rank 6: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.013108197948987335, 'reg___l2_leaf_reg': 30.585831347044437, 'reg___bagging_temperature': 4.990195806852486, 'reg___random_strength': 3.026093636184184, 'reg___subsample': 0.8482345981339361, 'reg___min_data_in_leaf': 39, 'reg___leaf_estimation_iterations': 9, 'reg___rsm': 0.760099706080316, 'reg___border_count': 37, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 259.4824642175213}

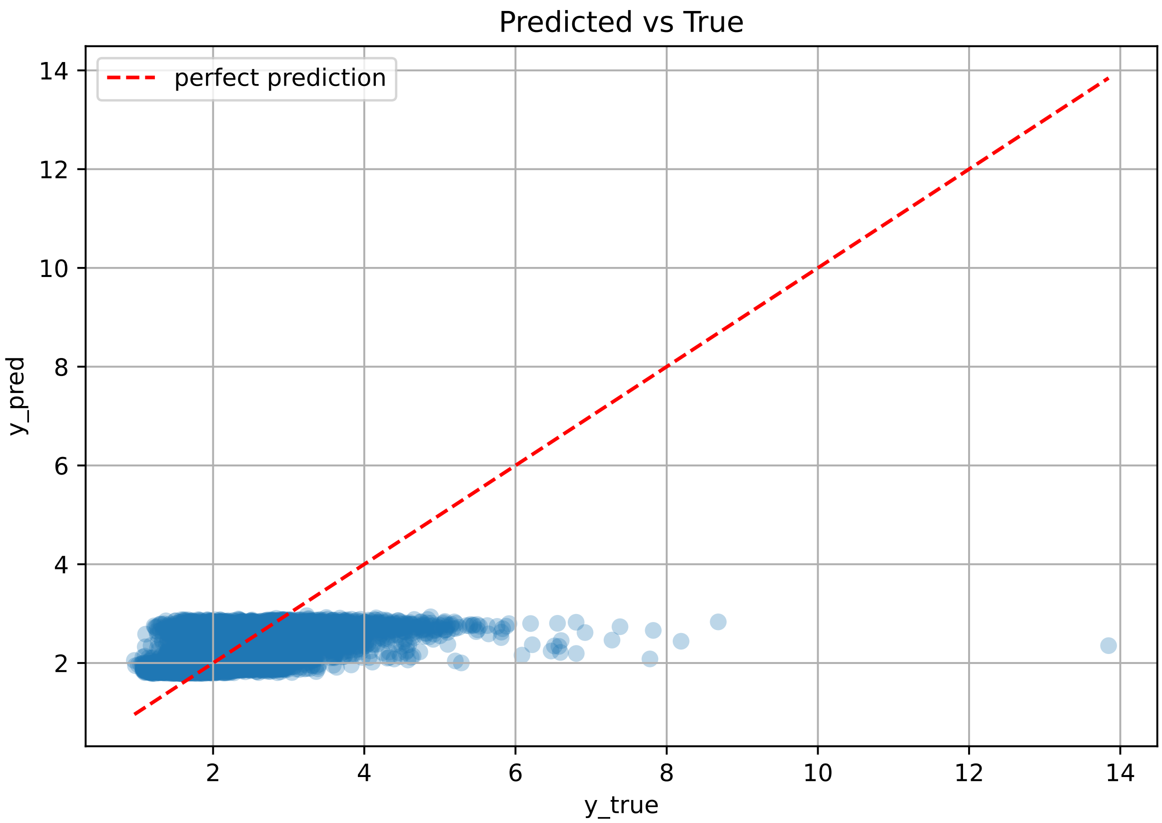
Rank 7: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.010868751910514335, 'reg___l2_leaf_reg': 28.650292256987612, 'reg___bagging_temperature': 4.92590060060266, 'reg___random_strength': 2.2539366835223307, 'reg___subsample': 0.8126510559696215, 'reg___min_data_in_leaf': 33, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.5369067362318989, 'reg___border_count': 38, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 1204.2413879926407}

Rank 8: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.013105216812221068, 'reg___l2_leaf_reg': 25.033276414890828, 'reg___bagging_temperature': 4.202403713504354, 'reg___random_strength': 3.073606169757184, 'reg___subsample': 0.8931291502141695, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 9, 'reg___rsm': 0.5996206251159875, 'reg___border_count': 40, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 286.0667295027407}

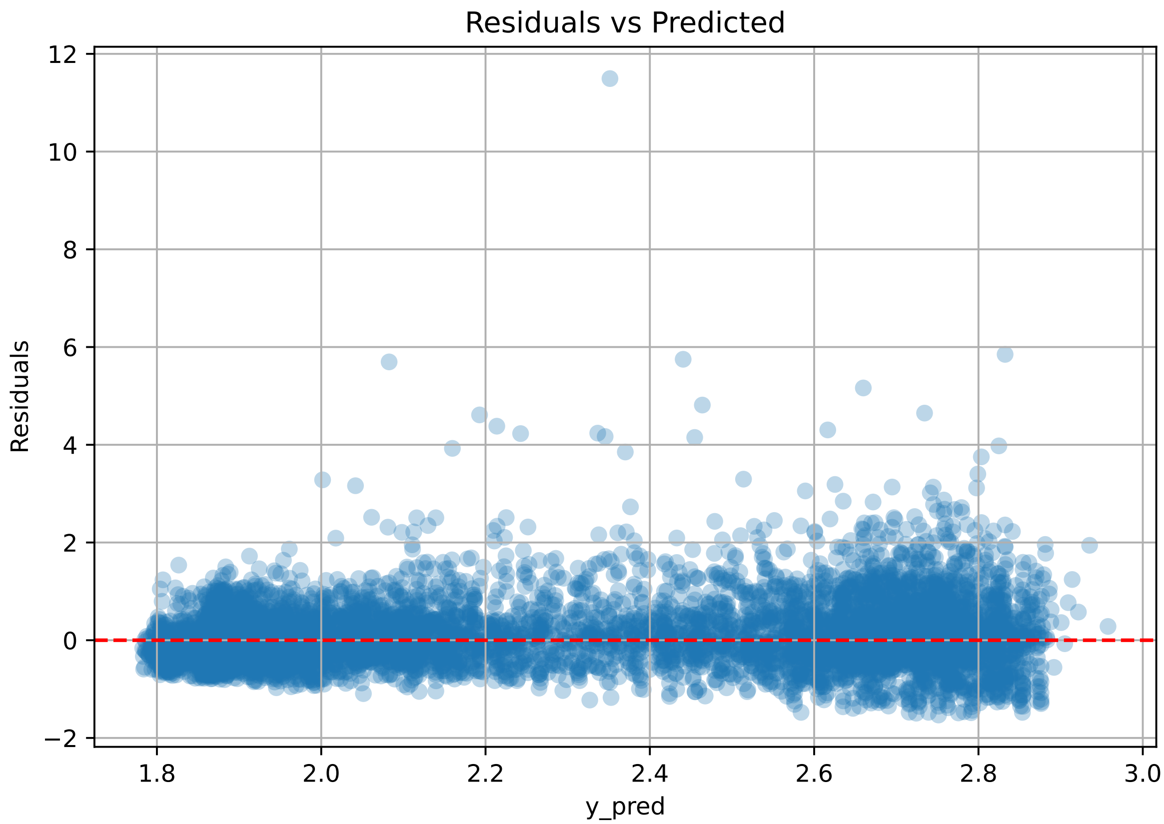
Rank 9: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.011233601128763525, 'reg___l2_leaf_reg': 14.708130773497306, 'reg___bagging_temperature': 4.419021676830827, 'reg___random_strength': 4.043303146009897, 'reg___subsample': 0.8173703976454358, 'reg___min_data_in_leaf': 28, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.7472571648316125, 'reg___border_count': 91, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 60.5414617643581}

Rank 10: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.014238278501211784, 'reg___l2_leaf_reg': 14.298361222907955, 'reg___bagging_temperature': 4.397960345635814, 'reg___random_strength': 2.661579213311445, 'reg___subsample': 0.8196607473668026, 'reg___min_data_in_leaf': 28, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.5542164741007503, 'reg___border_count': 89, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 25.18372719243348}

2.4.14 Scatter



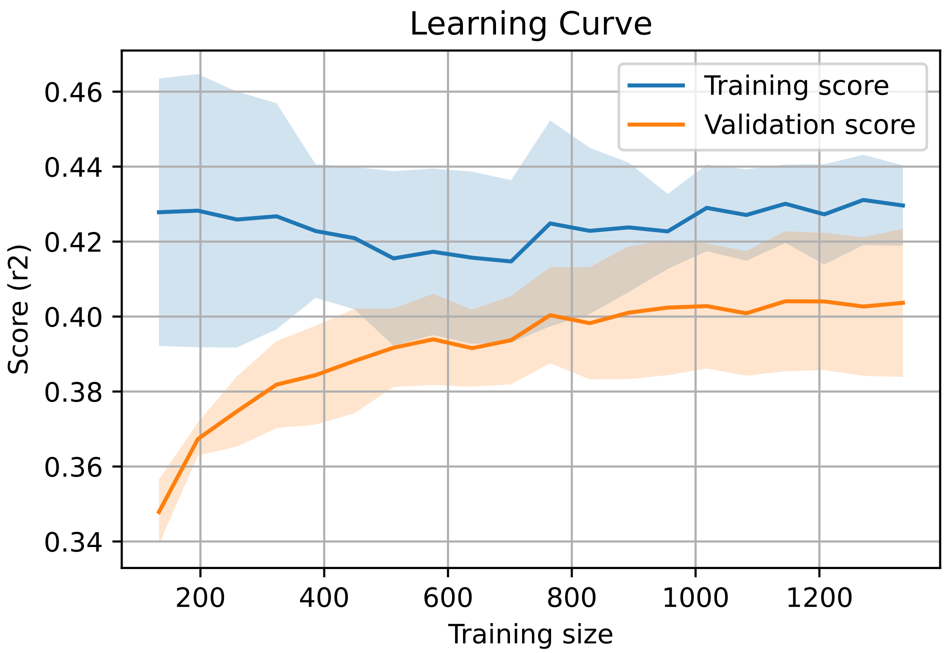
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 4

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817

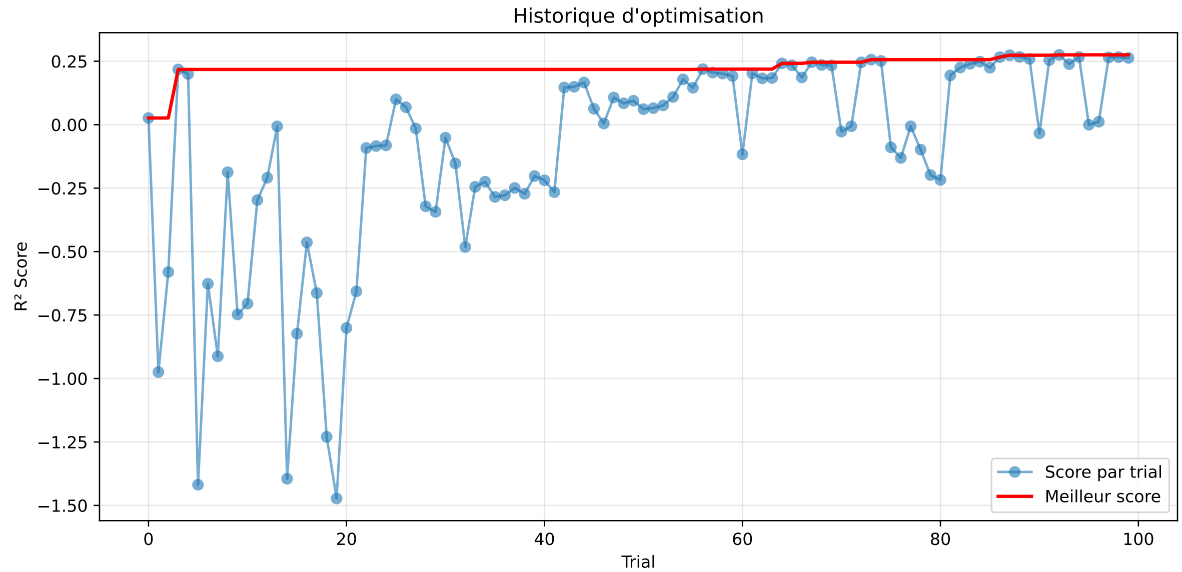


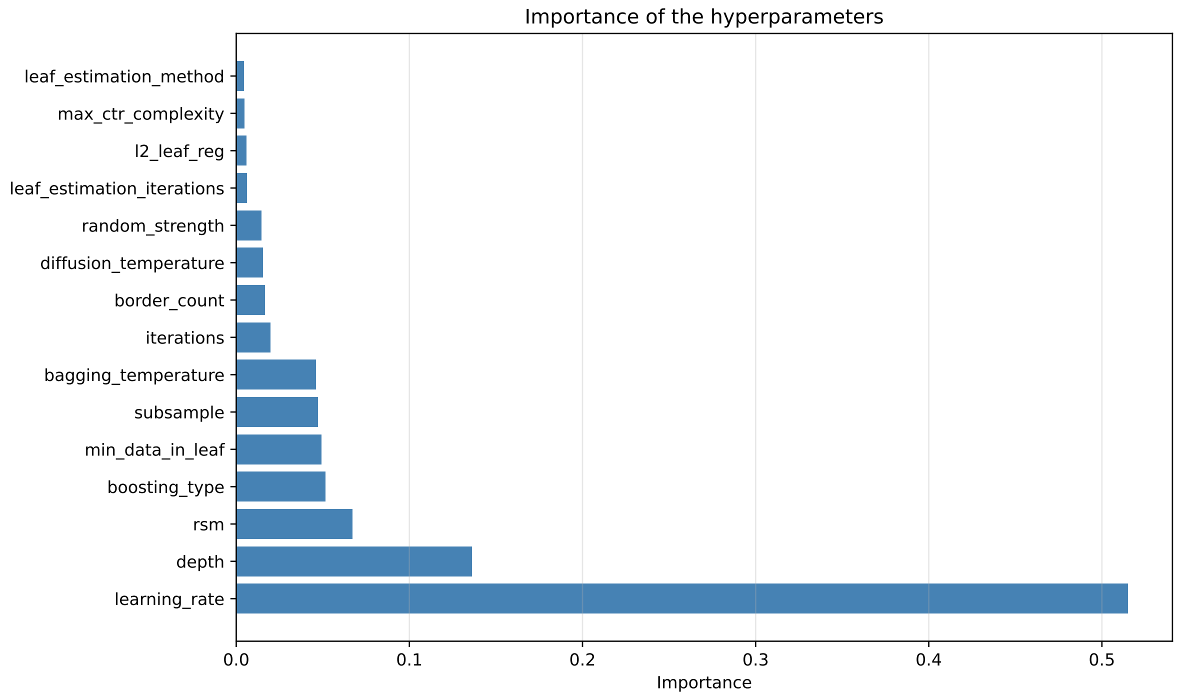
2.4.17 Gridsearch — Group: 6

Len(group) : 1353, **Len(training)** : 948, **Len(test)** : 405

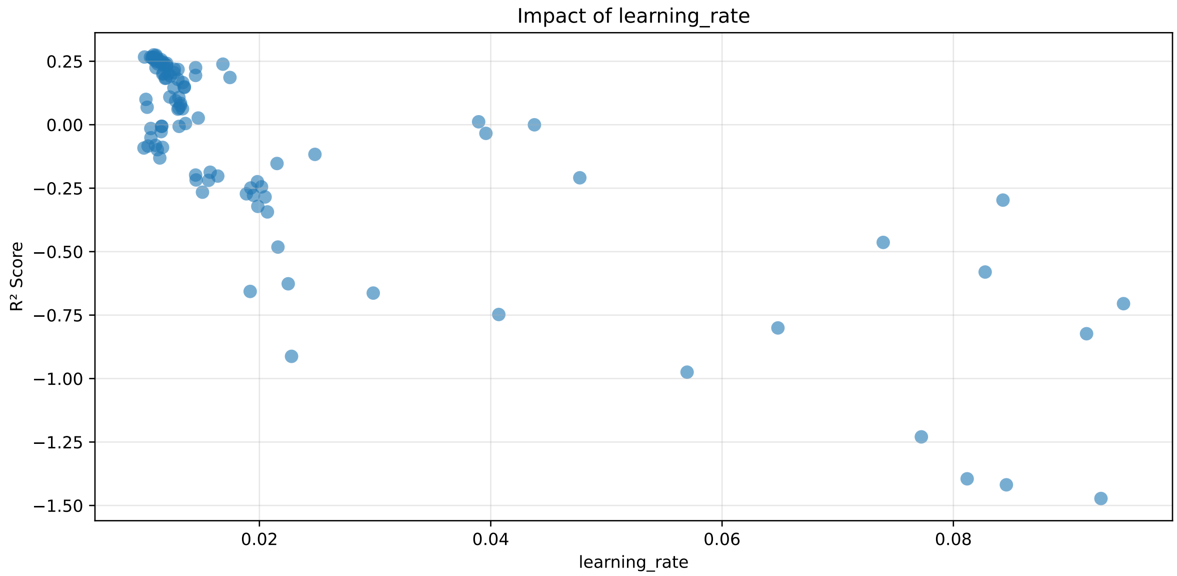
Distribution y_train vs y_test: y_train: mean=2.286, std=0.802, min=1.041, max=7.383 y_test: mean=2.187, std=0.733, min=1.078, max=5.525

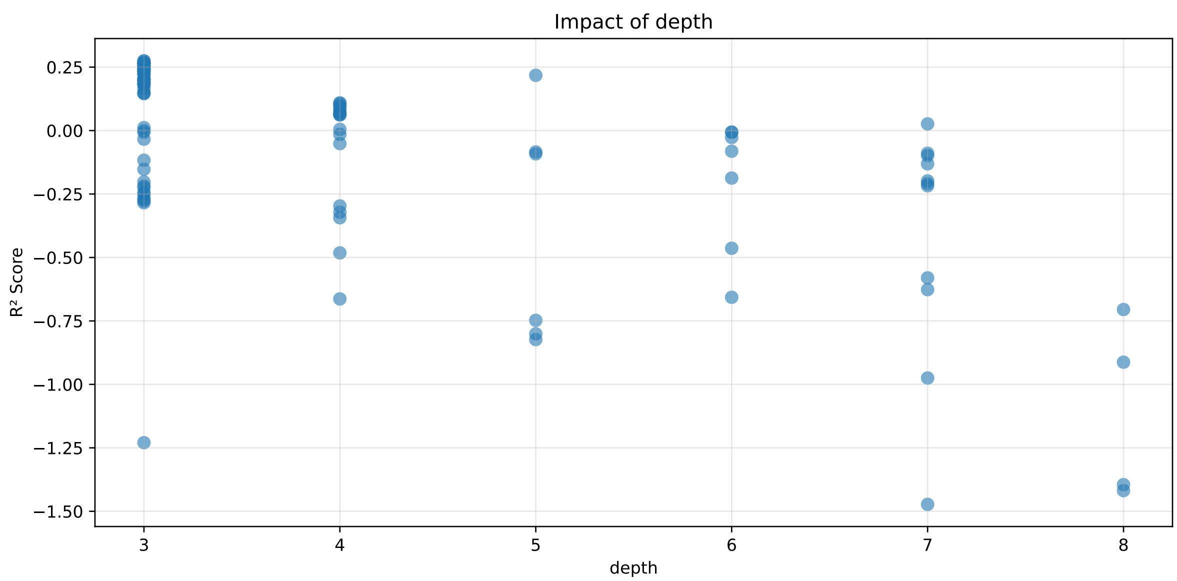
==================== TOP Importance FEATURES ==================== feature importance r_umidity 60.454900 r_NH3 12.358371 r_THI 11.999665 r_temperature 8.610977 r_CO2 6.576087

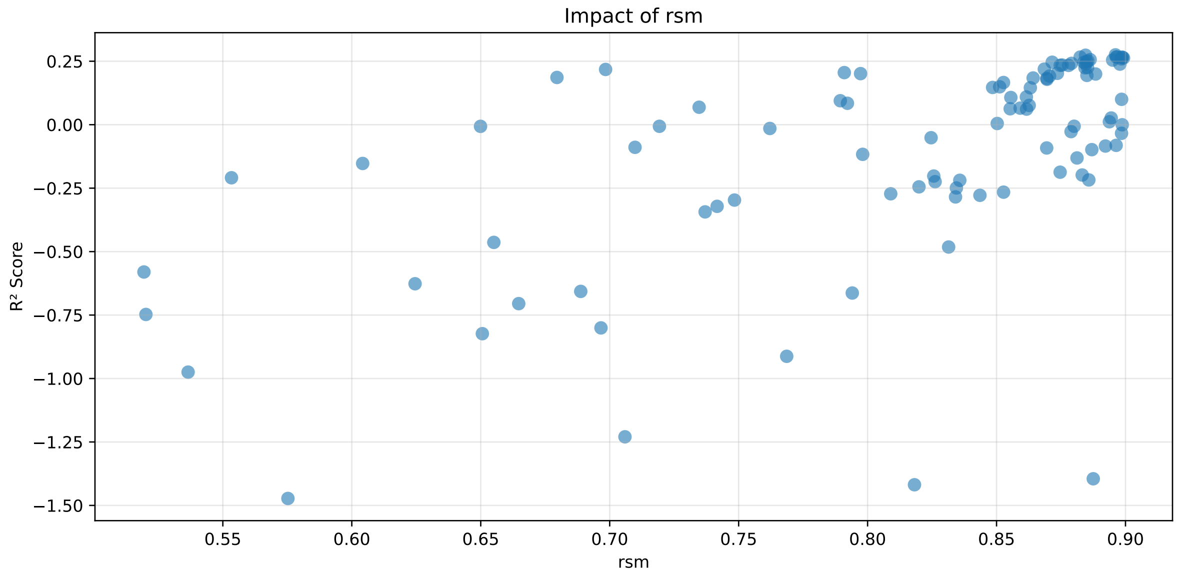


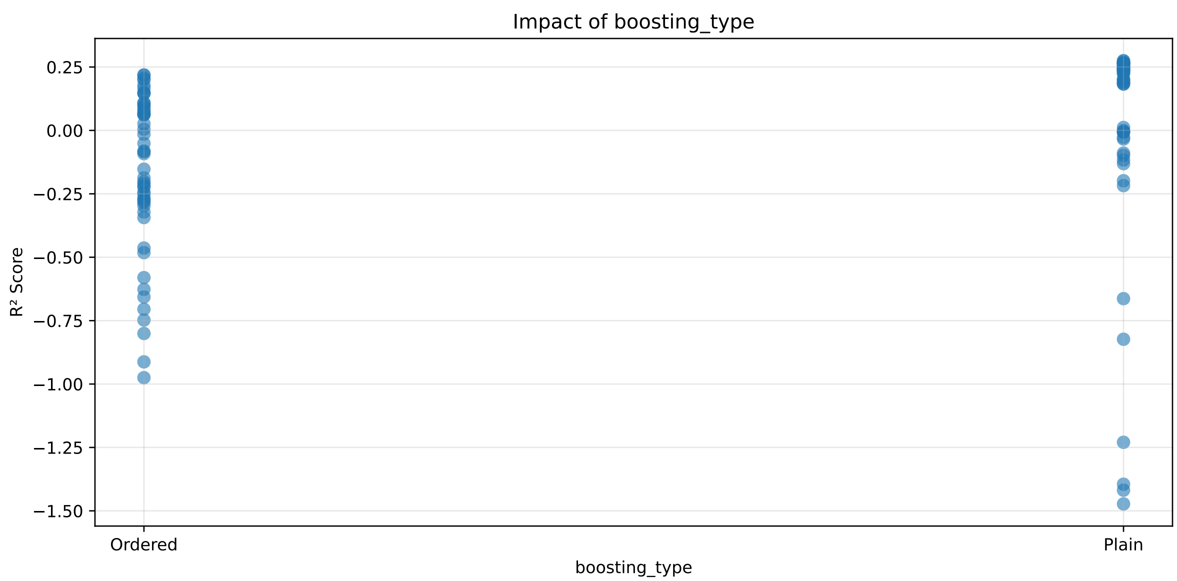


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'rsm', 'boosting_type']


Impact of learning_rate
R² Score
learning_rate
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
0.02
0.04
0.06
0.08


Impact of depth
R² Score
depth
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
3
4
5
6
7
8



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C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```
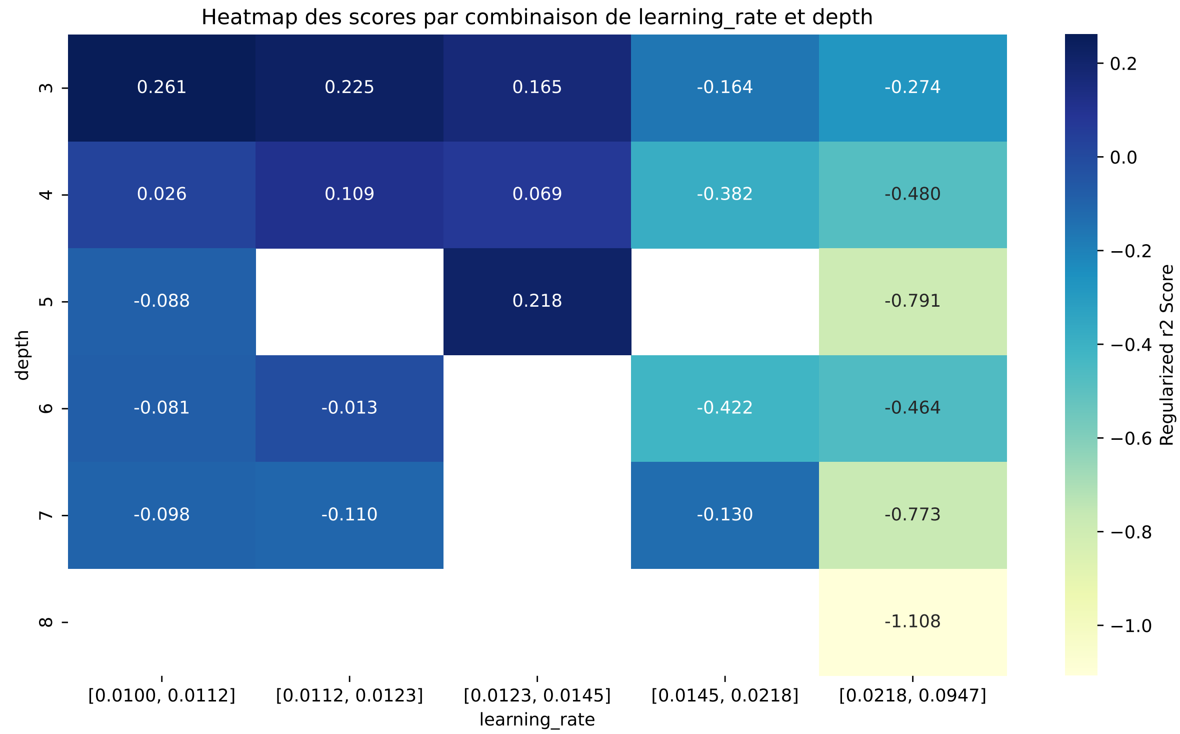



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4098	0.3873	0.4752	0.0516	0.2751	0.4826	0.0952	1.2459	1.0579
2	0.4187	0.3945	0.4768	0.0484	0.2736	0.4831	0.0885	1.2244	1.0613
3	0.4107	0.3867	0.4709	0.0489	0.2669	0.4826	0.0959	1.2479	1.062
4	0.3898	0.3693	0.4481	0.0438	0.2667	0.4578	0.0885	1.2397	1.0555
5	0.4198	0.3942	0.4795	0.0502	0.2663	0.4854	0.0912	1.2313	1.0649
6	0.4091	0.3852	0.4673	0.0491	0.2661	0.4702	0.085	1.2207	1.0619
7	0.416	0.3908	0.4743	0.0495	0.2651	0.4766	0.0858	1.2194	1.0644
8	0.4097	0.3852	0.4674	0.0504	0.2629	0.4722	0.0869	1.2257	1.0635
9	0.4087	0.3838	0.4678	0.0425	0.2593	0.4784	0.0946	1.2465	1.0649
10	0.4762	0.4395	0.5412	0.0561	0.2563	0.5431	0.1035	1.2355	1.0834
---	---	---	---	---	---	---	---	---	---
100	0.9904	0.5799	0.7211	0.0437	-1.4726	0.7626	0.1827	1.3151	1.7079

Hyperparameters:

Rank 1: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.010858266389137158, 'reg___l2_leaf_reg': 4.423434122973623, 'reg___bagging_temperature': 2.680189463768659, 'reg___random_strength': 4.820094975901698, 'reg___subsample': 0.6897825107975737, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.8961285758464161, 'reg___border_count': 33, 'reg___leaf_estimation_method': 'Gradient',

’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 4805.0087078851175}

Rank 2: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.011063544844015517, ’reg__l2_leaf_reg’: 5.3779046254182985, ’reg__bagging_temperature’: 1.713671454613603, ’reg__random_strength’: 4.804834238382126, ’reg__subsample’: 0.7265627606422173, ’reg__min_data_in_leaf’: 37, ’reg__leaf_estimation_iterations’: 10, ’reg__rsm’: 0.8845065528763569, ’reg__border_count’: 124, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 4636.106429623292}

Rank 3: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.011095569079279942, ’reg__l2_leaf_reg’: 4.451693081942035, ’reg__bagging_temperature’: 2.631167935890617, ’reg__random_strength’: 4.195810605904383, ’reg__subsample’: 0.6390414615844027, ’reg__min_data_in_leaf’: 39, ’reg__leaf_estimation_iterations’: 10, ’reg__rsm’: 0.896431618646648, ’reg__border_count’: 46, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 3610.6635874173785}

Rank 4: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.010067946876843224, ’reg__l2_leaf_reg’: 4.469292325077591, ’reg__bagging_temperature’: 2.7647034185222044, ’reg__random_strength’: 4.8300888220513905, ’reg__subsample’: 0.6886707636705756, ’reg__min_data_in_leaf’: 39, ’reg__leaf_estimation_iterations’: 10, ’reg__rsm’: 0.8967215470235348, ’reg__border_count’: 46, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 4752.39232667403}

Rank 5: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.01088534702958412, ’reg__l2_leaf_reg’: 4.155722353013002, ’reg__bagging_temperature’: 1.8122421204486905, ’reg__random_strength’: 4.434574729642349, ’reg__subsample’: 0.7275719820004367, ’reg__min_data_in_leaf’: 37, ’reg__leaf_estimation_iterations’: 10, ’reg__rsm’: 0.8824237357604652, ’reg__border_count’: 125, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 4584.941727246596}

Rank 6: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.010579014773671685, ’reg__l2_leaf_reg’: 4.376192844198021, ’reg__bagging_temperature’: 1.924130164850023, ’reg__random_strength’: 4.822151443870497, ’reg__subsample’: 0.5986323917092893, ’reg__min_data_in_leaf’: 39, ’reg__leaf_estimation_iterations’: 10, ’reg__rsm’: 0.8986751788175453, ’reg__border_count’: 125, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 5720.955028547543}

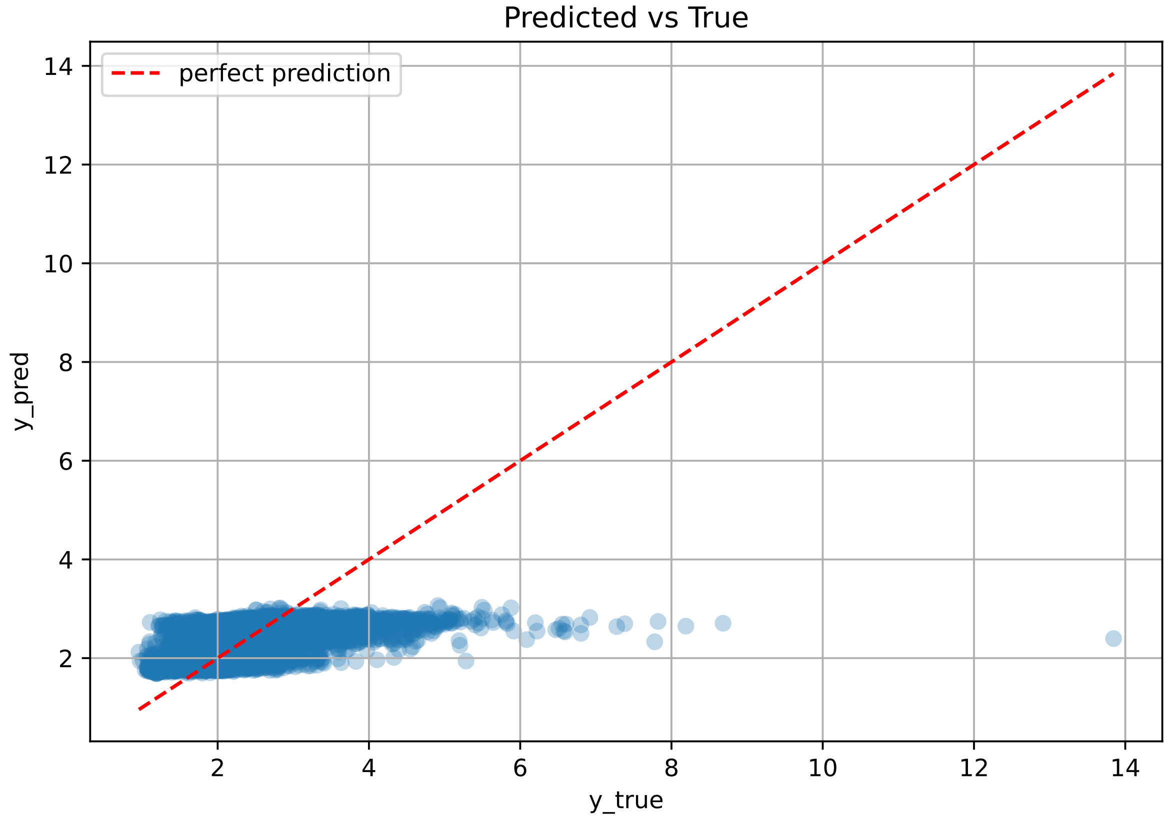
Rank 7: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.010728471265769896, ’reg__l2_leaf_reg’: 4.0149013770930875, ’reg__bagging_temperature’: 2.7531987006494765, ’reg__random_strength’: 4.184786623869173, ’reg__subsample’: 0.7361397538594427, 'reg___min_data_in_leaf': 39, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.8988298814875666, 'reg___border_count': 125, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 4698.692796561311}

Rank 8: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.01072771062964779, 'reg___l2_leaf_reg': 3.1407168368732536, 'reg___bagging_temperature': 1.917616032858762, 'reg___random_strength': 4.838238059544686, 'reg___subsample': 0.607577079860908, 'reg___min_data_in_leaf': 39, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.899237534251061, 'reg___border_count': 126, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 5808.955095599255}

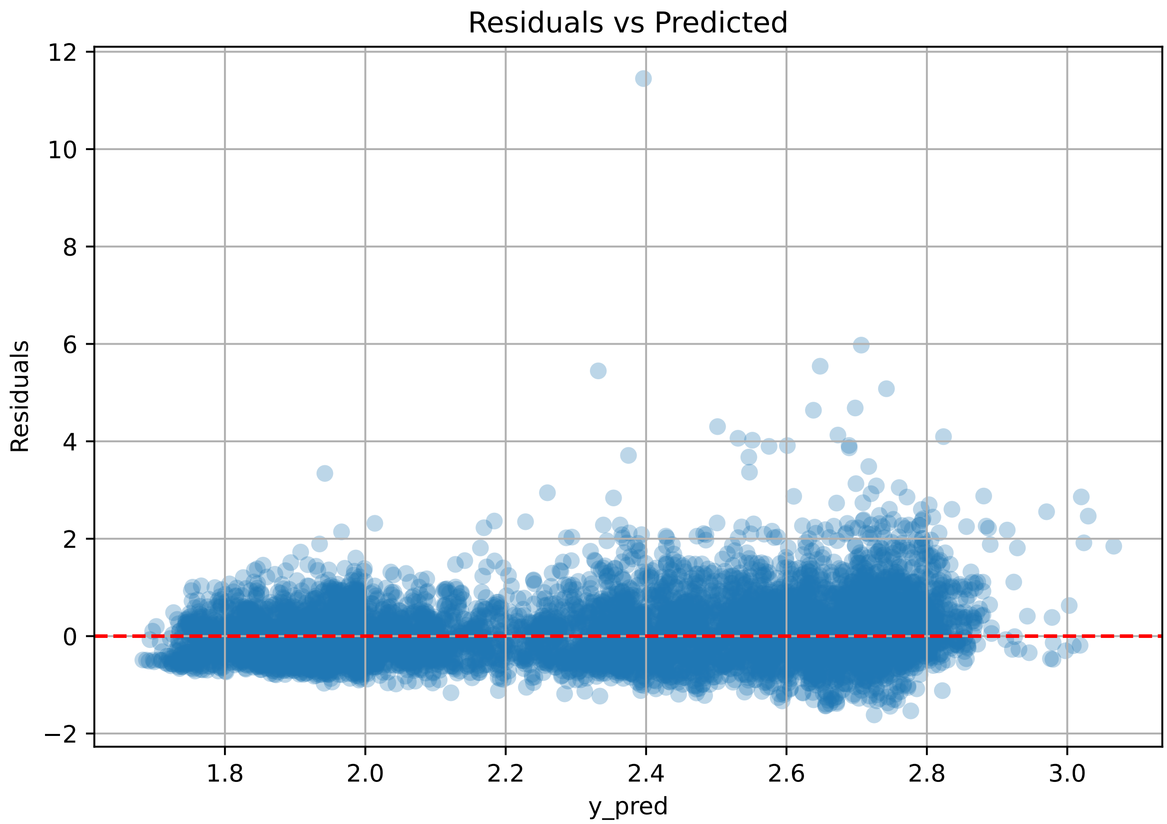
Rank 9: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.011088473921176472, 'reg___l2_leaf_reg': 3.8729675140315964, 'reg___bagging_temperature': 2.642051208844359, 'reg___random_strength': 4.765515248275194, 'reg___subsample': 0.646942253378352, 'reg___min_data_in_leaf': 47, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.8984415934667876, 'reg___border_count': 44, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 4678.816325042325}

Rank 10: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.011503904795215163, 'reg___l2_leaf_reg': 1.6480109457702028, 'reg___bagging_temperature': 1.538023707658678, 'reg___random_strength': 4.392831007098108, 'reg___subsample': 0.7049951890357491, 'reg___min_data_in_leaf': 42, 'reg___leaf_estimation_iterations': 9, 'reg___rsm': 0.886295390838298, 'reg___border_count': 40, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 4304.58664817035}

2.4.18 Scatter



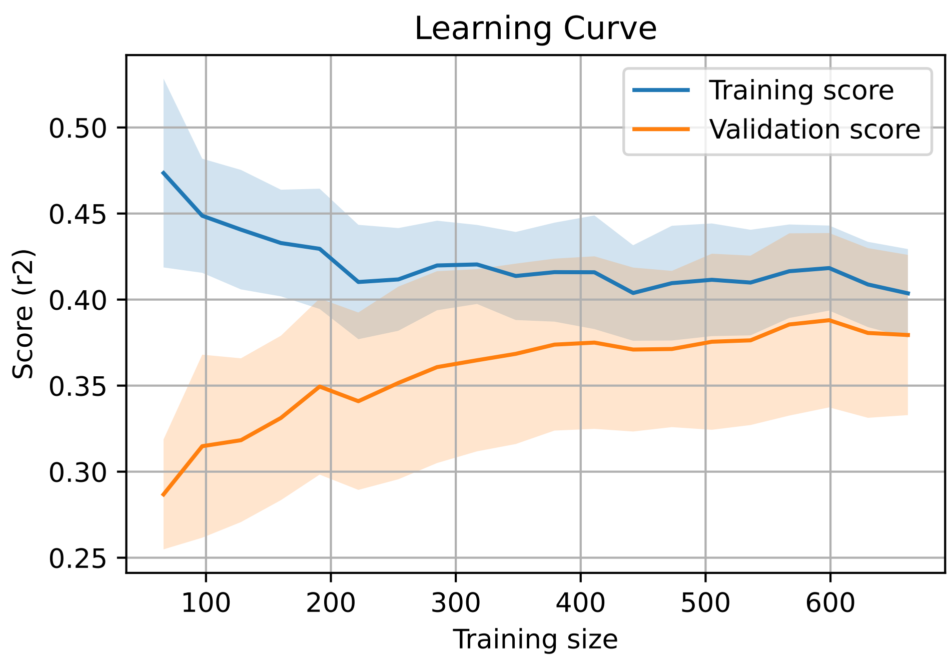
2.4.19 Scatter residuals



2.4.20 Learning curve — Group: 6

Len(group) : 1353, **Len(training)** : 948,

Len(training_real) : 664, **Len(test_of_training)** : 284, **Len(test)** : 405



2.5 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	test_cv	std_test_cv	pen_score	test	diff	r_test	r_train
LGBM	1	557	238	795	0.58	0.553	0.516	0.034	0.553	0.536	-0.016	0.97	1.05
LGBM	2	1909	817	2726	0.685	0.655	0.558	0.013	0.6553	0.574	-0.081	0.876	1.045
LGBM	3	398	170	568	0.167	0.158	0.052	0.044	0.1119	0.072	-0.086	0.456	1.058
LGBM	4	1909	817	2726	0.394	0.374	0.423	0.026	0.2734	0.436	0.062	1.165	1.054
LGBM	6	948	405	1353	0.357	0.34	0.404	0.055	0.3403	0.429	0.088	1.26	1.05
LGBM	MEAN	5721	2447	8168	0.437	0.416	0.391	0.034	0.387	0.409	-0.007	0.945	1.051
CatBoost	1	557	238	795	0.636	0.6	0.56	0.043	0.4169	0.582	-0.018	0.97	1.061
CatBoost	2	1909	817	2726	0.735	0.701	0.602	0.017	0.7012	0.609	-0.092	0.868	1.048
CatBoost	3	398	170	568	0.424	0.316	0.179	0.043	-0.2223	0.187	-0.129	0.591	1.341
CatBoost	4	1909	817	2726	0.432	0.408	0.467	0.02	0.2859	0.468	0.061	1.149	1.06
CatBoost	6	948	405	1353	0.41	0.387	0.475	0.052	0.2751	0.483	0.095	1.246	1.058
CatBoost	MEAN	5721	2447	8168	0.527	0.482	0.457	0.035	0.291	0.466	-0.017	0.965	1.113

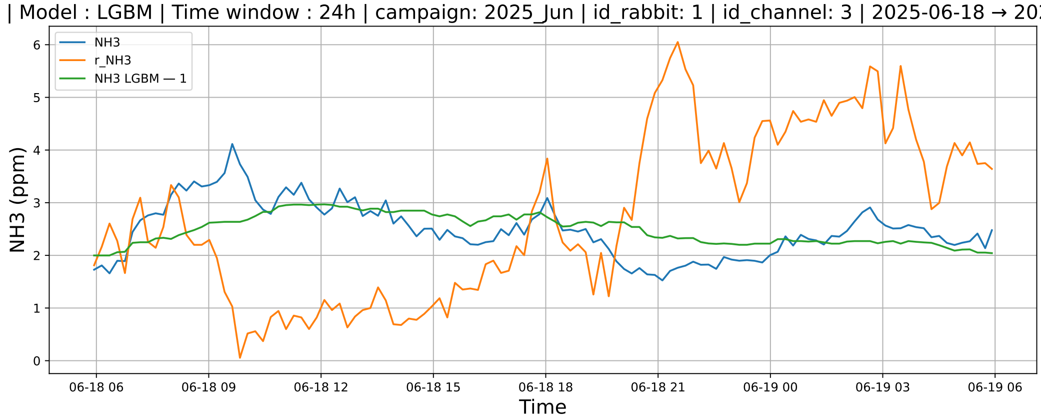
3 Plot

3.1 Standalone plots

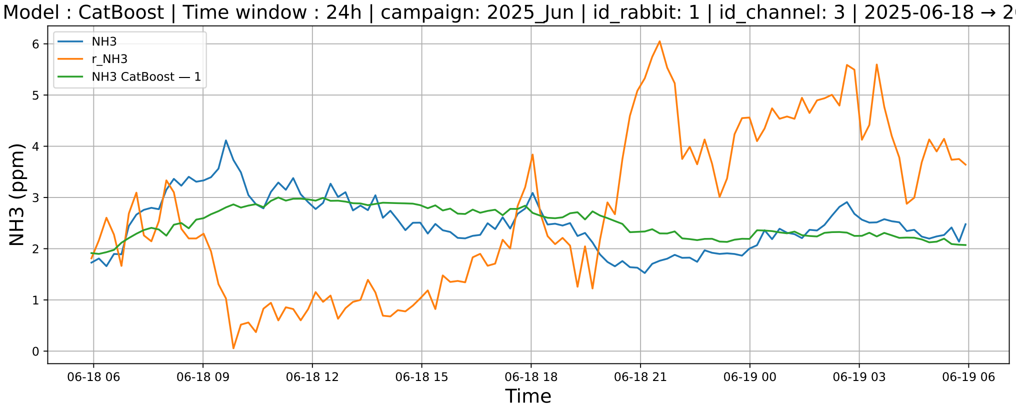
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

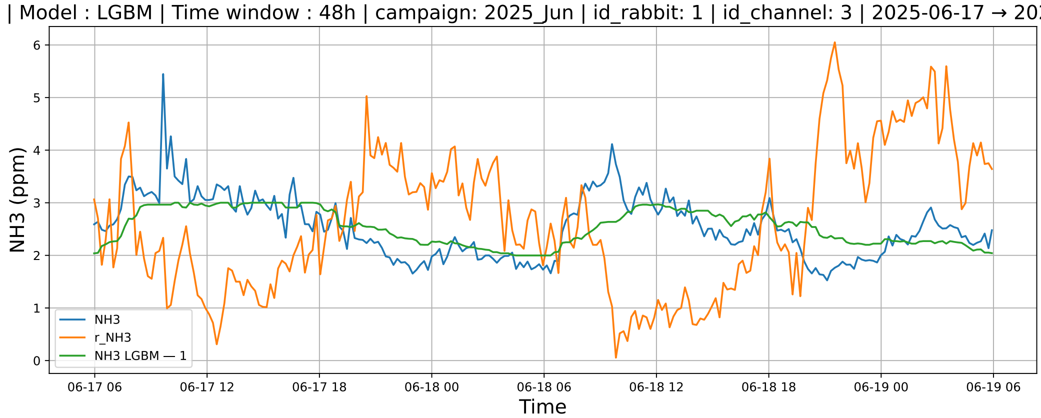


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

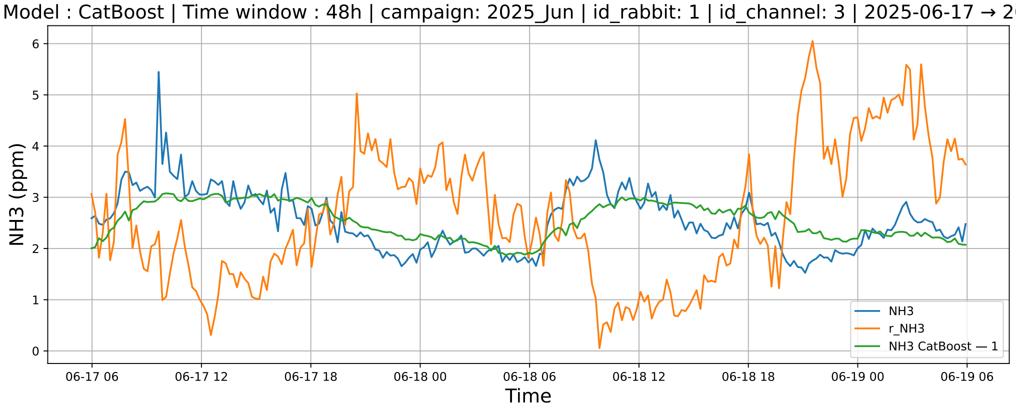


3.1.1.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

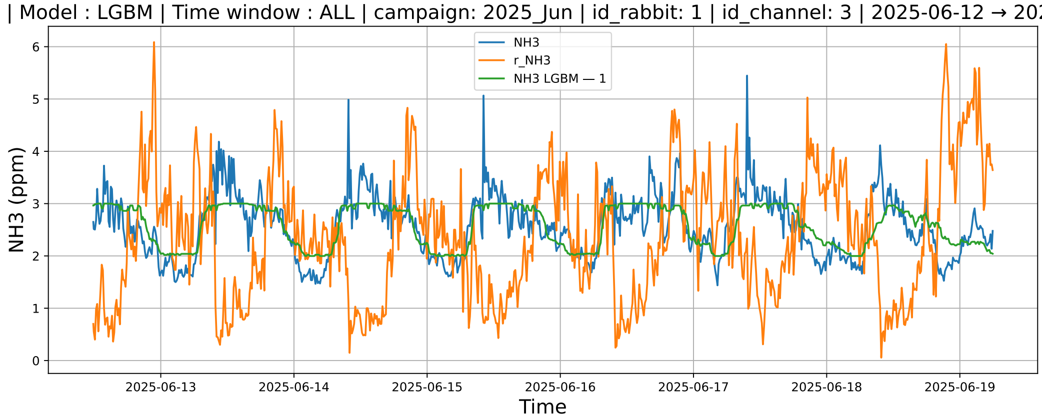


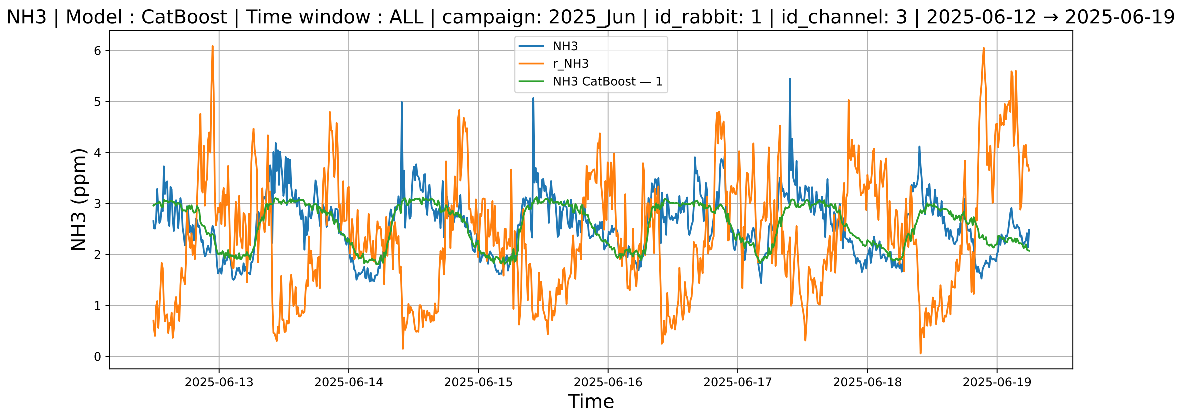
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.1.1.3 Time window : ALL

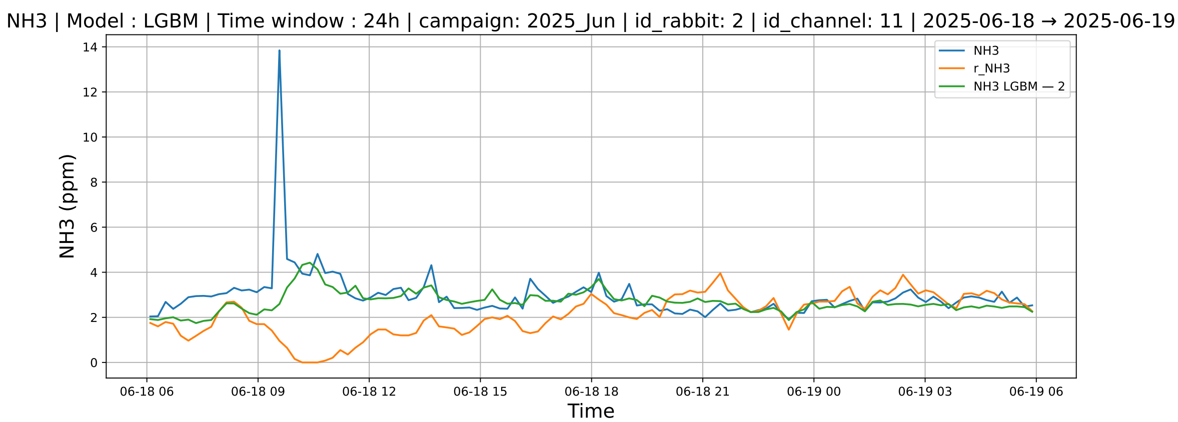
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

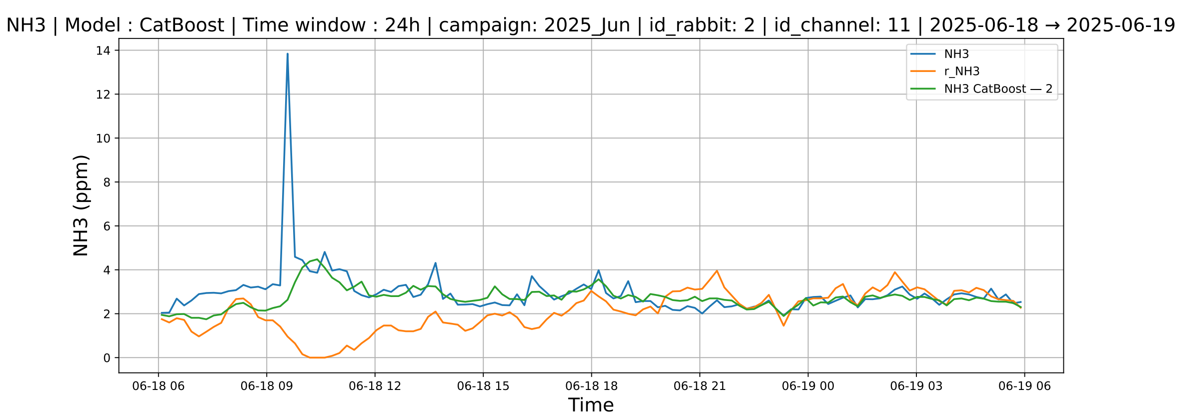




3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

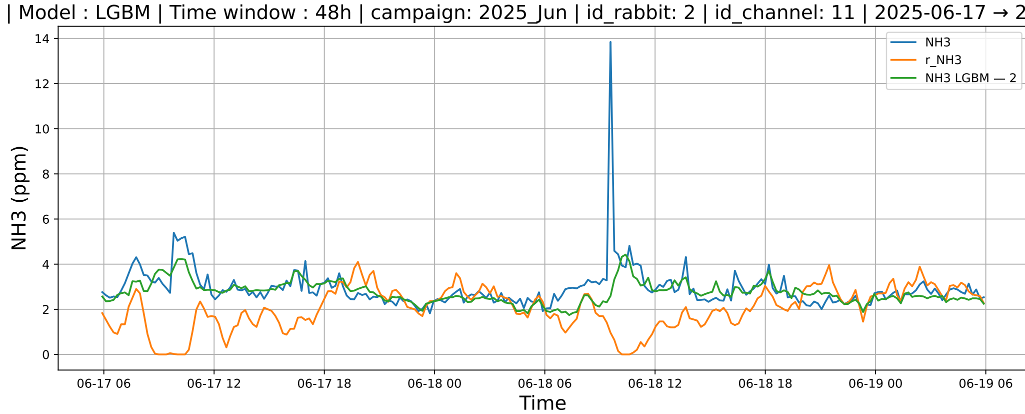
3.1.2.1 Time window : 24



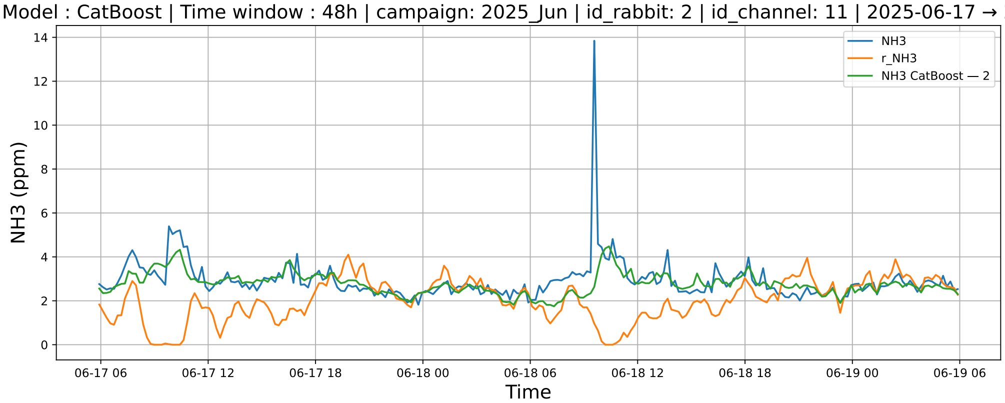


3.1.2.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

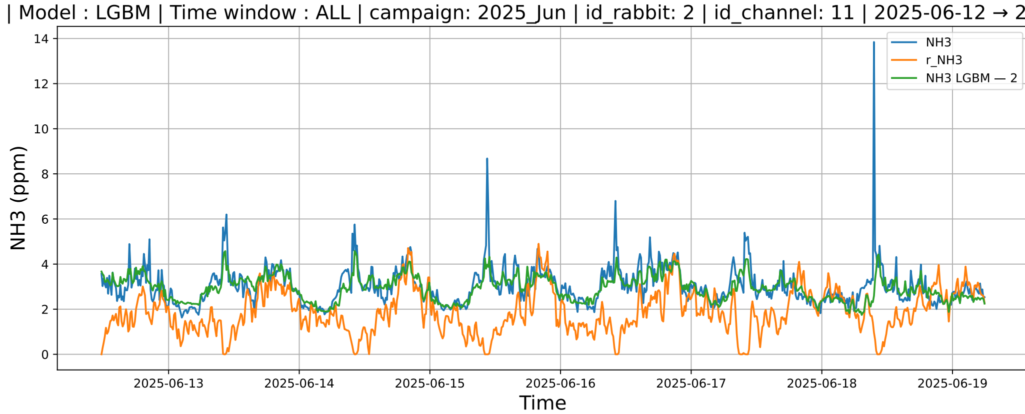


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

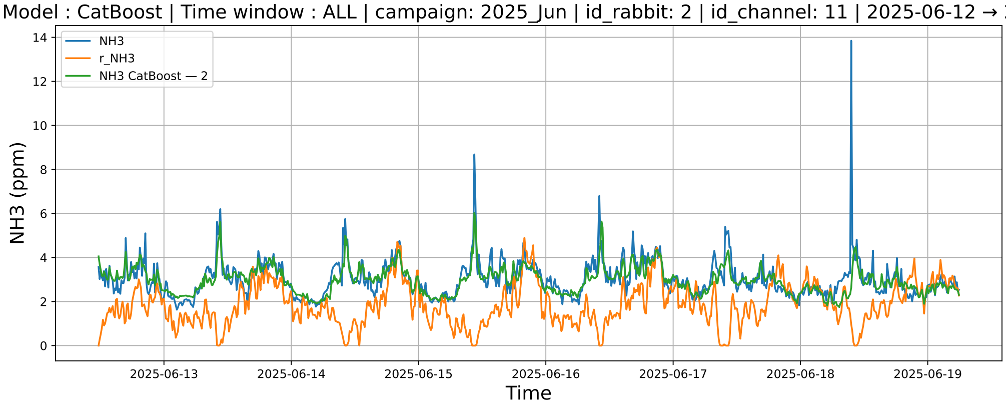


3.1.2.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



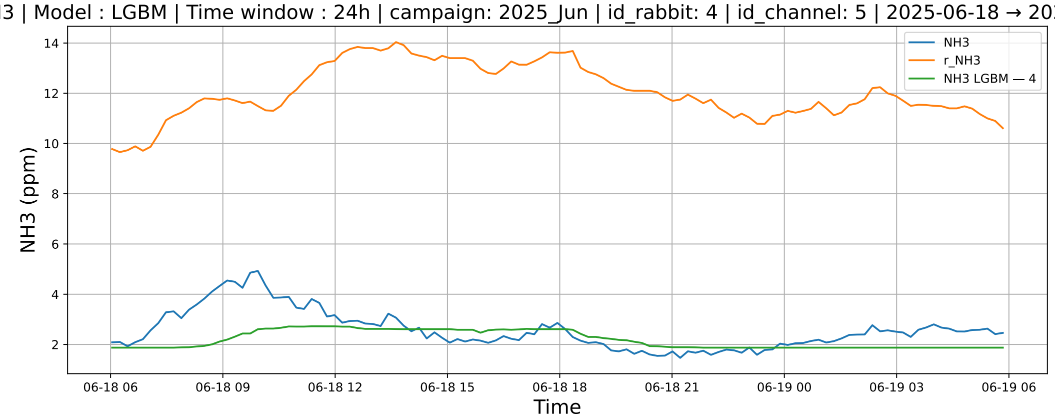
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



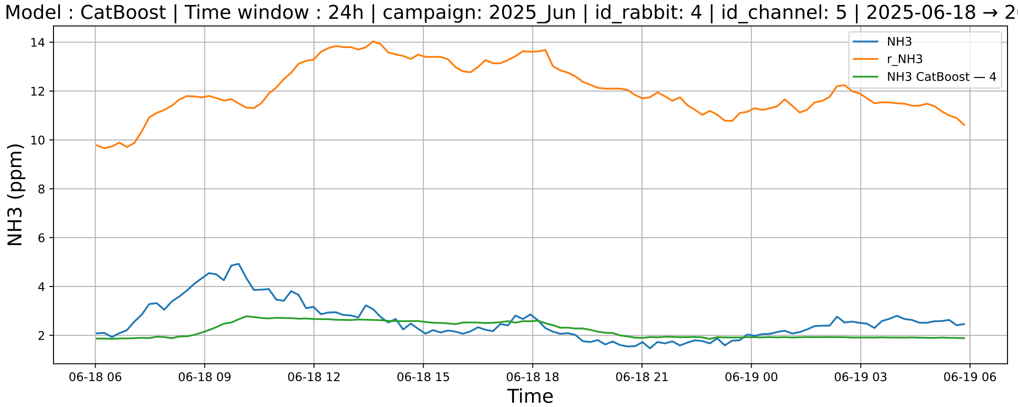
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

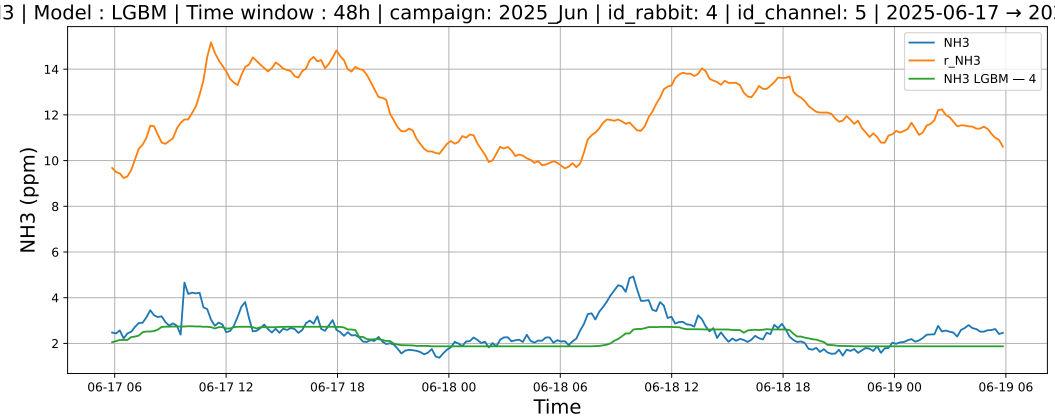


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

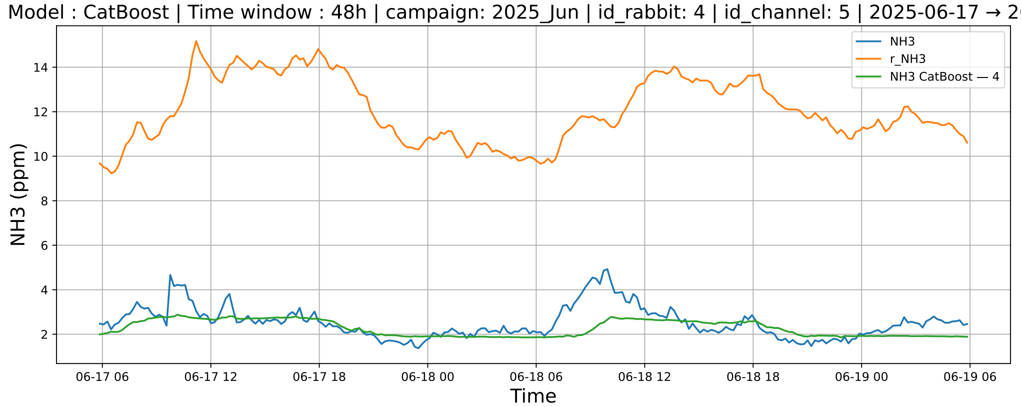


3.1.3.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

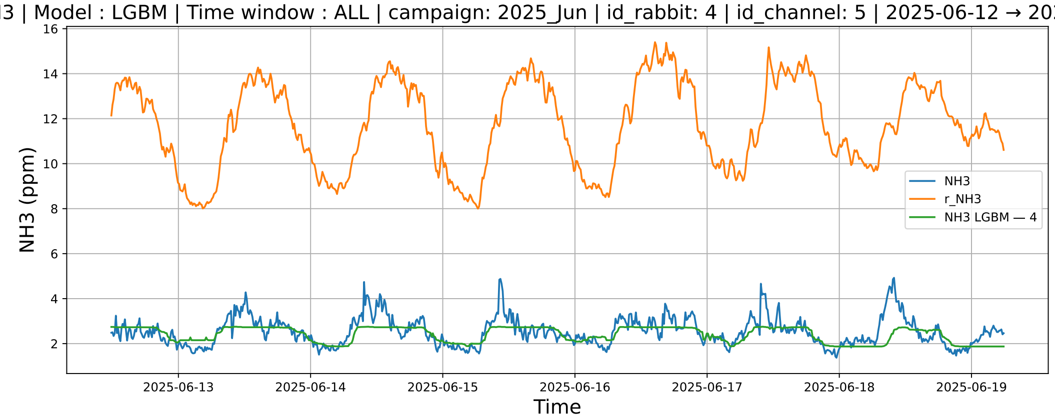


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

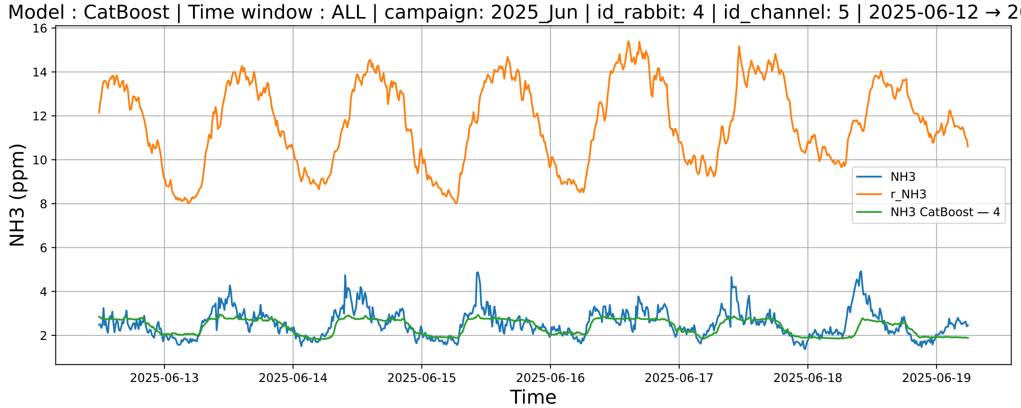


3.1.3.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



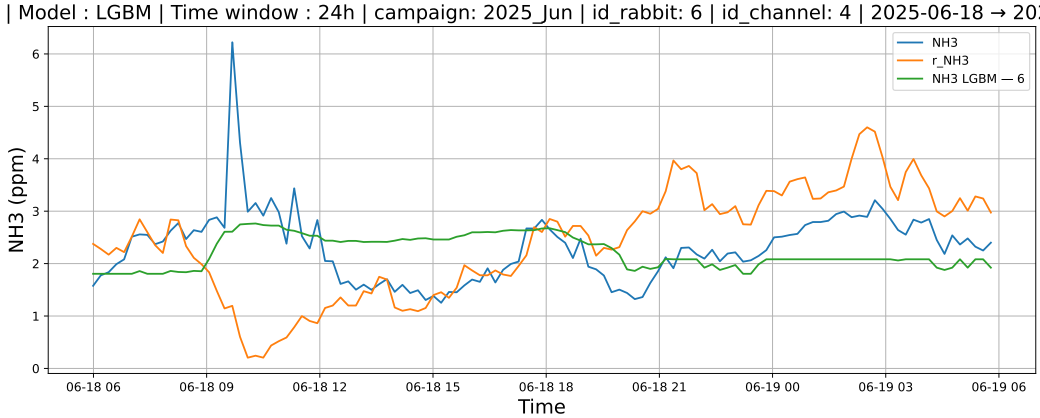
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



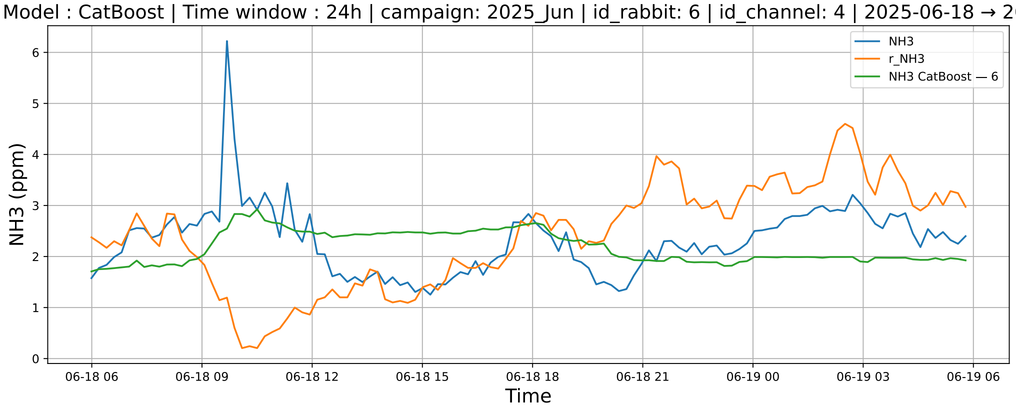
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

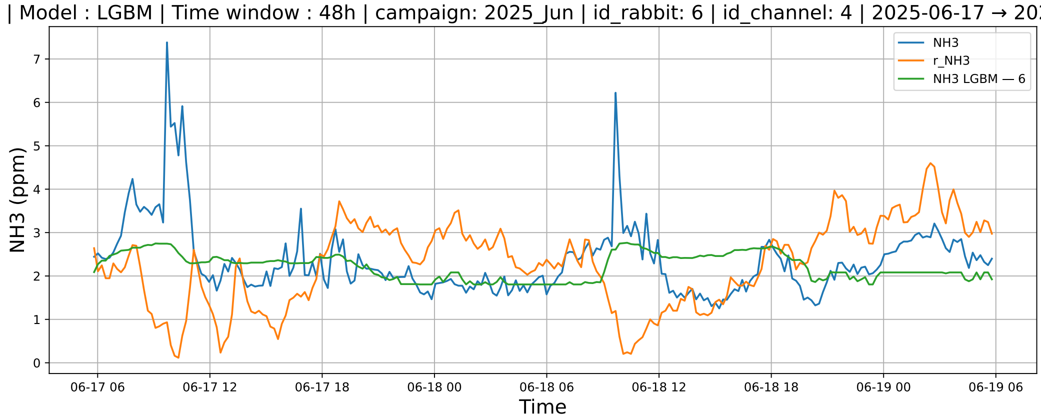


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

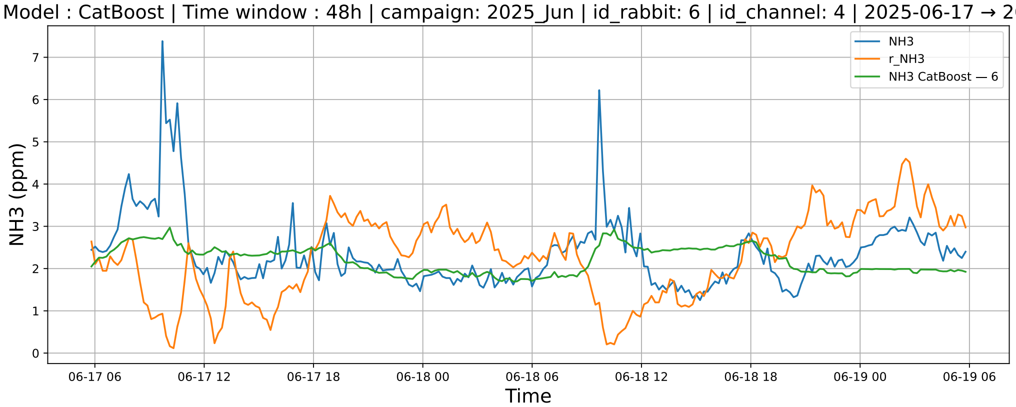


3.1.4.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

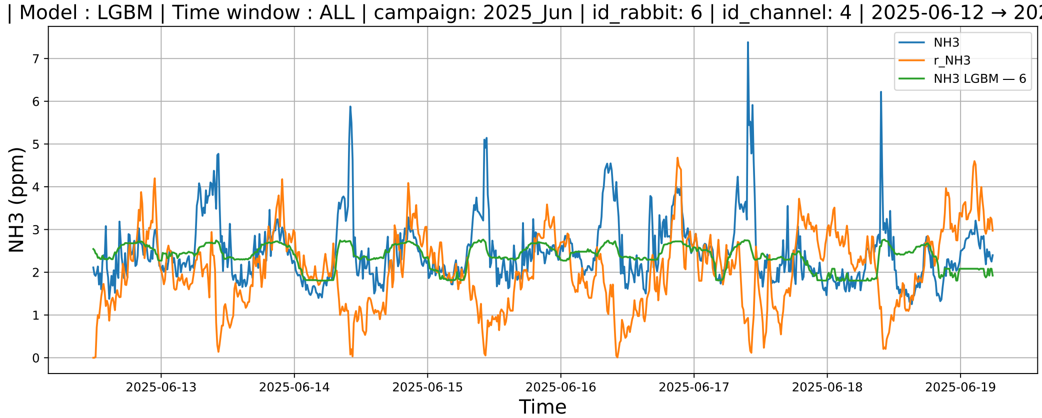


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

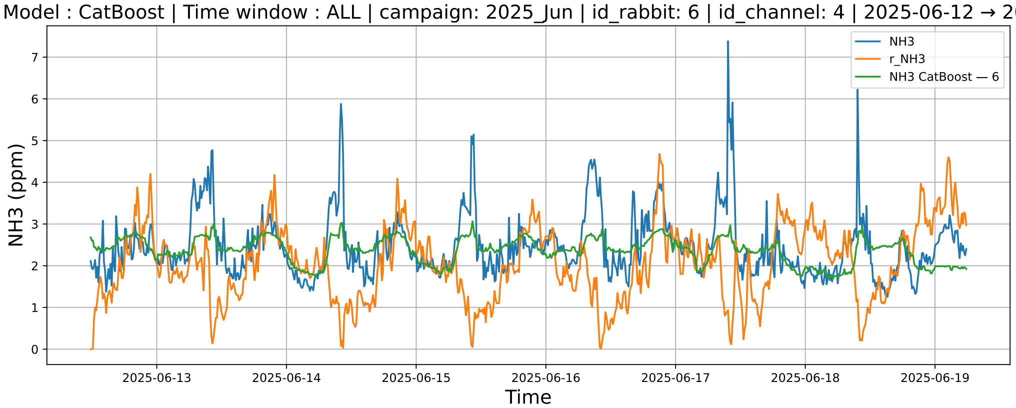


3.1.4.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



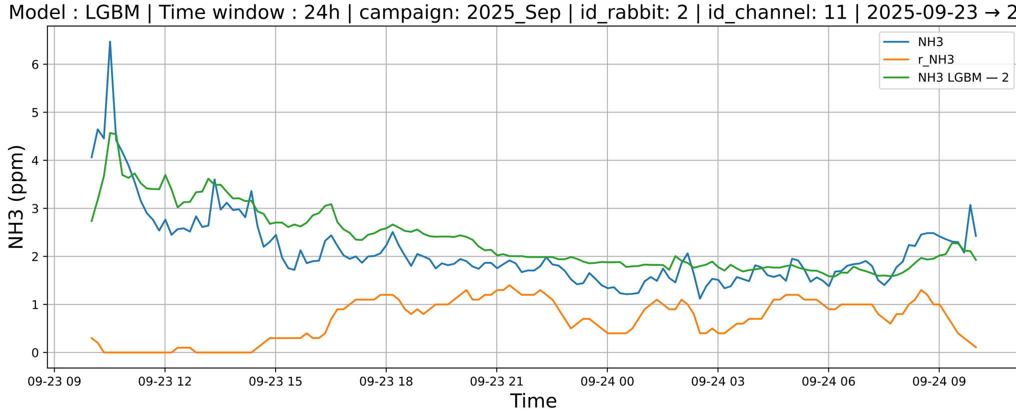
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



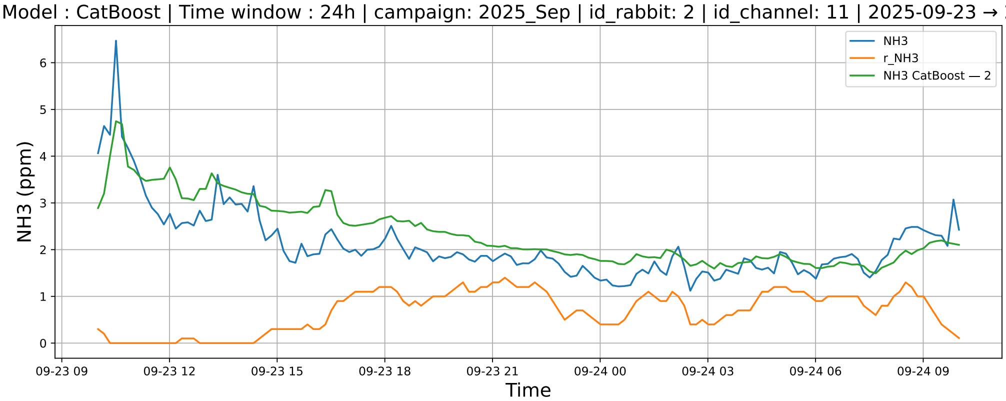
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

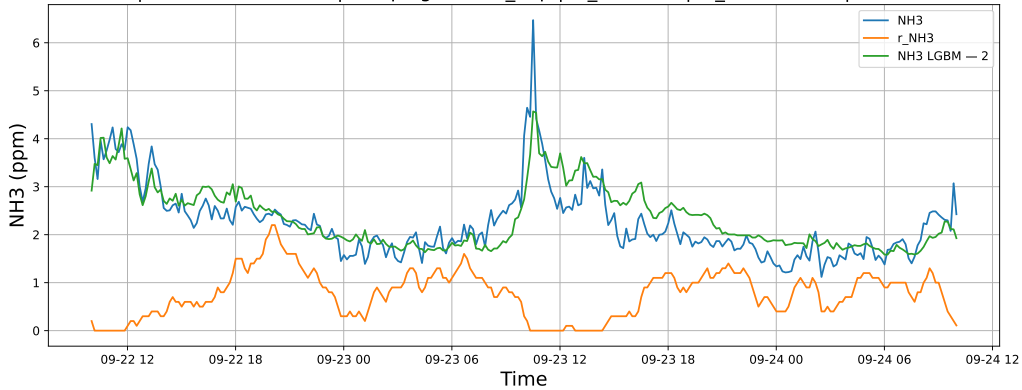


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

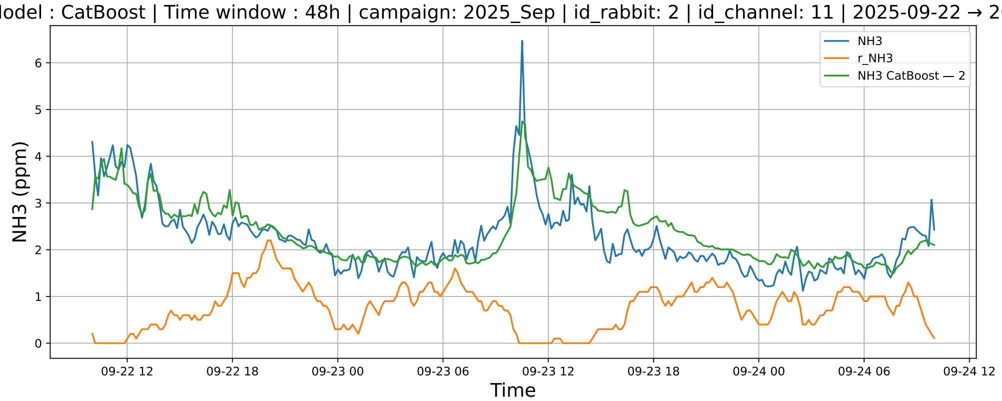


3.1.5.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

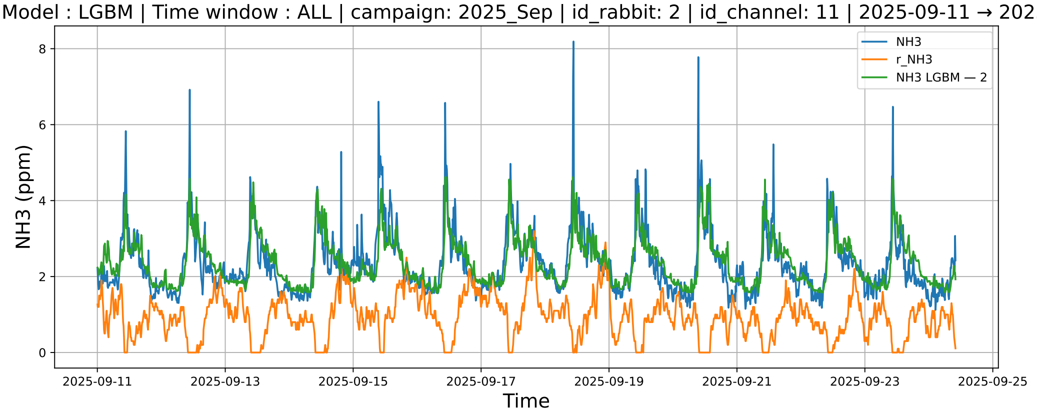


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

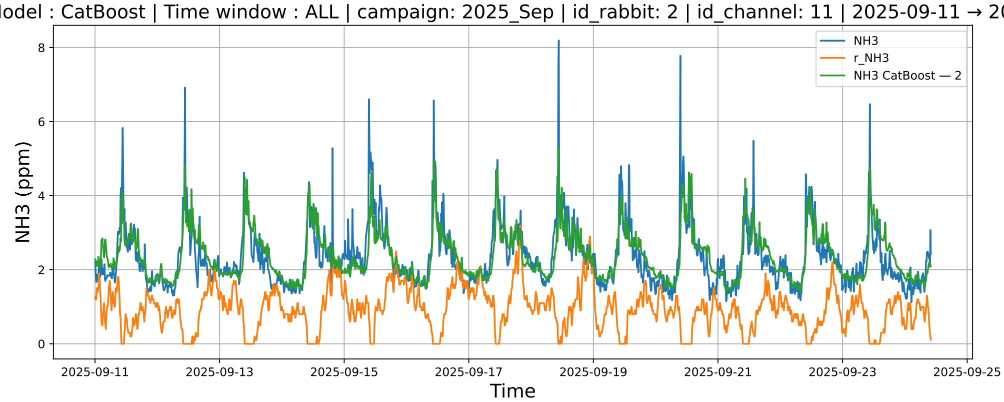


3.1.5.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



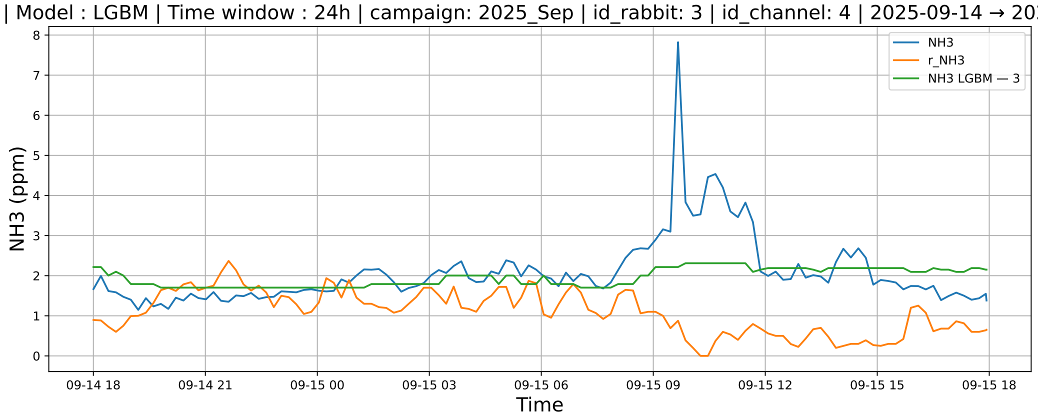
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



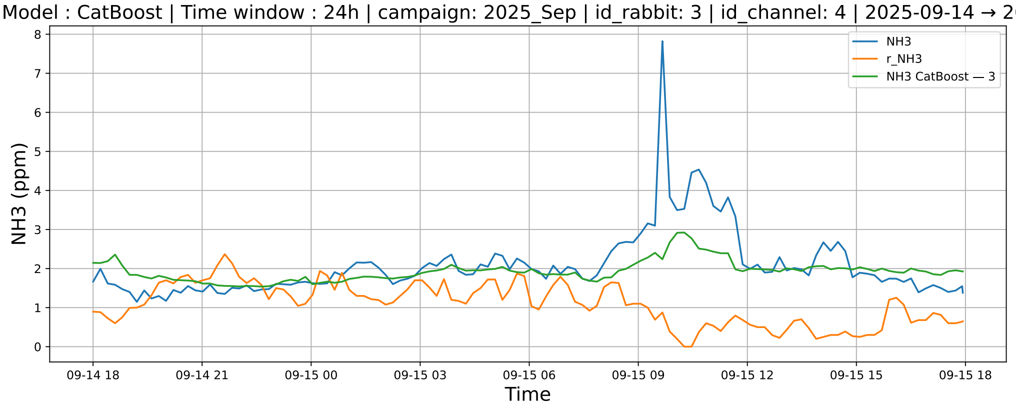
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

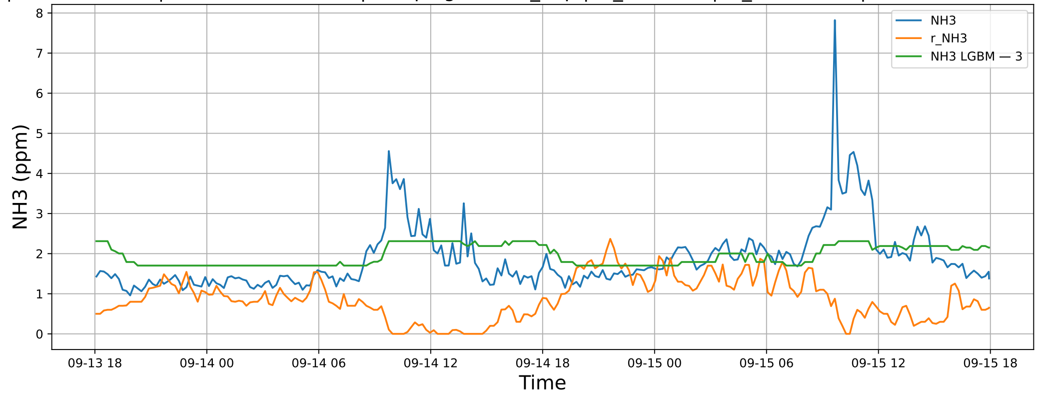


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

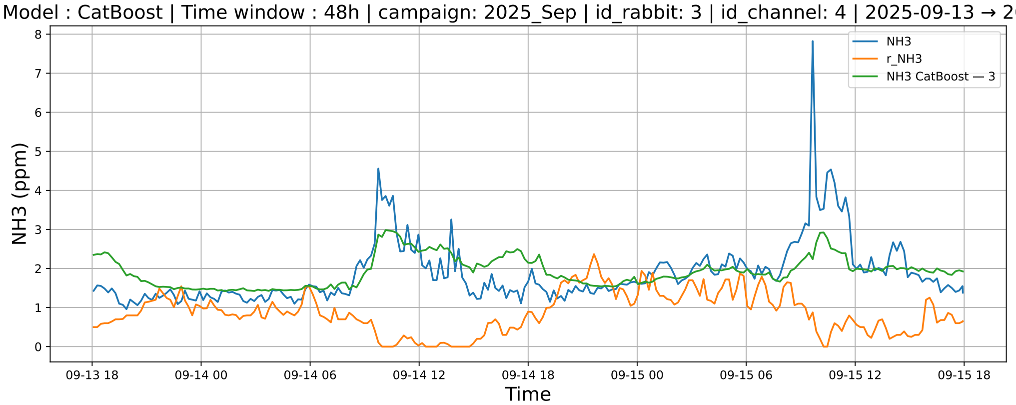


3.1.6.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

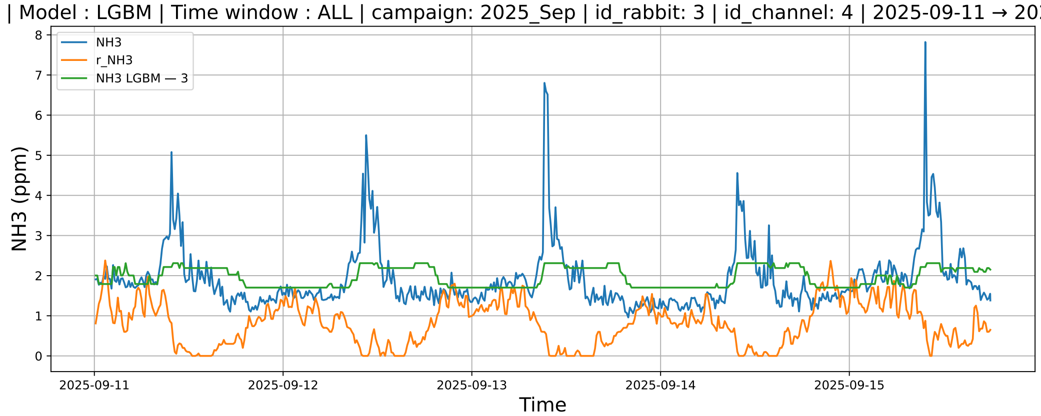


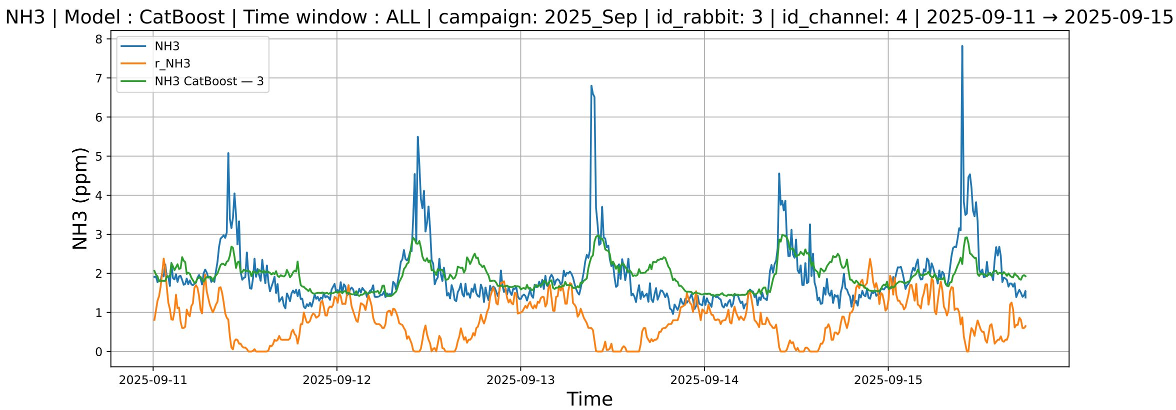
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.1.6.3 Time window : ALL

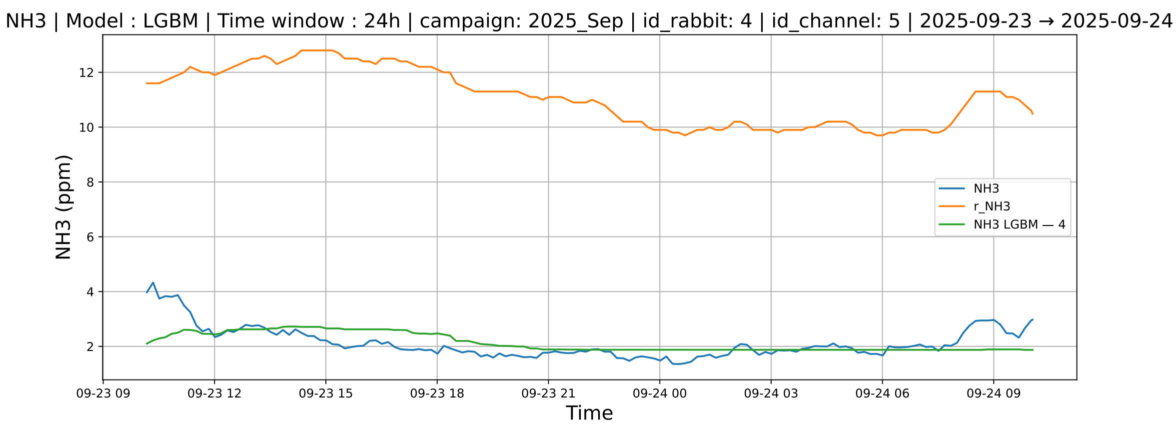
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

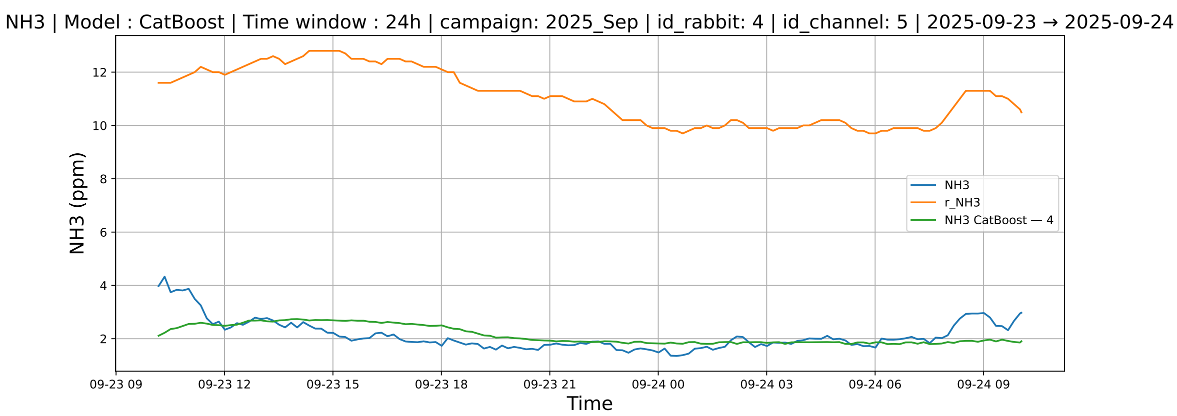




3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

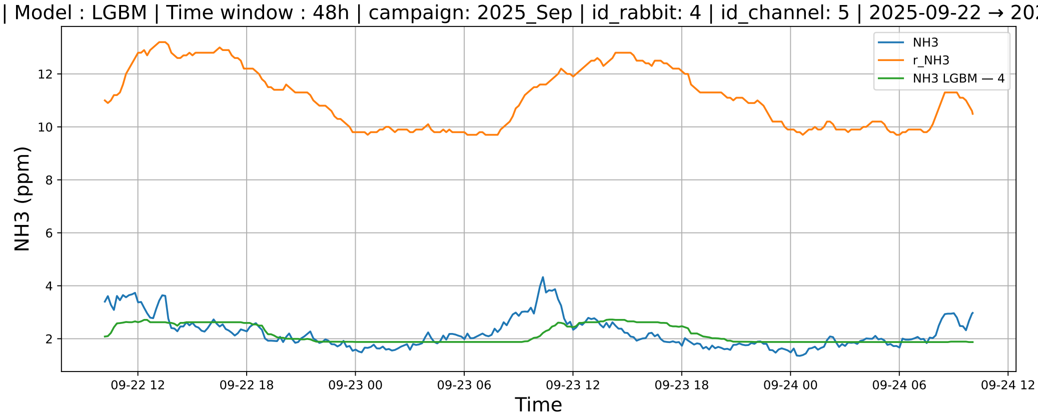
3.1.7.1 Time window : 24



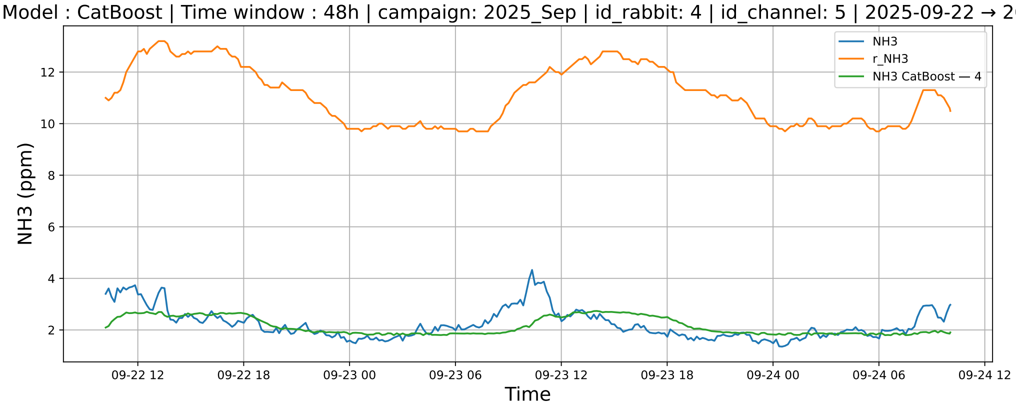


3.1.7.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

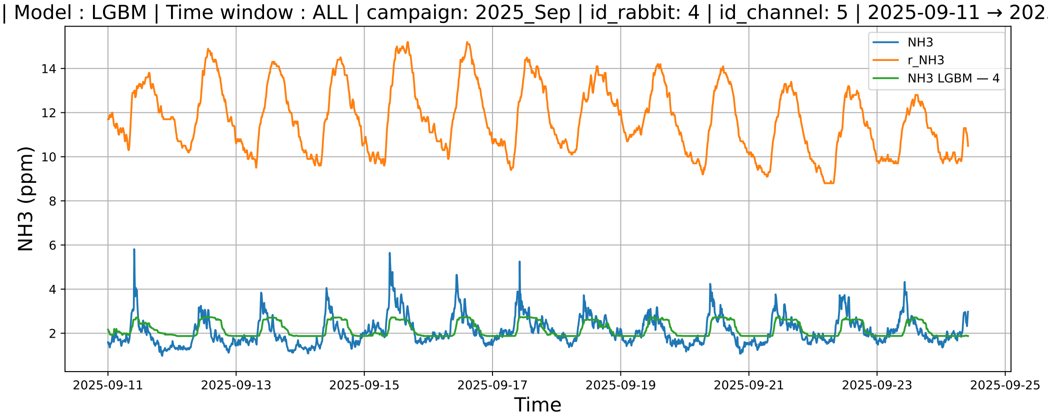


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

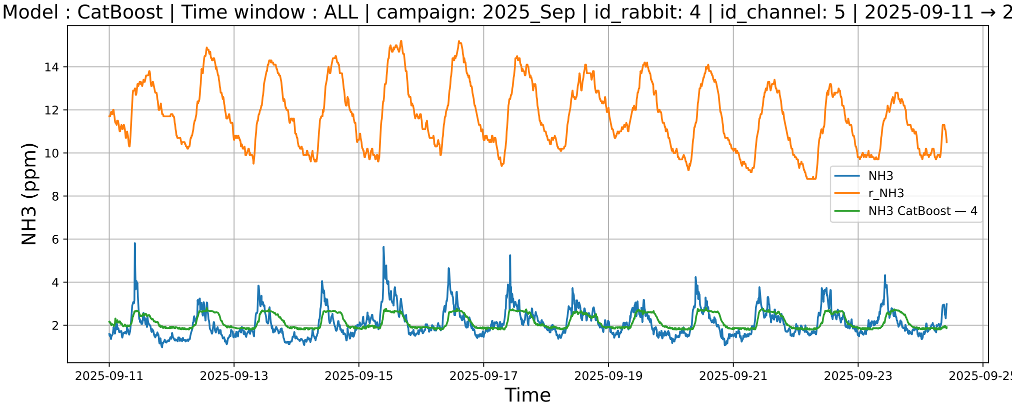


3.1.7.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



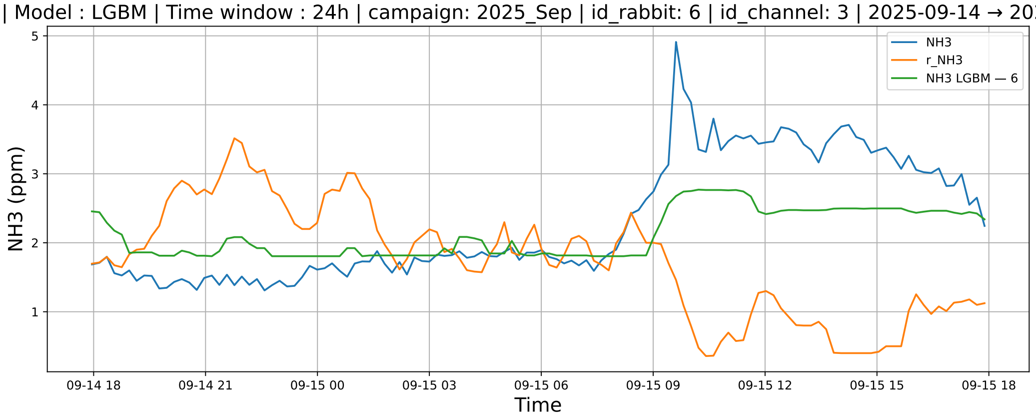
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



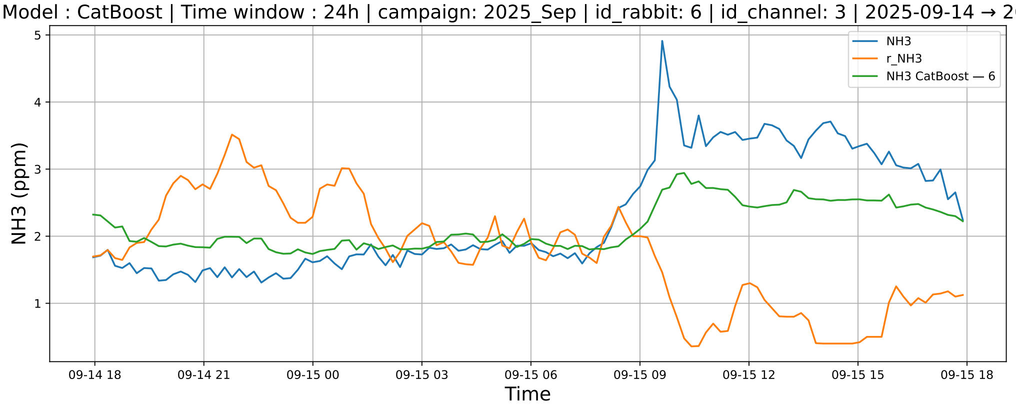
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

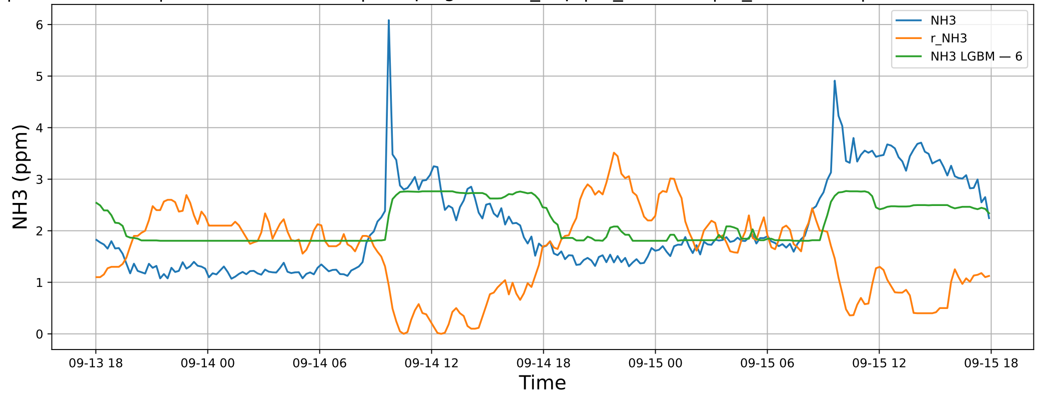


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

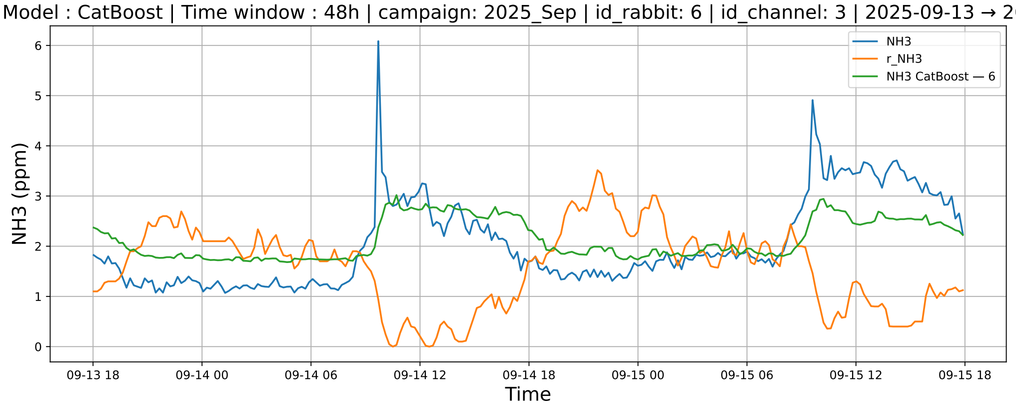


3.1.8.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

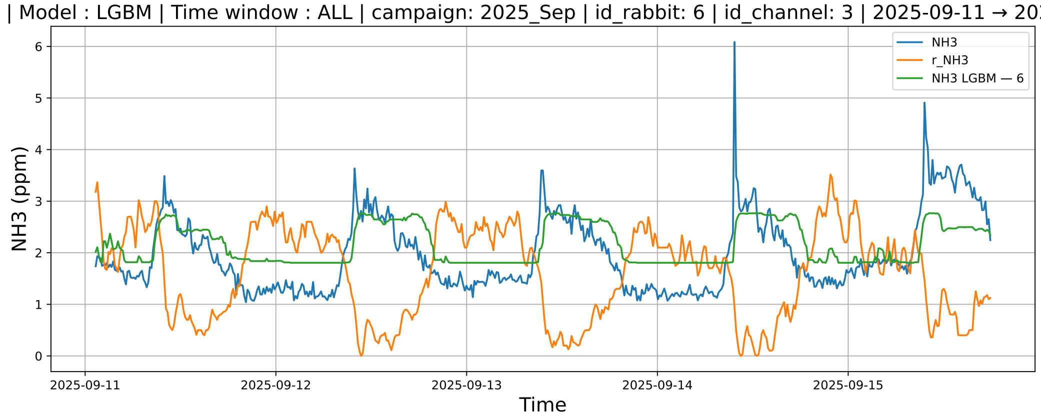


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

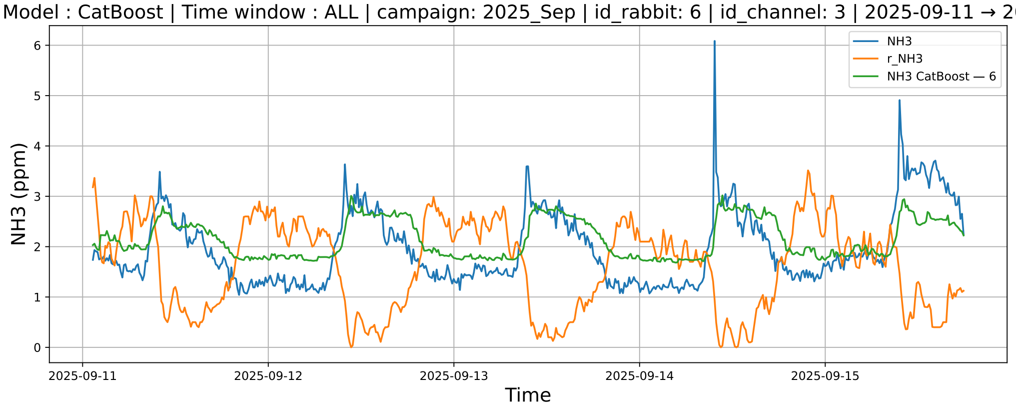


3.1.8.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

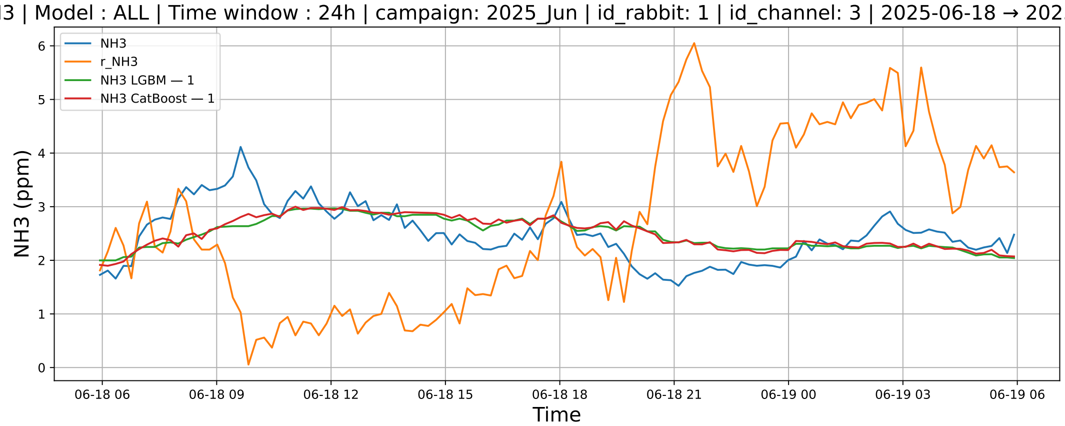


3.2 Compare plots

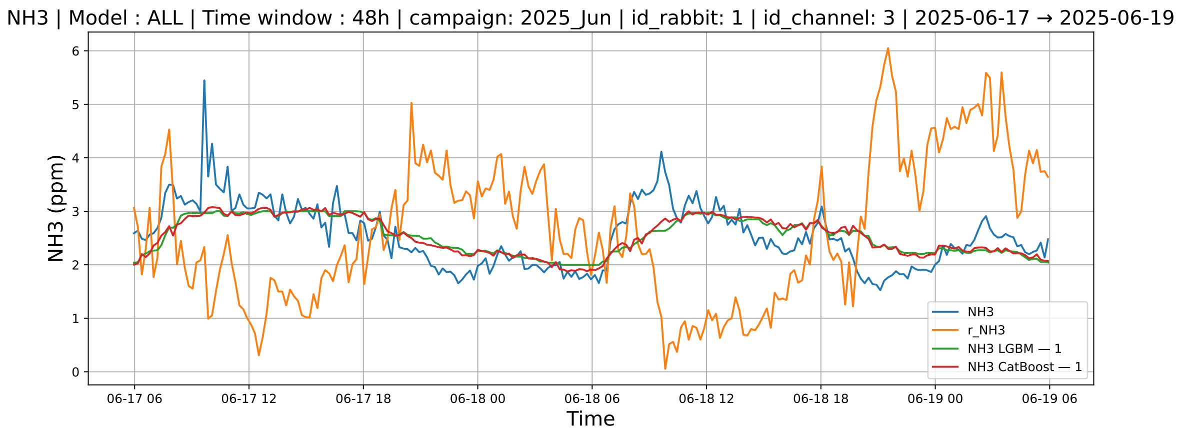
3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.2.1.1 Time window : 24

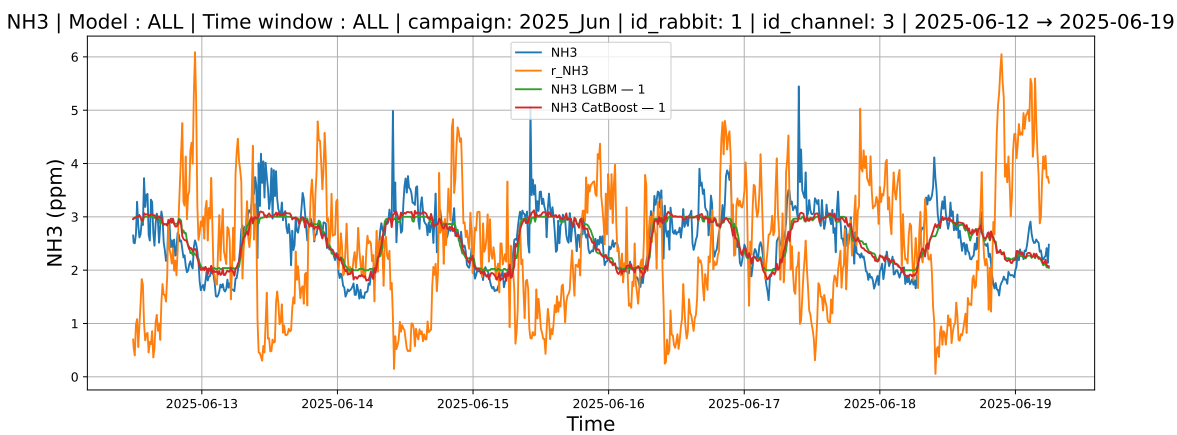
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



3.2.1.2 Time window : 48



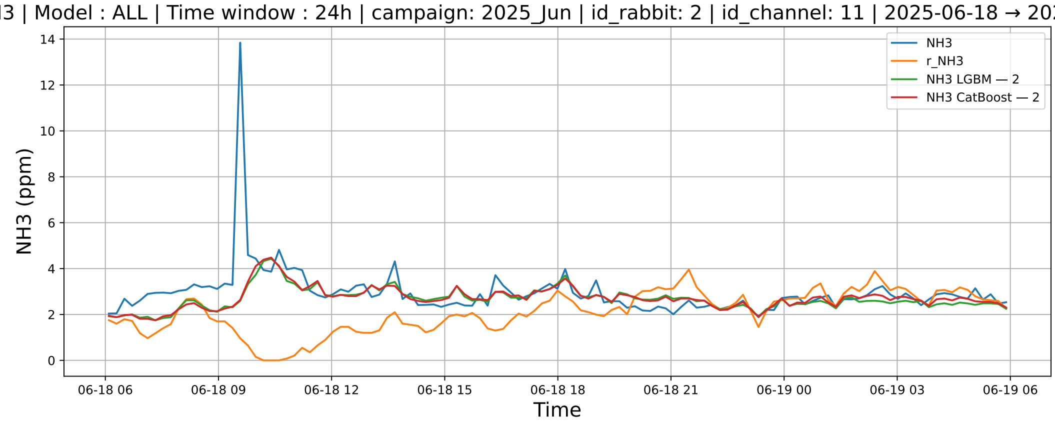
3.2.1.3 Time window : ALL



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

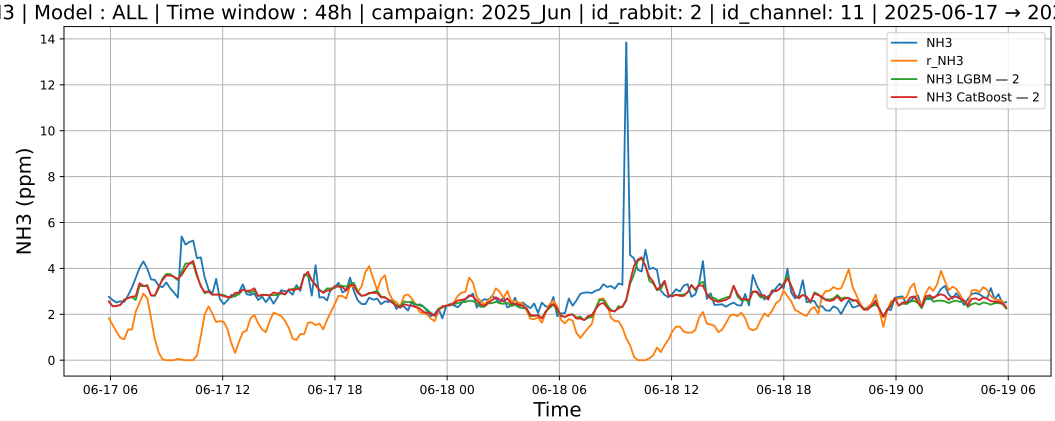
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



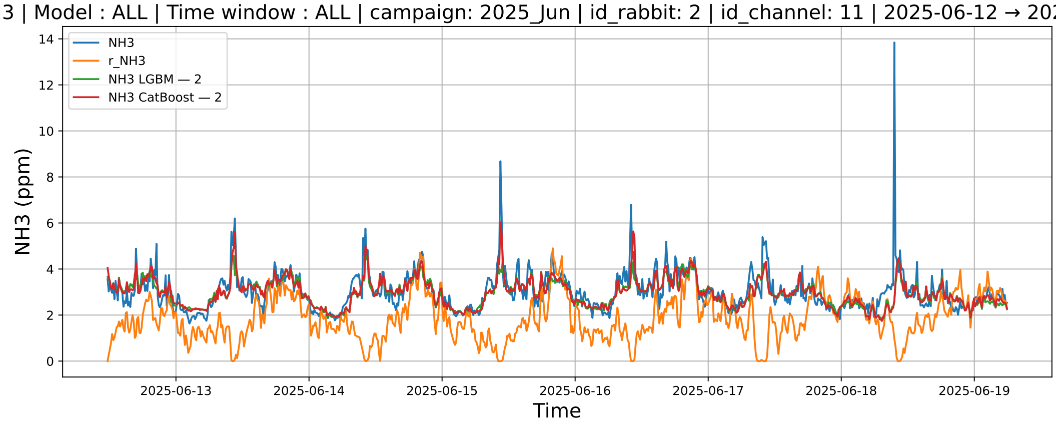
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



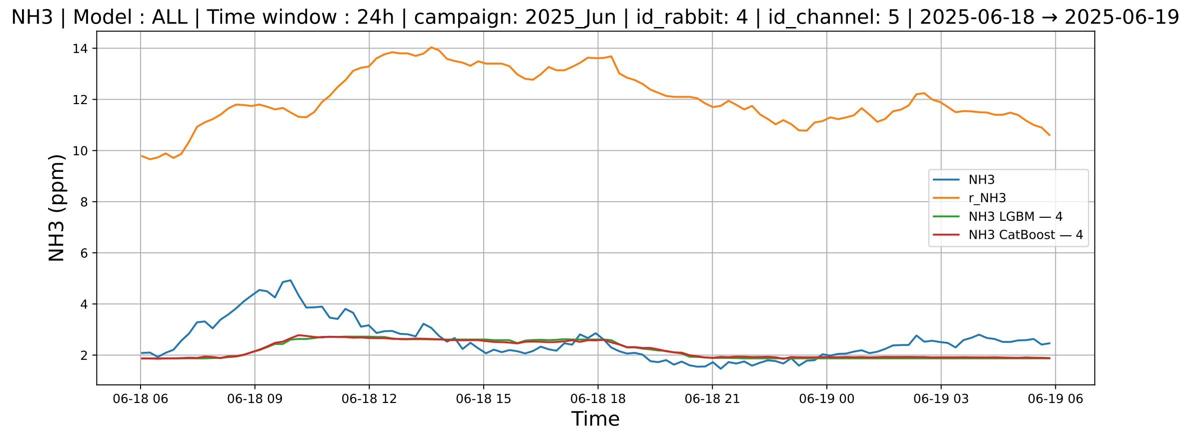
3.2.2.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

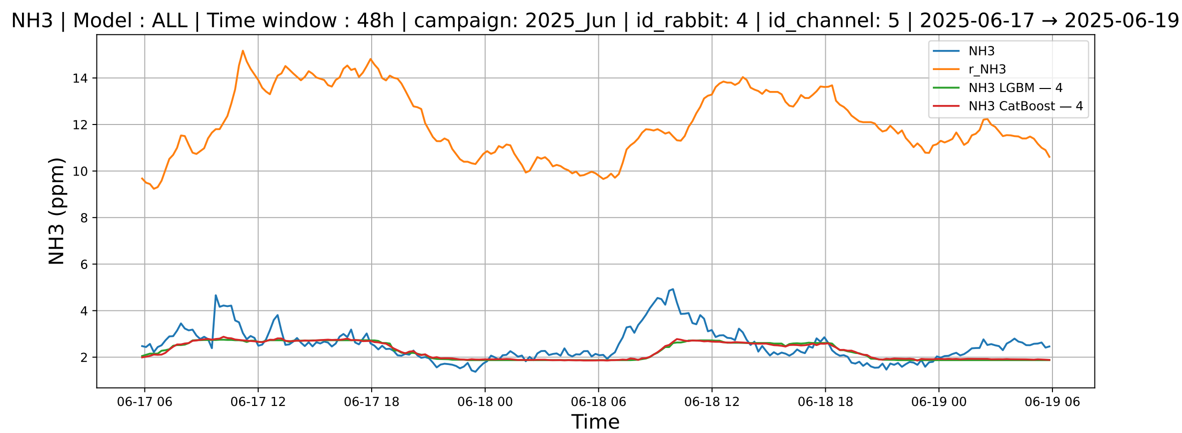


3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

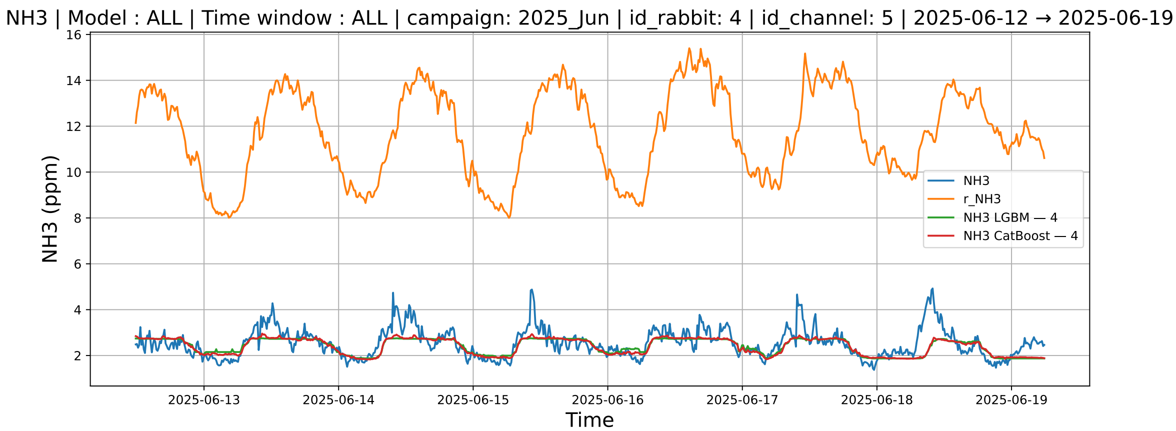
3.2.3.1 Time window : 24



3.2.3.2 Time window : 48

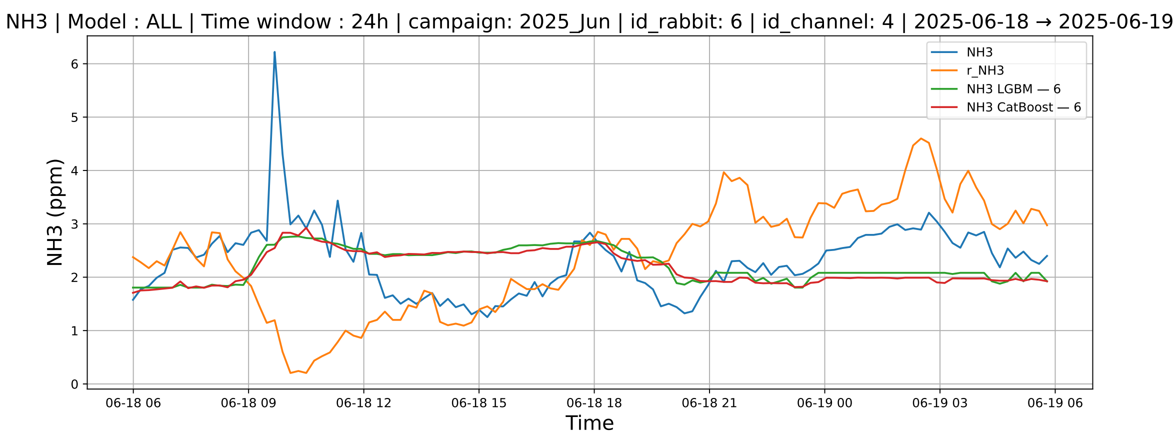


3.2.3.3 Time window : ALL

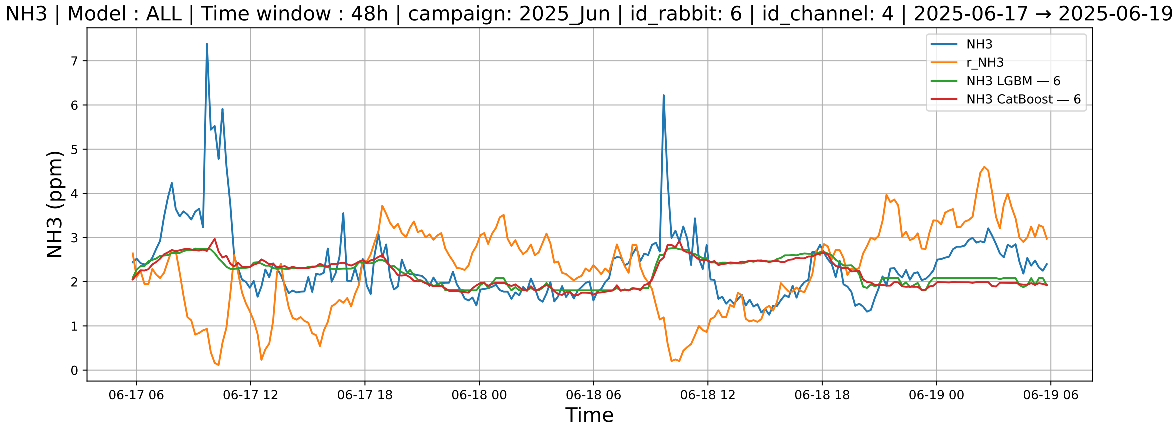


3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

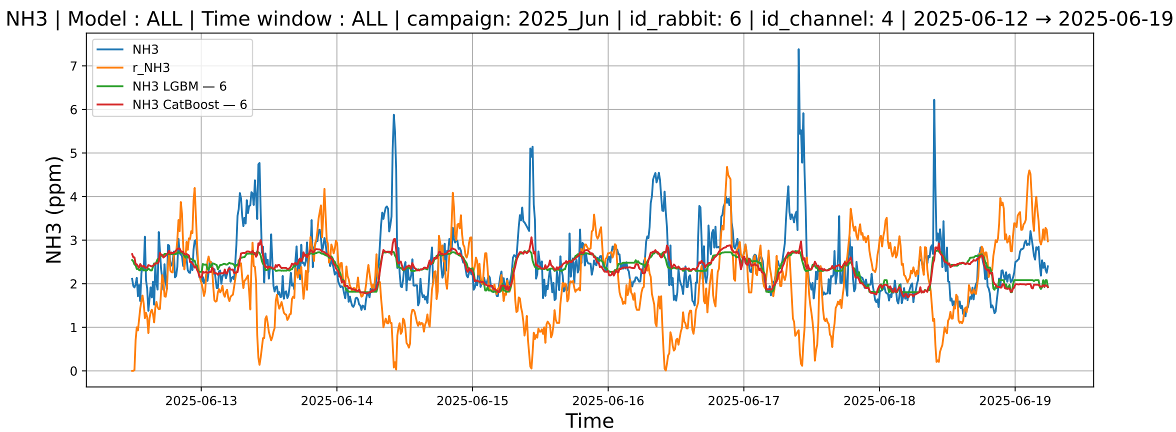
3.2.4.1 Time window : 24



3.2.4.2 Time window : 48



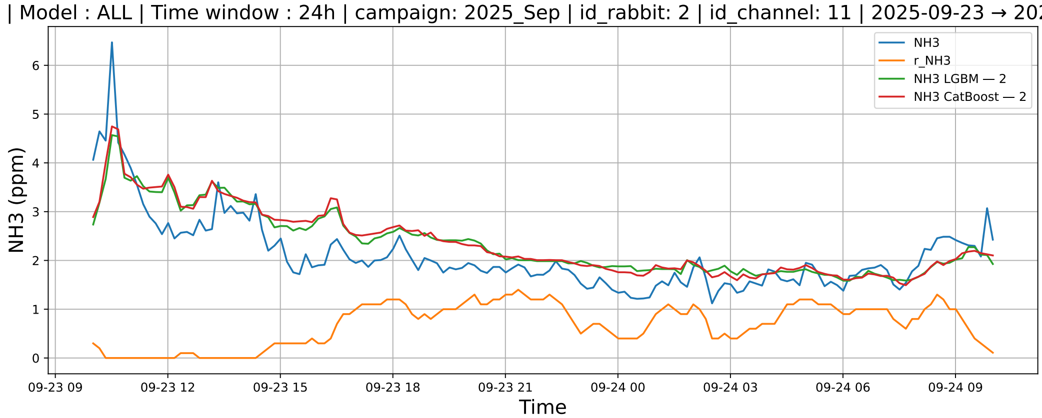
3.2.4.3 Time window : ALL



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

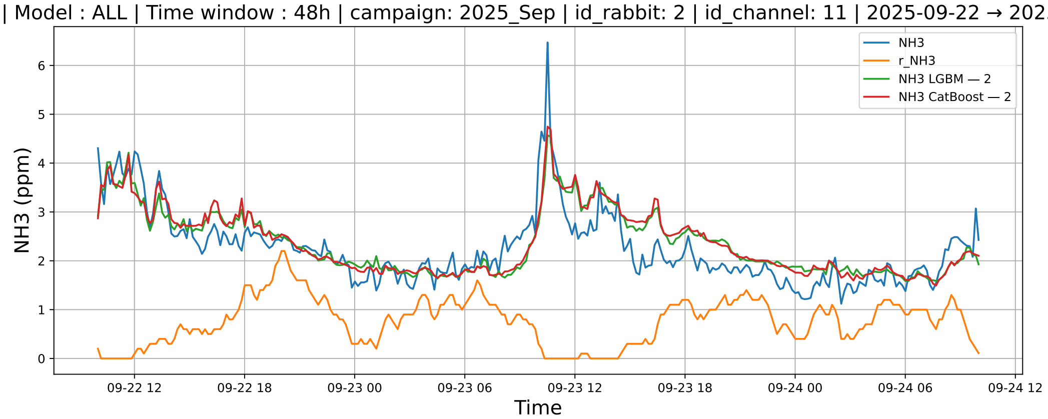
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



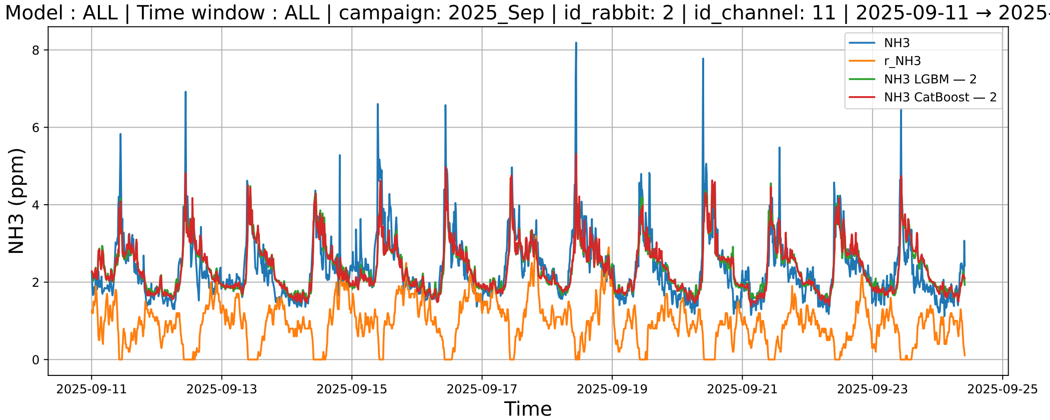
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



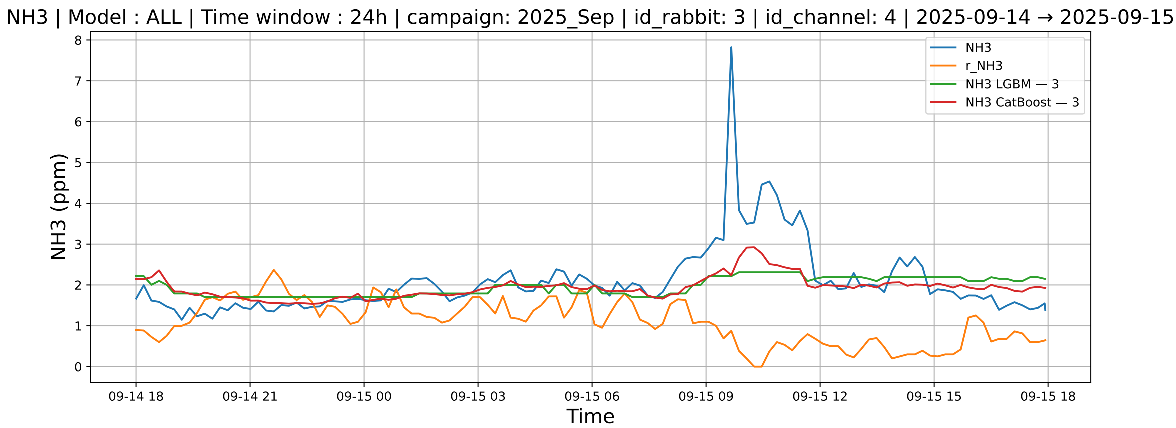
3.2.5.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

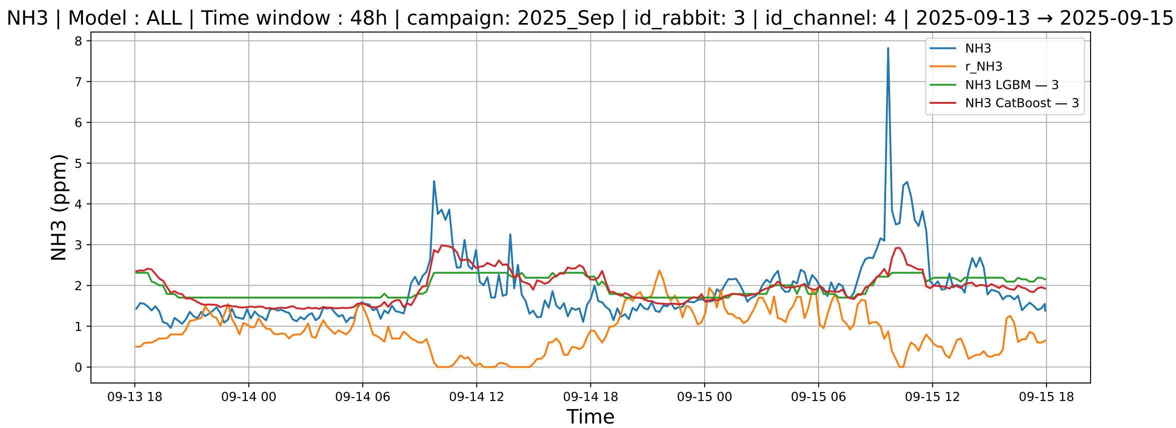


3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

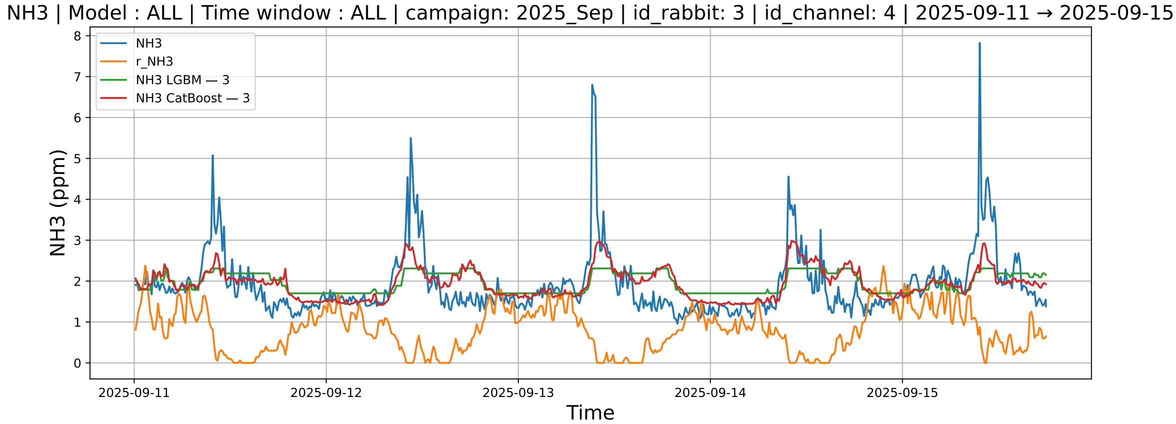
3.2.6.1 Time window : 24



3.2.6.2 Time window : 48

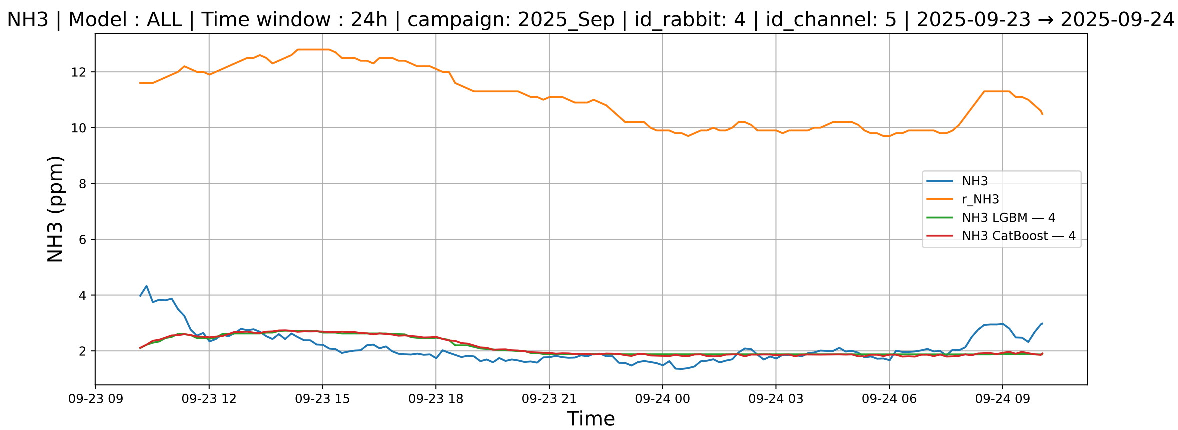


3.2.6.3 Time window : ALL

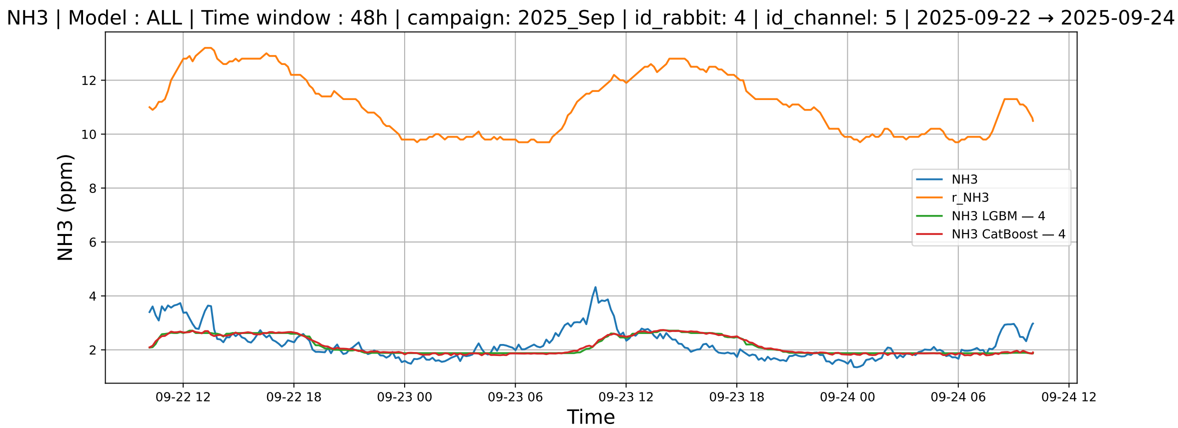


3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

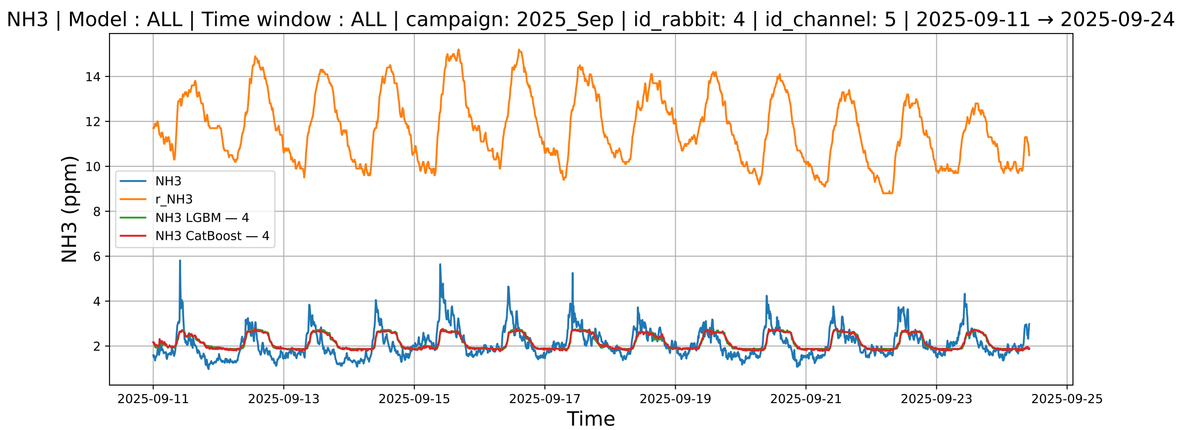
3.2.7.1 Time window : 24



3.2.7.2 Time window : 48



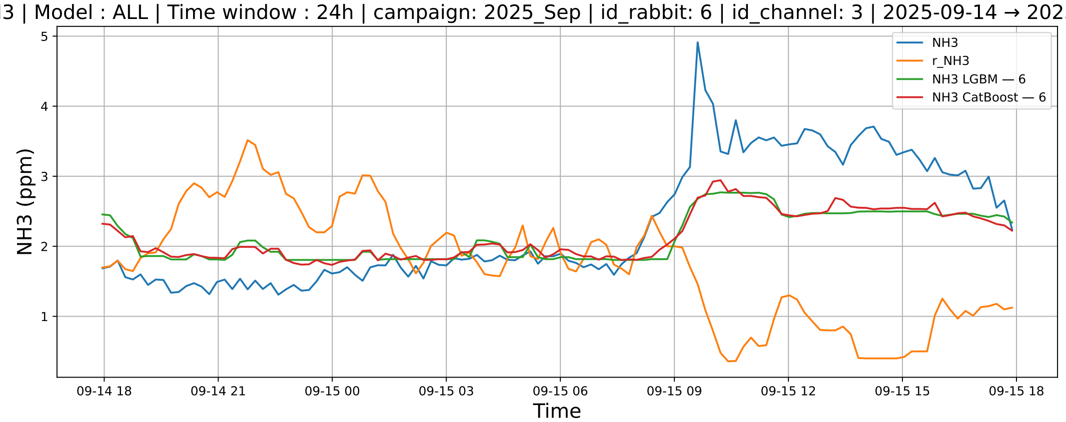
3.2.7.3 Time window : ALL



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

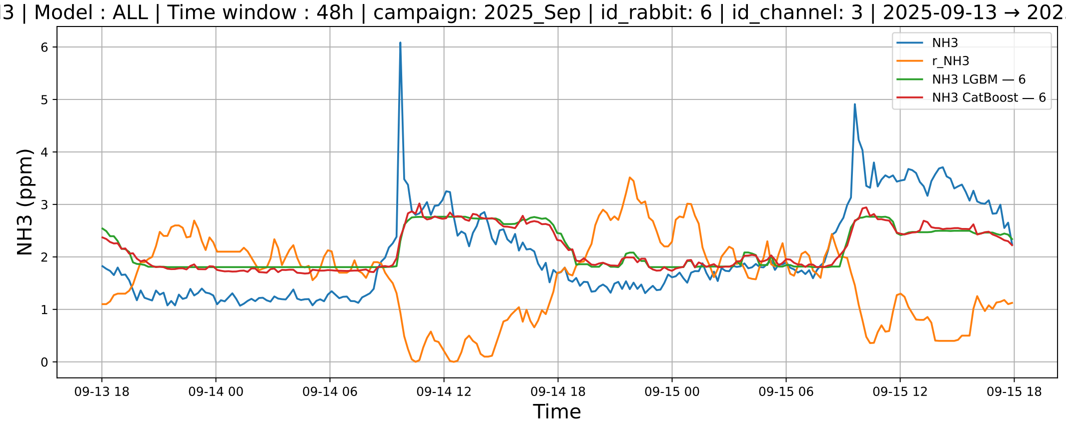
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15